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Audited Semantic Degradation in LLM-Agent Networks

Measurement Sensitivity, Persona Composition, and Network Conditions

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Abstract

This thesis examines how automated claim-recoverability scores change when large language model (LLM) agents transmit climate and geoengineering messages. Its primary contribution is a measurement-sensitivity audit of a prior auditor–node instrument. Secondary analyses describe persona compositions, eight bundled network conditions, and a separate 63-node literature-informed hashtag graph. The study does not test whether an online firestorm occurred.

The canonical Phase 2 analysis retained one run per configuration, comprising 288 configurations and 1,728 article cells over eight processing ticks. It classified 292 cells with at most one scored event as unscorable under the predefined hatch rule. Continuous and dual-discrete cell mean MI remained separate because missing items receive different penalties and a common threshold is not equivalent across the instruments. Dual-discrete scores exceeded continuous scores on every network-condition aggregate. This consistent difference indicates whole-pipeline sensitivity, not an increase on a common measurement scale or a causal instrument effect. Conspiracy-oriented personas and conspiracy-heavy mixes scored above scientific personas and the conspiracy-free mix in both instruments. Network conditions conditioned these contrasts, but no universal ranking emerged.

The D-net stress test used four configurations and eight article-level cells on a 63-node constructed hashtag graph. Conspiracy-only assignments exceeded mixed assignments in all four like-for-like comparisons. This recurrence is not a replication or digital-twin validation. Across the main-grid runs, 203,928 calls succeeded and client invocations recorded 64,270 HTTP-error responses; 121 of 288 runs were affected and 36,194 audit-eligible events remained unscored. No auditor parse/all-correct fallback was observed. The continuous H–He difference changed from +0.1549 canonically to -0.0295 in zero-error runs and is not robust. The dual-discrete difference remained positive (+0.3986 canonical; +0.4695 zero-error). These conditioned comparisons do not identify causal or instrument-independent effects. No human factuality, realism, or persona-validity evaluation was conducted.

Zusammenfassung

Diese Arbeit untersucht, wie sich automatisierte Kennwerte der Anspruchswiedergewinnbarkeit ändern, wenn Agenten auf Basis großer Sprachmodelle (LLM) Klima- und Geoengineering-Texte weitergeben. Der Hauptbeitrag ist eine messmethodische Sensitivitätsprüfung eines zuvor veröffentlichten Auditor-Knoten-Instruments. Nachgeordnete Analysen beschreiben Personenkompositionen, acht gebündelte Netzwerkbedingungen und einen gesonderten, literaturinformierten 63-Knoten-Hashtag-Graphen. Die Studie prüft nicht, ob ein Online-Firestorm stattgefunden hat.

Die kanonische Phase-2-Auswertung behielt einen Lauf je Konfiguration: 288 Konfigurationen und 1.728 Artikelzellen über acht Verarbeitungstakte. 292 Zellen mit höchstens einem bewerteten Ereignis wurden nach der vorab festgelegten Schraffurregel als nicht auswertbar klassifiziert. Stetige und dual-diskrete zellweise Mittelwerte des Misinformationsindex (MI) blieben getrennt, weil fehlende Items unterschiedlich bestraft werden und eine gemeinsame Schwelle über die Instrumente nicht äquivalent ist. Dual-diskrete Werte lagen auf jedem Aggregat der Netzwerkbedingungen über den stetigen Werten. Dieser durchgängige Unterschied zeigt eine Sensitivität der gesamten Pipeline, keinen Anstieg auf einer gemeinsamen Messskala und keinen kausalen Instrumenteneffekt. Konspirativ ausgerichtete Personas und konspirationslastige Mischungen lagen in beiden Instrumenten über wissenschaftlichen Personas und der konspirationsfreien Mischung. Netzwerkbedingungen modulierten diese Kontraste; eine universelle Rangfolge entstand nicht.

Der D-Net-Stresstest umfasste vier Konfigurationen und acht artikelbezogene Zellen auf einem konstruierten 63-Knoten-Hashtag-Graphen. Rein konspirative Zuweisungen lagen in allen vier paarweisen Vergleichen über gemischten Zuweisungen. Dieses Wiederkehren ist keine Replikation und keine Validierung eines digitalen Zwillings. Über die Hauptgitterläufe waren 203.928 Aufrufe erfolgreich; die Clientaufrufe verzeichneten 64.270 HTTP-Fehlerantworten. 121 von 288 Läufen waren betroffen, und 36.194 auditfähige Ereignisse blieben unbewertet. Ein Auditor-Parse- bzw. Alles-korrekt-Rückfall wurde nicht beobachtet. Der stetige H–He-Unterschied wechselte von +0,1549 im kanonischen Satz auf –0,0295 in fehlerfreien Läufen und ist nicht robust. Der dual-diskrete Unterschied blieb positiv (+0,3986 kanonisch; +0,4695 fehlerfrei). Diese bedingten Vergleiche identifizieren keine kausalen oder instrumentunabhängigen Effekte. Es wurde keine menschliche Bewertung von Faktentreue, Realismus oder Personengültigkeit durchgeführt. Die Misinformation Propagation Rate (MPR) bleibt die knoten- bzw. zweigweise Mittelung des MI und ist keine zeitnormierte Rate.

Individual contribution and prior-work boundary

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Prior co-authored work (excluded from the thesis originality claim). The auditor-node engine, the Misinformation Index (MI), the Misinformation Propagation Rate (MPR), and the general-news linear-chain study are co-authored prior work (Maurya et al., 2025). The bibliographic record lists Raj Gaurav Maurya as first author together with Vaibhav Shukla, Raj Abhijit Dandekar, Rajat Dandekar, and Sreedath Panat. **[CANDIDATE MUST CONFIRM: no additional unpublished co-author claims apply to that paper; first-author status matches the version submitted with this thesis.]** This thesis does *not* claim sole authorship of that paper, of the engine, of MI, or of MPR.

Traceable thesis-layer artefacts. The climate corpus, Phase 2 campaign, Pfeffer operationalisation, composition analysis, measurement-sensitivity audit, and D-net stress test form the thesis research layer. The following allocations are inferred from git history of https://github.com/RajGM/LLM_Society and from files under `thesisExperiment/`; they remain unconfirmed until the candidate and supervisors sign them.

Conception

Proposal extract names Raj Gaurav Maurya as candidate for a TUM M.Sc. Data & Society thesis under Prof. Dr. Jürgen Pfeffer and Dr. Ramit Debnath, with a climate/geoengineering simulation layer. **[CANDIDATE MUST CONFIRM: independent conception of the climate layer, factorial design, and estimands; record any supervisor-originated design decisions.]**

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Campaign execution

The canonical Phase 2 harvest (288/288 configurations; four D-net configurations) is archived under `thesisExperiment/runs_phase2/` and `thesisExperiment/results_phase2/`. Harvest-completion commits in this clone are predominantly Cursor Agent. **[CANDIDATE MUST CONFIRM: who launched, monitored, and accepted the live gpt-4o-mini campaign.]**

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Abbreviations

D-net	Separate 63-node constructed hashtag graph.
DTFS	Digital Twin Fidelity Score (implementation diagnostic).
H	Homogeneous persona assignment.
He	Heterogeneous persona assignment.
IFD	Information Fidelity and Distortion record.
LLM	Large language model.
MI	Misinformation Index (automated claim-recoverability score).
MPR	Misinformation Propagation Rate (node- or branch-level mean MI).
T2c	Phase 2 continuous-instrument campaign.
T2d	Phase 2 dual campaign with a discrete headline.
k^*	First scored tick above MI 3 sustained through the last later scored observation within eight ticks.

1 Introduction

1.1 Problem and motivation

Online misinformation changes as people repeat, reinterpret, and contest it. This process matters when a scientific proposal becomes attached to an established conspiracy narrative. Solar geoengineering provides such a case. Public discussion of stratospheric aerosol intervention and Harvard’s SCoPEX project has intersected with claims about chemtrails, weather control, and elite manipulation (Debnath et al., 2023). Their presence in public discourse raises a policy-relevant question about how scientific proposals become attached to narratives of weather control and elite manipulation. The relevant problem is therefore not only whether false content travels. It is also how network structure and social identity shape factual change during transmission.

Online-firestorm theory motivates attention to clustered, rapid, identity-structured communication. This thesis does not test whether a firestorm occurs, because it does not measure negative word-of-mouth, empirical communication speed, or cross-media escalation. It instead uses selected firestorm mechanisms as design prompts for a controlled study of how factual claims change during synthetic transmission. Pfeffer, Zorbach, and Carley describe an online firestorm as a sudden volume of negative word of mouth directed at a person, company, or group (Pfeffer et al., 2014). Their Outlook identifies seven interrelated factors: speed and volume, binary choices, network clusters, unrestrained information flow, lack of diversity, cross-media dynamics, and network-triggered decision processes. The theory explains why particular communication environments can intensify collective reactions. It does not specify a generative agent model or a quantitative tipping rule.

Geoengineering discourse supplies an empirical setting in which several of these mechanisms may matter. Debnath and colleagues analysed 814,924 English-language tweets containing the geoengineering hashtag between 2009 and 2021 (Debnath et al., 2023). They identified a prominent chemtrails discourse alongside climate-action and environmental-concern themes. They also documented temporal changes around solar-radiation-management projects and governance events. Their study maps observed discourse at scale. It does not provide controlled counterfactuals in which the same message enters different networks or identity compositions.

Agent simulation can provide such control, but it introduces a separate validity problem. Classical cascade models represent diffusion through states or activation probabilities (Kempe et al., 2003). They can isolate structural mechanisms, but they do not directly represent claim-level changes in natural language. Large language model agents can transform text while retaining persona and interaction context. Their outputs, however, reflect prompts, model behaviour, and scoring choices rather than observed human conduct. Generative social simulations must therefore expose their assumptions and restrict claims to the evidence that

their design supports (Larooij & Törnberg, 2025; Sargent, 2013).

Maurya and colleagues provide the immediate technical basis for this thesis. Their 2025 arXiv preprint introduced an auditor–node framework for persona-conditioned rewrites on linear chains (Maurya et al., 2025). The framework scores factual degradation through a misinformation index (MI) and aggregates scores through a misinformation propagation rate (MPR). That study used general-news domains, fixed chains of up to 30 rewrites, and a discrete question-answering auditor. Its framework and published findings are prior work. This thesis neither reintroduces those metrics nor reuses its crime and commercial results as new evidence.

1.2 Research gap

The three preceding research strands leave a specific methodological gap. Pfeffer et al. provide a theory of firestorm conditions without a generative testbed. Debnath et al. provide an observational account of geoengineering discourse without controlled interventions on network structure or identity composition. Maurya et al. provide a generative misinformation instrument without a climate corpus, a graph-topology comparison, or an explicit mapping to firestorm theory. The gap addressed here lies at their intersection. It concerns how to design and audit synthetic semantic propagation without treating simulation output as observed social behaviour.

The proposal framed this intersection as six factors: valence, surprise, identity alignment, network clustering, information echo, and temporal acceleration. Those labels are not Pfeffer et al.’s original seven-factor list. This thesis treats them as a computational remapping. It also records the parts that the simulation does not implement. Cross-media dynamics has no media-broadcast layer. Binary platform choices have no direct analogue because agents may forward, reinterpret, or drop a message. Surprise is not manipulated: every run uses the same single-node, single-time seed, and the temporal horizon remains fixed. The study therefore evaluates selected mechanisms rather than claiming to reproduce the complete firestorm theory.

Measurement creates a second gap inside the simulation. The prior discrete auditor counts omitted and contradicted answers. The present campaign also uses a continuous factual-recovery score. These instruments share an approximate range but not a common measurement scale. A difference between their outputs may reflect scoring design rather than a different social process. The analysis must therefore keep discrete and continuous MI separate.

Empirical grounding creates a third constraint. The public Debnath artefacts available to this study did not yield hydrated tweet text or conversation cascades. The thesis consequently uses theory-faithful behavioural-profile reductions. It also uses a 63-node hashtag co-occurrence graph as a structural fallback. Neither object is an empirical user cluster or a retweet network. This distinction prevents the simulation from being described as a digital twin of Twitter.

1.3 Aim and analytical approach

This thesis evaluates an exploratory testbed for audited semantic degradation in climate and geoengineering messages. The testbed extends the prior auditor–node framework with climate and geoengineering seeds, Debnath-grounded behavioural profiles, and configurable network topologies. It maps the firestorm theory to simulation controls, held conditions, and measured outcomes. Its primary question is whether conclusions about persona composition and network conditions remain stable under two auditor regimes. Composition and network comparisons are secondary descriptions of bundled configurations.

The main campaign crosses eight graph topologies with two identity designs and two scoring modes. The topologies are a linear chain, ring, Erdős–Rényi graph, small-world graph, scale-free graph, echo chamber, polarised graph, and hierarchy. The homogeneous arm assigns one of 12 behavioural profiles to all eight nodes. The heterogeneous arm assigns one of six eight-profile mixtures. Each configuration processes six climate or geoengineering articles over eight ticks. The resulting design contains 1,728 article-level cells. All cells use `gpt-4o-mini`. The parser retains one stochastic run per configuration. Although configurations record seed 42, several graph builders and action sampling use unseeded randomness.

The analysis reports MI and mean node MPR within each scoring mode. It defines k^* as the first tick at which mean MI exceeds 3 and remains above 3 for every later observed tick. This quantity is a right-censored simulator clock. It is not an estimate of ignition time on social media. The analysis also retains 292 cells with at most one scored event as hatched dead cascades. It excludes those cells from live-cell means but does not recode them as zero.

A separate D-net analysis places the same agents on the 63-node hashtag graph. This analysis compares simulated cascade shape with the documented fallback structure. It also compares homogeneous conspiracy profiles with a mixed profile assignment on the same edge set. The comparison concerns graph structure and simulated auditor scores. Debnath et al. report no empirical MI or MPR against which those scores could be validated.

1.4 Research questions

The study addresses four research questions. Each question follows from a design choice that the available evidence can evaluate.

RQ1 How sensitive are claim-recoverability levels and network-condition orderings to the continuous and dual-discrete auditor regimes?

RQ1 treats scoring mode as a measurement question. It compares modes without averaging their values or interpreting their gap as a third misinformation measure.

RQ2 Within each auditor regime, how do live-cell mean MI and within-window k^* occurrence vary across named persona families and heterogeneous compositions?

RQ2 compares persona families and named mixtures. It treats composition as substantive and H or He as an assignment description.

RQ3 How do the eight bundled network conditions differ in same-regime H/He contrasts, live-cell mean MI, and scorable-cell survival?

RQ3 compares the eight generated topologies. It reports dead-cell rates with live-cell means because cascade survival changes the observed population.

RQ4 What can the 63-node constructed hashtag-graph stress test establish about recurring composition patterns, and where does that comparison fail?

RQ4 separates constructed graph properties from simulated content scores. It tests a documented fallback rather than claiming reconstruction of Debnath et al.’s tweet cascades.

These questions support descriptive comparisons. They do not support population estimates or causal effects. The single stochastic run also precludes confirmatory inference about topology, identity, or scoring mode.

1.5 Contributions

This thesis makes four bounded contributions.

1. It provides an explicit mapping from Pfeffer et al.’s seven firestorm factors to the simulation’s varied, held, measured, and absent elements. The mapping corrects the proposal’s six-factor shorthand and retains cross-media dynamics as an unimplemented boundary.
2. It documents a climate and geoengineering simulation layer for the prior auditor–node framework. This layer contains six campaign articles, 12 homogeneous behavioural profiles, six heterogeneous mixtures, eight topologies, and reproducible configuration files. The profiles reduce themes reported by Debnath et al.; they are not clusters estimated from 814,924 tweets.
3. It evaluates identity composition and network topology under two distinct factual-degradation instruments. The analysis preserves dead cascades, separates continuous from discrete scores, and distinguishes persona composition from the broad homogeneous–heterogeneous contrast.
4. It links a 63-node hashtag co-occurrence structure to a custom simulation graph and audits the limits of that comparison. This analysis reports structural mismatch as well as similarity. It does not equate auditor MI with Twitter behaviour.

These contributions concern operationalisation, experimental design, and transparent measurement. They do not include the original auditor–node architecture, MI, MPR, or the published linear-chain findings. They also do not establish a validated model of public response to geoengineering.

1.6 Scope and limitations

The empirical scope is deliberately narrow. The campaign uses six authored or curated climate articles and a finite set of prompt-based behavioural profiles. It runs eight-node graphs for eight processing ticks with one model and one retained realisation per configuration. The design changes text through LLM role-play. It does not model stable human belief, individual psychology, or platform recommendation systems.

The data scope is narrower than the proposal anticipated. The study did not hydrate the 814,924 tweet identifiers. It did not estimate HDBSCAN profiles, reconstruct conversation trees, or reproduce Debnath et al.’s temporal series. The hashtag graph records co-occurrence structure, not retweet transmission. The simulation clock cannot be converted into minutes, days, or empirical hop distance.

The inferential scope follows these design limits. The results describe one stochastic campaign. They do not warrant significance claims, robust rankings, or general laws about homogeneous and heterogeneous groups. The two auditors remain model-based measurements. Their disagreement is evidence about the instrument, not an independent factual ground truth. The k^* measure is conditional on the eight-tick observation window. A value of “none” means that no crossing sustained through the final later scored observation appeared within that window.

1.7 Thesis roadmap

Chapter 2 develops the theoretical and empirical background. It covers online firestorms, climate and geoengineering discourse, diffusion models, LLM-agent simulation, and automated factuality assessment. Chapter 3 maps Pfeffer et al.’s seven factors to the thesis design. Chapter 4 separates the inherited system from the climate-specific design and documents the articles, behavioural profiles, hashtag graph, campaign, auditor regimes, hatch rule, and k^* . Chapter 5 answers RQ1 by establishing instrument sensitivity, RQ2 by comparing named persona compositions, RQ3 by examining topology-conditioned same-regime contrasts, and RQ4 through the separate D-net stress test. Chapter 6 interprets the findings in the same RQ1–RQ4 order. Chapter 7 addresses validity, ethics, and reproducibility. Chapter 8 answers the research questions and states the conditions required for stronger claims.

The thesis therefore proceeds from theory to operationalisation, from operationalisation to measurement, and from measurement to bounded interpretation. Its central claim is methodological. In this campaign, conclusions about semantic degradation depend on the auditor regime; persona-prompt composition is more informative than the pooled H/He label, and bundled network conditions alter survivor-only outcomes. The simulation does not represent the relevant human population.

2 Literature Review

This chapter locates the thesis at the intersection of misinformation diffusion, online-firestorm theory, geoengineering discourse, network science, and large-language-model (LLM) agent simulation. These research areas address different analytical objects. A model may reproduce cascade reach without representing changes in claims. Conversely, fluent agent output may appear socially plausible without providing a valid model of human behaviour. The review therefore connects substantive theory to the thesis’s observables and closes by stating the specific gap addressed by the experiments.

2.1 Misinformation as diffusion and semantic transformation

Misinformation is not a single technical object. The term can refer to false content, an actor’s belief, an exposure, a sharing decision, or a population-level diffusion pattern. Lazer et al. (2018) consequently call for research that joins individual cognition, institutions, and platform structure rather than treating falsity as a classification problem alone. Large-scale observational evidence confirms that diffusion is socially structured: Vosoughi et al. (2018) found that false news on Twitter travelled farther, faster, deeper, and more broadly than true news in their data. Novelty and emotional response helped distinguish false from true cascades. Bots accelerated true and false news at roughly the same rate; the differential spread was attributed primarily to humans. These findings motivate models with content and agent state, but they do not imply that every simulated cascade should reproduce the same ordering.

Classical diffusion models provide useful null formalisms. In the independent cascade model, an activated node receives one probabilistic opportunity to activate each neighbour (Kempe et al., 2003). Compartmental epidemic models (Kermack & McKendrick, 1927) and threshold models (Granovetter, 1978) reduce the state of a node to susceptible, active, adopted, or an opinion scalar. Their abstraction is an advantage for identifying conditions for reach or consensus, but an edge transmits a state rather than rewriting a proposition. This thesis instead studies a transmission in which the text itself can lose, invert, or acquire claims. The distinction is between *whether* something propagates and *what it becomes while propagating*.

Empirical cascade size is also not synonymous with virality. Goel et al. (2016) show that similarly popular diffusion events range from broadcast-like stars to multigenerational chains and define structural virality from pairwise distances in a diffusion tree. A scale-free contagion model did not recover the full empirical diversity they observed. This is a warning against explaining an outcome from a topology label alone. In the present study, event count, semantic degradation, and time to a severity threshold are therefore conceptually separate outcomes.

2.2 Online firestorms: theory and operationalisation

Pfeffer et al. (2014) define an online firestorm as a sudden discharge of large quantities of negative word-of-mouth and complaint behaviour directed at a person, organisation, or group. Their contribution is a qualitative synthesis of the mechanisms that make rapid, affective outrage possible. The paper’s outlook identifies *seven* interrelated factors: speed and volume of communication, binary choices, network clusters, unrestrained information flow, lack of diversity, cross-media dynamics, and network-triggered decision processes. It does not present an agent-based generative model, a semantic misinformation scale, or a tipping-time estimator.

The distinction is necessary because the thesis dimensions are not Pfeffer et al.’s original factors. The six simulation dimensions used in this thesis—valence, surprise, identity alignment, network clustering, information echo, and temporal acceleration—are a *computational remapping*, not Pfeffer et al.’s original list. Valence represents the affective character of firestorm messages. Identity alignment and information echo approximate lack of diversity and clustered reinforcement. Network clustering maps directly to local structure, while surprise and temporal acceleration separate seeding shocks from the speed of subsequent exchange. Cross-media dynamics has no equivalent in the present single-platform society and is a boundary condition. Nor does the simulator reproduce the original paper’s binary platform affordances in a literal way. The remapping is thus an operational hypothesis to be assessed, not a claim that all seven mechanisms have been implemented.

Later work has moved toward measuring observed firestorms. Nuortimo et al. (2020) propose a three-level scale based on a media-analytics case, and Nuortimo et al. (2025) develop dimensions of firestorm width, intensity, and peak duration with AI-assisted validation. These studies make event severity more explicit, but remain different from a claim-level instrument applied during a controlled generative cascade. The literature leaves open how a theory of rapid collective outrage should be connected to the semantic mutation of scientific information.

2.3 Geoengineering discourse as the substantive case

Solar geoengineering is an unusually demanding case for studying that connection. Discussion combines scientific uncertainty, climate politics, governance, distrust, and established conspiracy narratives. It therefore cannot be represented adequately as generic “fake news.” The principal empirical grounding is Debnath et al. (2023), who analysed 814,924 English-language tweets containing **#geoengineering** from 2009 to 2021. Using natural-language processing, toxicity analysis, word embeddings, and network analysis, they show that the false chemtrails narrative substantially shaped geoengineering discussion. Conspiracies spilled into debates in the United Kingdom, United States, India, and Sweden and connected with broader political themes. Positive emotions tended to increase following governance events, whereas negative and neutral emotions increased around project and experiment announcements; toxicity widened anti-solar-radiation-management spillovers.

Debnath et al. establish an observational association, not a causal diffusion model or an LLM-agent society. Their corpus is nevertheless valuable for defining plausible discourse tendencies and seed events. This thesis derives reduced belief profiles for conspiracy, climate-action, and

environmental-concern positions from that account. These profiles are *theory-grounded types*; they are not user clusters or fitted population proportions. The public data deposit contains tweet identifiers (Debnath, 2023). The thesis did not hydrate and recluster the 814,924 tweets and therefore does not claim to replicate Debnath et al.’s analysis, infer their social graph, or construct a Twitter digital twin. The DataCite record for version 1 of that deposit is Debnath’s 2022 “Geoengineering on Twitter” dataset (Debnath, 2022); version 2 is a 2023 update.

This separation supports a defensible use of the case. The observational study supplies the empirical phenomenon—chemtrails-related spillover, affect, and toxicity—while the simulation supplies counterfactual control over seeds, identity composition, and network structure. Agreement between the two would be evidence of external plausibility, not proof that the simulated agents are the users in Debnath et al.’s data.

2.4 Network topology, homophily, and echo

Topology changes who can expose whom and how many independent or reinforcing paths connect groups. Erdős–Rényi graphs provide a random-graph reference (Erdős & Rényi, 1960); Watts–Strogatz networks combine local clustering with short paths (Watts & Strogatz, 1998); and preferential attachment generates heavy-tailed degree distributions and hubs (Barabási & Albert, 1999). These models are not interchangeable proxies for real social media. They are controlled structural hypotheses whose degree distribution, clustering, path length, and seed position must be reported.

The content being transmitted also changes which topology should facilitate diffusion. Centola and Macy (2007) distinguish simple contagion, for which one contact may suffice, from complex contagion requiring reinforcement from multiple neighbours. Long weak ties can accelerate the former but fail to provide sufficiently wide bridges for the latter. Firestorm participation and conspiracy reinterpretation may involve social affirmation and are therefore not safely assumed to behave like infection over a single edge.

Agent composition and topology are joined by homophily. Similarity structures social ties and thereby the information, attitudes, and interactions available to people (McPherson et al., 2001). Empirically, selective exposure and homogeneous interaction can support polarised information communities: Del Vicario et al. (2016) relate emotional contagion and group polarisation in Facebook communities, while Törnberg (2018) models viral misinformation as a complex contagion. A high-level “echo chamber” label, however, is not a measurement. This thesis therefore distinguishes treatment variables (network generator and persona assignment) from observables such as edge homophily and community separation. Modularity compares within-group edge density with a degree-preserving null expectation (Newman, 2006); it indicates structural segregation, not the truth or extremity of group content.

2.5 Homogeneous and heterogeneous agent populations

Homogeneous-agent models isolate a response rule and simplify attribution, but they can turn an imposed persona into a population-wide mechanism. Heterogeneous populations permit cross-cutting exposure, specialist roles, and different propensities to reinterpret or resist a message. At the same time, heterogeneity is not intrinsically realistic. Results depend on which types are included, their prevalence, their network positions, and whether prompts create genuine behavioural variation or merely labels.

The immediate methodological predecessor is the auditor–node system of Maurya et al. (2025), available as a 2025 arXiv preprint. It compares persona-homogeneous and persona-heterogeneous linear rewrite chains and introduces the discrete Misinformation Index (MI) and branch-level Misinformation Propagation Rate (MPR). Those general-news linear-chain findings are *prior work*, not results of this thesis. The present work retains the auditor–node idea but changes the empirical object to geoengineering discourse, uses Debnath-grounded profiles, and introduces non-chain network treatments.

The broader literature demonstrates both the promise and the unresolved status of LLM societies. Social Simulacra populates prototypes of online communities (Park et al., 2022), while Generative Agents combines memory, reflection, and planning to create believable local interactions (Park et al., 2023). S³ simulates information, attitude, and emotion propagation against social-network data (Gao et al., 2023). OASIS adds platform actions, recommendation mechanisms, dynamic networks, and very large populations (Yang et al., 2024); AgentSociety similarly prioritises large-scale social environments (Piao et al., 2025). These systems expand scale and behavioural surface area, but scale does not by itself validate an agent as a model of a person.

Most directly, Li et al. (2026) condition LLM-agent forwarding on psychological traits and compare diffusion under different network structures, including an empirical Higgs community. Adjacent LLM-enhanced influence simulators such as LAID also treat information propagation as the focal process (Hu et al., 2024). Their focal state remains sharing and diffusion rather than audited claim-by-claim mutation in geoengineering text. The literatures are complementary: one asks whether an item is forwarded; this thesis asks how the item’s factual content changes as a function of identity mix and topology.

Persona conditioning also creates a validity problem. Language models can reproduce some subgroup response patterns when conditioned on demographic backstories (Argyle et al., 2023), and they can recover qualitative patterns from selected economic experiments (Horton, 2023). Such results support LLMs as inexpensive theory probes, not synthetic replacements for human participants. Training-data contamination, prompt sensitivity, stochasticity, stereotype amplification, limited within-group variance, and obedience to experimenter instructions all threaten behavioural interpretation (Argyle et al., 2023; Larooij & Törnberg, 2025). In this thesis, “homogeneous” and “heterogeneous” consequently describe prompt conditions in a computational experiment. They do not denote representative human populations.

2.6 Measurement validity: discrete and continuous instruments

The thesis’s central dependent variable is produced by an instrument, not observed directly. Construct validity concerns whether evidence supports the interpretation and use of a score. It cannot be established by naming a quantity “misinformation.” Cronbach and Meehl (1955) require a measure to sit in a nomological network connecting a construct to observable consequences, and Messick (1995) treats validity as an integrated argument about score meaning, generalisation, external relations, and consequences. For MI, relevant evidence would include item coverage, human agreement, expected relations with known distortions, stability across judges, and sensitivity to irrelevant wording.

QA-based factuality evaluation supplies a methodological warrant. Fabbri et al. (2022) show that question answering can assess factual consistency in summarisation more directly than lexical overlap. LLM judges can also agree substantially with human preferences, but exhibit position, verbosity, and self-enhancement biases (Zheng et al., 2023). Neither QA-based factuality evaluation nor reported LLM-judge agreement validates the instrument used here. This instrument asks an LLM judge to answer fixed binary questions about geoengineering and therefore measures a different construct in a different setting.

The distinction between discrete and continuous scoring must therefore remain visible. In discrete MI, each item is classified as recovered, missing, or contradicted and the item outcomes are aggregated into a bounded score. This format is inspectable and corresponds to explicit propositions, but it collapses uncertainty and treats all items as equally diagnostic. A continuous instrument can retain graded evidence or confidence and may detect smaller changes, but its numeric precision is not automatically substantive precision. It requires calibration and a defensible interpretation of distances between scores.

The statistical literature cautions that converting a genuinely continuous quantity into categories discards information, reduces power, and creates artificial discontinuities around a cut point (Altman & Royston, 2006; MacCallum et al., 2002). That warning applies to the thesis’s severity bands: the boundary at $MI > 3$ is an operational decision inherited from prior work, not a psychometrically established natural kind. Conversely, the warning does not prove that a continuous LLM score is valid. Discrete and continuous MPR are different instruments; they should not be pooled or treated as interchangeable observations merely because both have numeric output.

Shared generator and judge families add common-method dependence. A fluent rewrite may be rewarded because it resembles the judge’s preferred formulation, and a failed parse can masquerade as low misinformation if the implementation fails open. Credible validation therefore requires frozen test cases, fail-closed parsing, deterministic judging where possible, manual inspection, human coding of a sample, and agreement reported at the decision boundary used for the substantive claim. Until such evidence exists, MI and MPR are best described as *simulation audit scores*, not validated measures of human misinformation belief (Lewandowsky et al., 2012).

2.7 Validation of generative social simulation

Verification asks whether the program implements the intended model; validation asks whether the model is adequate for its intended purpose. Sargent (2013) separates data validity, conceptual-model validity, computerised-model verification, and operational validity. For agent-based social simulation, Richiardi et al. (2006) further stress transparent model description, sensitivity analysis, replication, and a validation test matched to the claim. Windrum et al. (2007) show that empirical validation can involve several strategies and that no single fitted trajectory settles model adequacy.

Applied here, validation has at least four levels. First, software checks must establish that topology, persona assignment, intervention timing, and auditor aggregation behave as specified. Second, internal validation asks whether changing a theoretically relevant input produces stable, interpretable changes rather than parser artefacts or dead cascades. Third, output validation compares patterns with held-out empirical regularities, such as plausible cascade structure or Debnath’s discourse themes. Fourth, measurement validation asks whether humans and independent instruments support the MI interpretation. Face-plausible agent prose is evidence for none of these levels by itself.

The intended use defines the attainable claim. A small, replicated, audited model may function as a mechanism probe: it can test whether a proposed combination of topology and identity is sufficient to produce a pattern under specified prompts. Without calibrated agent distributions and reconstructed empirical cascades, it cannot estimate the prevalence or causal effect of that mechanism among Twitter users. Failed or near-empty cascades are execution or design outcomes to diagnose, not evidence that a topology prevents misinformation. Stochastic LLM output also requires multiple seeds or models before a directional contrast can be considered stable.

2.8 Synthesis and research gap

The literature leaves a precise intersection underdeveloped. Firestorm theory identifies structural conditions for rapid, affective outrage but does not provide a generative, claim-level model (Pfeffer et al., 2014). Network diffusion research explains activation, reinforcement, and cascade shape but usually treats content as a binary or fixed state (Centola & Macy, 2007; Kempe et al., 2003). Debnath et al. provide a large observational map of geoengineering conspiracy spillovers but not controlled counterfactual societies (Debnath et al., 2023). LLM societies generate heterogeneous behaviour at increasing scale, yet the systems reviewed here commonly emphasise sharing, aggregate plausibility, or believability rather than a validated trajectory of proposition-level change (Larooij & Törnberg, 2025). Finally, the prior auditor–node study supplies MI/MPR and homogeneous versus heterogeneous chains, but not a geoengineering-grounded network experiment (Maurya et al., 2025).

This thesis neither proposes a general-purpose agent architecture nor replicates the 814,924-tweet study. Within the literature reviewed here, prior studies do not combine geoengineering-specific persona prompts, multiple small-network generators, and claim-recovery auditing

in one controlled campaign. The contribution is this documented combination and its measurement-sensitivity analysis.

The review therefore leaves three requirements for the analytical framework: separate content change from cascade reach, model persona composition jointly with network position, and treat auditor output as a measurement rather than ground truth. Chapter 3 turns these requirements into a generative process, five propositions, and an explicit evidence boundary.

3 Theoretical Framework and Operationalisation

3.1 Conceptual framework

3.1.1 Purpose and analytical levels

This thesis explains how factual content changes while it moves through a synthetic social network. The explanation distinguishes three levels: persona composition determines which agents transform a message, topology determines potential exposure, and propagation links agent transformations and exposure over time. The resulting texts are then scored with an explicit measurement instrument.

This sequence separates mechanism from measurement. The simulated mechanism produces texts, actions, and exposure paths. The auditor converts those texts into misinformation scores. The scores are observations of the simulation. They are not the mechanism itself.

The framework also separates misinformation from a firestorm. Misinformation concerns departure from factual source claims. An online firestorm concerns a rapid and large discharge of negative word-of-mouth (Pfeffer et al., 2014). The two phenomena may coincide. They need not coincide. Accurate criticism can form a firestorm. Severe misinformation can also diffuse slowly and quietly. Therefore, Misinformation Index (MI) and Misinformation Propagation Rate (MPR) do not by themselves identify a firestorm.

Three kinds of statement follow from this distinction. *Theory* states a proposed relation among composition, topology, exposure, and content change. *Operationalisation* states how the relation is represented in the simulation. *Evidence* states what an observed source or experiment can support. The distinction is maintained throughout this chapter.

3.1.2 The integrated generative argument

Let $G = (V, E, W)$ be a directed network. The node set V contains synthetic agents. The edge set E defines possible transmissions. The weights W represent relational conditions such as trust. Each node i receives a persona p_i . The complete assignment is denoted by $\mathbf{p} = (p_1, \dots, p_{|V|})$.

Let X_e be the text carried by propagation event e . When node i processes the event at time t , it applies a persona-conditioned transformation:

$$X_{e'} = T_{p_i}(X_e, H_i(t), R_i(t)). \quad (3.1)$$

Here, $H_i(t)$ is the information available to the agent. $R_i(t)$ denotes local relational conditions. The output may be forwarded, reinterpreted, or dropped. Thus, propagation is a sequence of selective text transformations. It is not the movement of a fixed information token.

The theoretical object is the joint process

$$\mathcal{Y} = F(\mathbf{p}, G, S, A, \mathcal{T}, \varepsilon), \quad (3.2)$$

where S is the seed message. A is the action rule. $\mathcal{T}$ is the exposure schedule. ε collects stochastic and model-specific variation. $\mathcal{Y}$ contains generated texts, transmissions, and cascade structure. An auditor J then maps the generated texts to observed scores:

$$\widehat{\mathcal{M}} = J(\mathcal{Y}; Q, G_Q, \theta_J). \quad (3.3)$$

The question set Q , gold answers G_Q , and auditor specification θ_J are part of the measurement model. They are not neutral bookkeeping choices.

The framework therefore has a clear order. Composition and topology shape exposure. Exposure selects which persona transformations occur. Repeated transformations produce factual retention or drift. The auditor then measures that drift. Pfeffer’s firestorm theory interprets the broader communication conditions. Debnath’s belief profiles ground the identities and issue domain.

3.1.3 Persona composition as a transformation mechanism

Theory. A persona is a structured prior over attention, interpretation, and expression. It makes some source claims salient. It makes other claims easy to omit or recode. Repeated use of the same prior can compound a stable distortion. Mixed priors can interrupt that process. They can also introduce new distortions. Heterogeneity is therefore not intrinsically corrective.

Composition has at least three dimensions. The first is the share of each persona family. The second is the diversity of those families. The third is their placement in the graph. Two societies can have the same shares but different outcomes. One may place a conspiracy persona at a hub. The other may place the same persona at the periphery. The label “heterogeneous” does not capture this difference.

Results from the prior auditor–node preprint illustrate this dependence on composition and order. In homogeneous 30-node branches, identity- and ideology-laden personas often accelerated drift. Expert and neutral personas often stabilized factual content (Maurya et al., 2025). Yet the controlled-random heterogeneous branches reached the high-loss band ($MI > 3$) in about 85% of branch–domain pairs. The prior study therefore does not establish a general “diversity buffers misinformation” law. It shows that persona effects depend on order, content, and repeated interaction.

Operationalisation. The thesis assigns persona identifiers and prompts to nodes. Homogeneous conditions repeat a focal identity. Heterogeneous conditions combine several identities. The relevant treatment is the complete assignment $\mathbf{p}$, not the H/He label alone. Composition should therefore be reported as persona-family shares and node positions. Results should also be conditioned on persona family. A lower network mean may otherwise reflect fewer conspiracy seats rather than corrective interaction.

Evidence. The prior arXiv preprint provides simulation evidence for persona-conditioned drift on linear branches. It does not provide human behavioural validation. It also does not provide evidence about complex-network placement. The present experiments can provide evidence about synthetic agents under fixed prompts and models. They cannot establish that real users with similar labels would behave identically.

3.1.4 Topology as an exposure mechanism

Theory. Topology affects factual content indirectly by determining which agents can receive and transform a message. Degree controls how many transmissions can follow an event. Path length controls how many sequential transformations can occur. Clustering creates repeated exposure from connected neighbours. Community structure concentrates exposure within groups. Bridges permit a frame to cross between groups. Hubs place selected agents in high-leverage positions.

Topology and composition are therefore non-separable. A clustered graph does not create an identity echo unless identities align with its clusters. A homogeneous graph produces perfect identity agreement by construction. That agreement is not evidence of an emergent echo chamber. Likewise, a scale-free graph is primarily a hub contrast. It is not automatically a clustering contrast.

Operationalisation. The topology generator defines E and initial edge conditions. Persona assignment then defines the relation between social position and identity. The design includes chains, rings, random graphs, small-world graphs, scale-free graphs, echo chambers, polarised graphs, and hierarchical graphs. Each topology represents a distinct exposure opportunity structure. No generator is treated as a complete model of a platform.

The linear chain is the direct link to the prior arXiv preprint. The small-world graph is the clearest named operationalisation of local clustering (Watts & Strogatz, 1998). The scale-free graph represents unequal connectivity and hub exposure (Barabási & Albert, 1999). The random graph provides a stochastic baseline (Erdős & Rényi, 1960). The echo and polarised graphs encode pre-existing segregation. They do not demonstrate that segregation emerged during the run. The hierarchical graph represents asymmetric transmission. It does not operationalise legacy-media feedback.

Evidence. Differences between topology cells are evidence about the implemented generators. They are not evidence about all networks in the same named class. One archived realisation per cell does not identify a topology-wide effect. Dead cascades also carry information about reachability. They must not be recoded as zero misinformation.

3.1.5 Propagation as a coupled selection process

Theory. Propagation determines which potential transformations become realized. At each event, an agent may forward, reinterpret, or drop a message. Forwarding preserves more of the received text. Reinterpretation invokes another persona-conditioned rewrite. Dropping

ends that path. The observed cascade is therefore selected by both network access and agent action.

This creates two distinct margins. The extensive margin concerns whether content reaches another node. The intensive margin concerns how much the content changes when it is processed. A topology can increase reach without increasing factual drift per rewrite. A persona can increase drift while reducing onward transmission. These mechanisms should not be collapsed into one mean.

Repeated exposure adds a further mechanism. Several neighbours may deliver related frames to the same node. The node can then receive social confirmation as well as content. Pfeffer describes this local echo as a consequence of network clusters (Pfeffer et al., 2014). In the simulation, however, repeated receipt is an implemented exposure pattern. It is not direct evidence of human persuasion.

Operationalisation. The simulator records event time, hop count, sender, receiver, action, and rewritten text. Ticks and hops answer different questions. A tick is a simulation update. A hop is a position along a transmission path. Neither is a unit of wall-clock time. The eight-tick horizon is a compressed experimental horizon. It is not an estimate of a Twitter half-life. It is also not Debnath’s eight-word skip-gram window.

The action set is ternary. It includes forwarding, reinterpretation, and dropping. This differs from Pfeffer’s binary-choice factor. No result from the ternary mechanism can confirm a binary-choice proposition without a dedicated ablation.

Evidence. Propagation logs support claims about simulated reach, depth, breadth, and rewriting. They do not support claims about platform speed in minutes or hours. They also do not identify causal persuasion by repeated exposure. Such claims would require a matched manipulation of exposure while other conditions remain fixed.

3.1.6 MI and MPR as a measurement model

Theory. Factual drift is latent until a measurement rule is applied. The auditor–node framework makes that rule explicit (Maurya et al., 2025). The source article defines a set of factual questions. The auditor evaluates whether each answer remains recoverable from a rewrite. MI summarizes deviation at one scored event. MPR summarizes MI over a specified collection of events.

In the prior arXiv preprint, the answer vector has ten binary items. MI is the Hamming count between the reference and audited vectors. Branch MPR is the mean MI across a 30-node branch. Despite its name, MPR is not a temporal derivative. It is a branch-level average severity measure. This matters when the concept is transferred to a graph.

Operationalisation. For discrete scoring with m questions, let $z_{ej} = 1$ when answer j is not preserved at event e . Then

$$\text{MI}_e^{(d)} = \sum_{j=1}^m z_{ej}. \quad (3.4)$$

The present climate instrument uses $m = 5$. Its numerical range therefore differs from the prior arXiv preprint’s ten-item instrument.

On a graph, there is no unique branch average. The aggregation set must be named. A node summary can be written as

$$\text{MPR}_i = \frac{1}{|E_i^{\text{scored}}|} \sum_{e \in E_i^{\text{scored}}} \text{MI}_e. \quad (3.5)$$

A network-time summary can instead average scored events at tick t :

$$\overline{\text{MI}}(t) = \frac{1}{|E_t^{\text{scored}}|} \sum_{e \in E_t^{\text{scored}}} \text{MI}_e. \quad (3.6)$$

These quantities answer different questions. Neither should be labelled simply “MPR” without its aggregation rule.

The high-loss threshold is also operational. The prior arXiv preprint labels averages at or below 1 as factual error. It labels values above 1 and at or below 3 as lies. It labels values above 3 as a high-loss or “propaganda” band in that ten-item instrument (Maurya et al., 2025). The present five-item reuse of > 3 is an arbitrary sensitivity cut, not a calibrated propaganda category. The thesis reuses these cuts. They are design thresholds. They are not validated psychometric boundaries.

Evidence. MI supports a claim about auditor-detected recoverability of selected source claims. It does not directly measure belief, intent, toxicity, affect, or public harm. MPR supports a claim about the average of those event scores over a declared set. It does not show that misinformation propagated faster. The prior arXiv preprint assumes a perfect auditor in its limitations. The present thesis must instead treat auditor error as a validity threat.

3.1.7 Instrument dependence

Theory. Observed misinformation is jointly produced by the generated text and the instrument. Formally,

$$\widehat{M}_{e,r} = J_r(X_e, Q, G_Q), \quad (3.7)$$

where r indexes an auditor mode. Changing r can change levels, rankings, and threshold crossings. The latent text X_e may remain unchanged. Therefore, a score difference across instruments is not automatically a content difference.

Operationalisation. The continuous campaign and the dual-discrete campaign are separate measurement regimes. The continuous auditor returns graded claim-level assessments. The discrete headline counts claim failures. The continuous sidecar in a dual run is a diagnostic attached to that regime. It is not a third ground truth. Scores from different regimes must not be averaged. Threshold crossings must be computed within one regime.

Instrument dependence has four sources. First, question selection determines which facts can be lost. Second, response coding determines whether partial preservation receives credit.

Third, the auditor model introduces judgement variance and bias. Fourth, aggregation determines which nodes and events receive weight. Topology can affect the fourth source by changing event counts. An event-weighted mean can therefore partly reflect cascade selection.

Evidence. Agreement across instruments strengthens robustness. Disagreement identifies a measurement sensitivity. It does not reveal which instrument is true. External validation requires human coding or another justified reference. LLM-as-judge studies motivate automated assessment, but they do not validate this climate-specific instrument by transfer alone (Zheng et al., 2023).

3.1.8 From branches to complex networks

Theory. The prior arXiv preprint studies sequential transmission chains. Each node has one predecessor in the realized branch. Depth and exposure order are therefore aligned. A complex network breaks this alignment. A node can receive several messages. A message can follow several paths. Communities, bridges, hubs, and cycles can alter both reach and rewrite order.

The extension is conceptual, not merely technical. In a chain, composition is an ordered list. In a graph, composition becomes a joint distribution over identities and positions. In a chain, MPR has one natural branch denominator. In a graph, several valid denominators exist. In a chain, a late node implies many prior rewrites. In a graph, a late tick need not imply a long path.

The generalization therefore preserves the event-level MI concept. It does not preserve every branch-level interpretation. Graph analyses must report path, node, event, and time aggregation separately.

Operationalisation. The chain serves as a bridge to prior work. The topology grid then changes the adjacency structure while retaining the same basic rewrite and audit process. Persona composition is crossed with topology where the design permits. This crossing tests whether identity effects depend on exposure structure. It does not constitute a literal replication of the 30-hop study. The climate articles, persona inventory, model, horizon, and graph process differ.

Comparisons also require common support. Reach, event count, and dead-cell status should accompany MI summaries. A live-only mean describes surviving cascades. It does not estimate the mean over cascades that failed to propagate. Similarly, a 63-node imported graph should not be pooled with an eight-node generated graph. The opportunity structures differ too sharply.

Evidence. The complex-network experiments can show whether a prior pattern survives a domain and structure shift. They cannot reproduce the prior arXiv preprint merely by reusing MI and MPR. Evidence of generalization requires directional stability under the changed object. Evidence of failure is also informative. It places a boundary on the chain result.

3.1.9 Debnath belief profiles as empirical grounding

Theory. Debnath et al. describe geoengineering discourse as an identity-structured information environment (Debnath et al., 2023). Chemtrails conspiracy content dominates the observed corpus. Smaller streams concern climate action and environmental risk. These streams provide a domain-specific basis for synthetic belief profiles. They also motivate identity alignment between a message and a persona.

The belief profiles are mechanisms of selective interpretation. A conspiracy profile is expected to assimilate ambiguous geoengineering information into a chemtrails frame. A climate-action profile may foreground governance and justice. An environmental profile may foreground ecological risk. Added scientist, journalist, and moderation roles supply institutional accuracy norms. These added roles support the experimental contrast. They are not empirical Debnath clusters.

Operationalisation. The persona library translates discourse types into prompts, tags, tones, and expertise fields. This is a theory-faithful reduction. It is not a reconstruction of individual Twitter users. The full tweet corpus was not hydrated. HDBSCAN profiles were not re-estimated. The imported 63-node structure is a hashtag co-occurrence graph. It is not a retweet cascade.

Debnath’s skip-gram context window concerns words inside tweets. It does not define simulation hops. Debnath also reports no empirical MI or MPR. Twitter toxicity, semantic proximity, and simulated factual drift remain different constructs.

Evidence. Debnath provides observational evidence about the issue domain, discourse streams, and platform corpus. It supports the content validity of the persona families. It does not validate their exact prompts. It also does not validate simulated propagation rates. Claims about the synthetic profiles must therefore remain claims about grounded experimental agents.

3.1.10 Pfeffer firestorms as the macro-theoretical frame

Theory. Pfeffer, Zorbach, and Carley define online firestorms as sudden, large discharges of negative word-of-mouth (Pfeffer et al., 2014). Their Outlook identifies seven interrelated factors:

1. speed and volume of communication;
2. binary choices;
3. network clusters;
4. unrestrained information flow;
5. lack of diversity;
6. cross-media dynamics; and

Table 3.1: Communication escalation and factual degradation are distinct outcomes.

	Low factual degradation	High factual degradation
Low communication escalation	Limited, largely faithful diffusion	Localized misinformation without a firestorm
High communication escalation	Viral but factually stable communication; it may still be an affective firestorm	A misinformation-amplifying firestorm candidate, if negative affect is also present

7. network-triggered decision processes.

This list is the theoretical source. It is not a list of simulator parameters.

The framework maps these factors onto different analytical levels. Speed and volume concern propagation intensity. Binary choices concern the action architecture. Clusters and unrestrained flow concern topology. Lack of diversity concerns composition and homophily. Network-triggered decisions concern repeated social exposure. Cross-media dynamics concerns an external feedback layer. The affective character of firestorms belongs to the definition. It is not an eighth numbered factor.

Operationalisation. The thesis uses a computational remapping. Identity composition operationalises part of lack of diversity. Graph generators operationalise selected cluster and flow conditions. Seed fan-out can operationalise shock versus drip. Ticks and activity schedules can operationalise simulated opportunity rates. These mappings are partial.

Two factors remain especially constrained. The action architecture is ternary rather than binary. The model has no legacy-media broadcast layer. Cross-media dynamics is therefore held outside the simulation. The hierarchical topology cannot substitute for that missing layer.

Valence, surprise, identity, clustering, echo, and temporal acceleration are thesis-level knobs. They are not Pfeffer’s original factor names. Their value lies in making parts of the theory manipulable. Their limitation lies in incomplete construct coverage.

Evidence. Pfeffer’s article provides qualitative and case-based theoretical evidence. It does not provide an LLM model, MI, MPR, or a numerical tipping threshold. An MI value above 3 is therefore not a Pfeffer firestorm criterion. Simulation evidence can test patterns implied by the remapping. It cannot confirm all seven factors when some factors are held or absent.

3.1.11 A two-axis outcome model

The integrated framework uses two outcome axes. The first axis is communication escalation. It includes reach, volume, breadth, depth, and simulated speed. The second axis is content degradation. It includes event MI and declared MPR aggregates. Firestorm interpretation also requires negative or affective content.

This yields four conceptual cases (Table 3.1).

The final cell is the thesis’s central concern. It is still a candidate category. The simulation must show escalation and degradation separately. It must also justify the affective

interpretation. No single mean MI can establish all three conditions.

3.1.12 Derived propositions

The framework yields five propositions. P1 and P5 are examined directly in the present campaign. P2–P4 identify mechanisms that the current bundled design can describe only indirectly and that require matched ablations for a direct test. They do not report findings.

P1: composition–alignment proposition. Factual drift should increase when a persona’s interpretive prior aligns with a misinformation frame. The effect should be strongest under repeated use of that persona. However, heterogeneous composition may either interrupt or compound drift. Its direction depends on order and role composition.

P2: position–composition proposition. The effect of a persona depends on network position. Placing an aligned persona at a hub should alter more downstream opportunities than placing it at the periphery. Composition shares alone are therefore insufficient.

P3: cluster–echo proposition. Clustering should increase repeated local exposure. Identity echo should increase only when persona placement aligns with the clustered structure. Structural clustering and identity homophily are distinct conditions.

P4: reach–severity proposition. Topology may change cascade reach without changing drift per scored rewrite. Persona transformations may change drift without increasing reach. Communication escalation and content degradation should therefore be analysed separately.

P5: instrument–conditional proposition. The magnitude and threshold status of measured drift depend on the auditor regime. A substantive conclusion is robust only when its direction survives justified measurement alternatives. Instrument disagreement is a result about measurement. It is not evidence of two underlying cascades in the same run.

3.1.13 Theory, operationalisation, and evidence ledger

The ledger in Table 3.2 separates theoretical constructs, operational choices, and admissible evidence.

3.1.14 Scope of inference

The framework supports mechanism-oriented simulation claims. It can show that a specified persona, graph, seed, and auditor produce a specified pattern. It can compare matched cells within one measurement regime. It can test whether chain patterns persist on more complex networks.

The framework does not support direct population inference. The agents are not sampled humans. The graphs are not representative platform samples. The auditor is not an error-free observer. The imported hashtag graph is not a reconstructed diffusion cascade. Accordingly, the evidence is computational and conditional.

3 Theoretical Framework and Operationalisation

Table 3.2: Separation of theoretical constructs, operational choices, and admissible evidence.

Construct	Theory	Operationalisation	Evidence boundary
Persona com-position	Distribution and placement of interpretive priors shape text transformation.	Persona prompts; homogeneous and heterogeneous assignments; family shares; node positions.	Synthetic-agent evidence only. H/He alone does not identify diversity effects.
Topology	Adjacency, clustering, hubs, and bridges shape exposure opportunities.	Eight graph generators plus an imported custom graph.	Generator-specific evidence. One archived realisation does not identify a topology class.
Propagation	Selective forwarding, rewriting, and dropping jointly determine realized paths.	Ternary actions; ticks; hops; trust; inbox and horizon constraints.	Simulation time is not wall-clock time. Ternary actions do not test binary choice.
Factual drift	Source claims may be omitted, contradicted, or transformed.	QA-based discrete or continuous auditor.	Measures selected claim recoverability, not belief, intent, toxicity, or harm.
MI	Event-level observed factual deviation.	Five-item climate score within a declared auditor mode.	Scale depends on question count, coding, and auditor.
MPR	Aggregate severity over a declared propagation unit.	Branch, node, event, or time mean with an explicit denominator.	Not a physical rate. Graph aggregates are not automatically comparable with chain MPR.
Debnath pro-files	Issue-specific identities condition interpretation of geoengineering claims.	Theory-faithful persona prompts and coarse discourse families.	Grounded reductions, not recovered user clusters or digital twins.
Pfeffer firestorm	Rapid, high-volume, negative word-of-mouth under seven interrelated conditions.	Partial mapping to composition, topology, seeding, and schedule.	MI/MPR alone cannot identify a firestorm. Cross-media remains untested.

This limitation does not remove the theoretical contribution. It clarifies it. The contribution is an explicit bridge across three literatures. Pfeffer supplies the macro conditions of online firestorms. Debnath supplies the climate-conspiracy identities and domain. Maurya et al. supply the persona rewrite and QA-auditor instrument. The thesis combines these elements in a complex-network design and tests which findings persist across network structures and measurement instruments.

4 System, Data, and Methods

Chapter 3 specified the relations that the campaign seeks to observe and the claims it cannot support. This chapter translates that framework into the executed persona assignments, network conditions, propagation protocol, auditor regimes, and analysis rules. It first separates inherited components from the thesis-specific work.

4.1 Prior-work boundary

The co-authored arXiv preprint introduced the auditor–node architecture, the Misinformation Index (MI), and the Misinformation Propagation Rate (MPR) (Maurya et al., 2025). This thesis uses that system as an inherited instrument. Its thesis-specific layer comprises the climate corpus, the Pfeffer mapping, the Phase 2 factorial design, the composition analysis, and the D-net stress test. A conservative, artefact-linked contribution draft appears in the front matter. Each allocation is labelled for candidate and supervisor confirmation; the repository does not contain a certified CRediT split.

4.2 Simulation design

This study used the repository’s agent-based information-diffusion engine to compare the propagation of climate and geoengineering claims across network structures and persona compositions. The design adapts the homogeneous and heterogeneous conditions of the prior auditor–node preprint (Maurya et al., 2025); it does not reproduce that study’s crime-news corpus. The present campaign instead uses climate-specific articles and belief profiles. All values reported in this section are taken from the executed Phase 2 configurations, engine source, campaign plan, hand-off record, and raw run schemas.

4.2.1 Experimental units and factorial design

The principal experimental unit was one configuration–article cell. The factorial grid crossed eight network topologies with either 12 homogeneous persona conditions or six heterogeneous persona mixes, two scoring modes, and six seed articles:

$$\underbrace{8 \times 12 \times 2 \times 6}_{\text{homogeneous}} + \underbrace{8 \times 6 \times 2 \times 6}_{\text{heterogeneous}} = 1,728 \text{ cells.} \quad (4.1)$$

The parser retained one analytical run per configuration ($N = 1$); 77 experiment names had multiple archived directories. It selected the most recently modified complete directory with at least one successful call, a rule that is not equivalent to one execution per configuration. Accordingly, article cells within a configuration are repeated observations from one simulation

Table 4.1: Provenance of the six campaign articles. Each record has five yes/no questions and Boolean expected answers in `articles/merged.json`. Access date: 2026-09-19.

Identifier	Principal sources
<code>scopex_2017</code>	Keutsch Group SCoPEX documentation (Keutsch Group, 2024); Debnath et al. on the 2017 volume peak (Debnath et al., 2023).
<code>chemtrails_gates_2018_2021</code>	Debnath et al. on chemtrails spillover (Debnath et al., 2023).
<code>sai_geoengineering</code>	IPCC WG I assessment of solar radiation modification (Intergovernmental Panel on Climate Change, 2021).
<code>paris_agreement</code>	UNFCCC Paris Agreement text (United Nations Framework Convention on Climate Change, 2015).
<code>climate_consensus</code>	IPCC WG I (Intergovernmental Panel on Climate Change, 2021); Cook et al. consensus analysis (Cook et al., 2013).
<code>polar_bears</code>	IPCC WG I Arctic/sea-ice assessment (Intergovernmental Panel on Climate Change, 2021).

run, not independent campaign replicates. The design supports descriptive comparison of the executed cells but not seed-to-seed uncertainty estimation.

The two scoring modes were continuous-only (T2c) and dual (T2d). In a dual run, the discrete score is the headline score and a continuous score is stored as a sidecar for the same event. The two modes are separate measurement instruments and were not averaged. Phase 1 discrete-only results were likewise not inserted into the Phase 2 grid.

4.2.2 Articles and ground truth

The merged article library contains 24 records, but every cell in the main grid uses the same six campaign articles: `scopex_2017`, `chemtrails_gates_2018_2021`, `sai_geoengineering`, `paris_agreement`, `climate_consensus`, and `polar_bears`. The set spans geoengineering research, a chemtrails conspiracy narrative, and established climate-science claims. Each article record contains an identifier, title, source text, five yes/no audit questions, the corresponding Boolean ground-truth answers, and source notes. These structured question–answer pairs provide the reference against which propagated text is scored. The remaining 18 records in the merged library are not additional observations in the 1,728-cell grid. Table 4.1 records the scholarly or institutional sources used to author the six campaign summaries and Boolean keys. The executable stimulus is the checked-in JSON; the table does not claim that the summaries are verbatim reprints of those sources.

Articles were processed sequentially within a configuration. Before processing each article, the simulation cleared the article’s prior history and all node inboxes. The complete source text was then inserted into the designated seed node at tick zero. No competing article group or intervention was enabled in the Phase 2 configurations considered here.

4.2.3 Persona conditions

The merged persona library contains 17 belief profiles. A profile provides at least an identifier, a display name, and a system prompt; the richer campaign records also contain tags, emotional tone, ideological bias, topic expertise, Debnath-type annotations, stylistic attributes, hashtag repertoires, and provenance. These profiles are theory-informed reductions of discourse types,

not HDBSCAN centroids reconstructed from the full Debnath Twitter corpus (Debnath et al., 2023). Their purpose is controlled role conditioning, not demographic representation.

The homogeneous arm (H) uses 12 profiles, one profile per configuration and the same profile at all eight nodes. They comprise four conspiracy profiles (`conspiracy_believer`, `conspiracy_haarp_weather`, `conspiracy_depopulation`, and `conspiracy_climate_piggyback`); three climate-action profiles (`climate_action_advocate`, `climate_justice_youth`, and `mitigation_first_policy`); and five science/environment profiles (`environmental_concern`, `ozone_stratosphere_specialist`, `biodiversity_food_security`, `climate_scientist`, and `science_journalist`).

The heterogeneous arm (He) declares six ordered mixes of eight identifiers. The sequential generators realise these orders directly. In particular, `mix_00` contains four conspiracy and four climate/environment profiles, `mix_01` contains two conspiracy and six climate/environment profiles, and `mix_02` contains no conspiracy profile. Mixes `mix_03`–`mix_05` introduce the additional moderator, SCoPEX climate-action, and peripheral-skywatcher profiles recorded in the merged library. Node assignment follows the order in each mix for the linear, ring, ER, small-world, and scale-free generators. Echo-chamber, polarised, and hierarchical configurations instead use cluster assignment. For five of six mixes this filtering and cycling realises only four or five unique personas, repeated across eight seats; `mix_02` is the only declared order realised as listed. Polarised persona assignment alternates by node index and therefore does not align with the graph’s two contiguous halves. Consequently, H versus He is a bundled assignment contrast in persona prevalence, diversity, placement, and sometimes assignment procedure, not a controlled test of diversity.

4.2.4 Network topologies

Every main-grid graph contains eight directed nodes. The eight topology families and their executed parameters are summarised in Table 4.2. Edge trust is stored on the source node. For generators that use persona homophily, identical persona identifiers keep the unadjusted base trust. Otherwise the source code sets `clip(baseTrust – 0.2 + 0.5 · Jaccard(tags), 0.05, 0.95)`.

The linear chain is retained as the direct structural link to the earlier prior linear-chain design. The other generators test cyclic, random, clustered, hub-dominated, ideologically partitioned, and authority–audience structures. The eight network conditions in Table 4.2 are experimental mechanisms. Echo-chamber, polarised, and hierarchical generators are bespoke treatments; their names do not imply validated platform models. Linear-chain and ring configurations also use a more rewrite-heavy action policy and a lower trust threshold than the other six generators, so chain- or ring-versus-graph differences estimate a bundled protocol contrast.

4.2.5 Seeding and propagation protocol

Each article is seeded once into `node_0` at tick zero. The seed message has source `ORIGIN`, hop count zero, tick zero, the unmodified article text, and an empty provenance chain. Nodes are always active and have an inbox cap of four messages per article. Processing then proceeds

Table 4.2: Network topologies in the Phase 2 main grid.

Topology	Construction and configured parameters
Linear chain	Directed path $v_i \rightarrow v_{i+1}$, initial edge trust 0.7.
Ring	Directed cycle $v_i \rightarrow v_{(i+1) \bmod 8}$, initial edge trust 0.7.
Random ER	Directed Erdős–Rényi graph with edge probability $p = 0.42$. The configuration records <code>minSeedOutDegree=2</code> and a seeded RNG, but <code>buildRandomER</code> ignores both and samples unseeded <code>Math.random</code> ; edge trust is $0.3 + U(0, 0.5)$ (Erdős & Rényi, 1960).
Small world	Watts–Strogatz-style directed ring lattice with $k = 4$ and rewiring probability $\beta = 0.1$ (Watts & Strogatz, 1998).
Scale free	Preferential attachment with $m = 2$, beginning from a fully connected seed of $m + 1$ nodes (Barabási & Albert, 1999).
Echo chamber	Two alternating chambers; within-/between-chamber edge probabilities 0.75/0.08, base trusts 0.85/0.15, and minimum seed out-degree 2.
Polarised	Two halves; within-/between-group edge probabilities 0.75/0.05, base trusts 0.88/0.10, and minimum seed out-degree 2.
Hierarchical	Bidirectional rooted 3-ary tree with downward/upward base trusts 0.85/0.30.

synchronously for at most eight ticks. Messages produced during tick t are delivered only after all nodes have completed that tick and are therefore available from the next tick. The simulation stops early when a tick produces no outgoing message.

For a received message x , the node first rejects the message if its hop count has reached the configured limit or if effective source trust is below the node’s threshold. The receiver looks up its reverse relation to the source and otherwise uses the hard-coded default 0.5; this need not equal trust on the edge that carried the message. It then samples one of three configured actions:

$$A(x) \in \{\text{forward}, \text{reinterpret}, \text{drop}\}.$$

Linear-chain and ring configurations use

$$P(A) = (0.10, 0.85, 0.05), \quad \tau = 0.08, \quad (4.2)$$

where the tuple is ordered as forward, reinterpret, and drop, and τ is the trust threshold. The other six main-grid configurations use

$$P(A) = (0.35, 0.50, 0.15), \quad \tau = 0.15. \quad (4.3)$$

A forward action preserves the received content. A reinterpret action sends the content to the node’s persona-conditioned language model with an instruction to rewrite it in 100–200 words and return only the rewritten text. A drop action records no outgoing text. Forwarded and reinterpreted messages are broadcast to all of the node’s outgoing neighbours with hop count incremented by one. The four-message inbox cap silently discards excess deliveries; capacity loss is not logged as a sampled drop.

Figure 4.1 summarises the executed protocol.

The configuration includes `relationEvolution=true` and `trustDelta=0.05`. In the executed source, however, propagation of all six articles is completed before the audit phase begins. The auditor then raises or lowers stored source trust by 0.05 according to whether the

Input: one configuration and the six fixed seed articles.

1. For each article a , clear inboxes and prior history for a , then deliver a to `node_0` with $h = 0, t = 0$.
2. For $t = 1, \dots, 8$, process every message at every active node. If $h \geq 8$ or source trust is below τ_v , record DROP. Otherwise sample $A \in \{\text{FORWARD}, \text{REINTERPRET}, \text{DROP}\}$. For REINTERPRET, rewrite the message using the node persona and `gpt-4o-mini`. For either non-drop action, queue the resulting text for all out-neighbours with $h \leftarrow h + 1$.
3. Deliver all queued messages synchronously. Stop early if the queue is empty; otherwise continue with the next tick.
4. After propagation of all six articles, score every forward or reinterpret output against its article’s five ground-truth items.

Figure 4.1: Synchronous propagation of six articles under an eight-tick budget, followed by post-hoc auditing. Messages delivered after tick eight are not processed. The figure shows the executed protocol, not a validated social process.

event’s misinformation index is at most or above 3. Thus these post-hoc trust updates cannot alter the already-completed propagation trajectories in the same run. Belief-state updates, frame analysis, strategic-agent behaviour, network evolution, opinion dynamics, institutional trust, competitive groups, and interventions are disabled in the raw Phase 2 metadata.

4.2.6 Agent and auditor models

Both rewriting agents and the auditor use the mutable alias `gpt-4o-mini`. Chat Completions requests used temperature 0.7, `max_tokens=700`, a 120-second socket timeout, no API seed, no response schema, and no retry or backoff. The runner normally used concurrency two. Response identifiers, model fingerprints, and raw auditor responses were not archived. The agent is conditioned by the assigned persona’s system prompt. The auditor receives the propagated text, each of the five article questions, and the expected yes/no answers. Scores are therefore LLM-as-judge measurements rather than human annotations (Zheng et al., 2023). Auditing occurs after propagation and only for events whose action is forward or reinterpret and whose output text is present.

For discrete scoring, question j receives $d_j \in \{-1, 0, 1\}$, denoting contradiction/distortion, missing information, or a correct statement, respectively. For $m = 5$ questions, the discrete misinformation index is

$$\text{MI}_{\text{disc}} = \sum_{j=1}^m \mathbb{I}(d_j \leq 0) = n_{\text{missing}} + n_{\text{incorrect}}, \quad \text{MI}_{\text{disc}} \in \{0, \dots, 5\}. \quad (4.4)$$

For continuous scoring, the auditor returns $c_j \in [0, 1]$, where 1 denotes fully correct, 0.5 missing or neutral information, and 0 contradiction. The implemented score is

$$\bar{c} = \frac{1}{m} \sum_{j=1}^m c_j, \quad \text{MI}_{\text{cont}} = m(1 - \bar{c}). \quad (4.5)$$

The two nominal 0–5 ranges are not commensurate. A missing discrete item adds 1, whereas a neutral or missing continuous score of 0.5 adds 0.5. Five missing items therefore yield 5

discretely but 2.5 continuously. The common $MI > 3$ cut has different meanings under the two instruments.

In dual mode, the discrete and continuous auditor calls are issued in parallel for the same text. The event’s top-level `misinfoIndex` is MI_{disc} . The raw event additionally stores both score objects, the absolute diagnostic gap

$$\Delta_{\text{dual}} = |MI_{\text{disc}} - MI_{\text{cont}}|, \quad (4.6)$$

and Pearson agreement after mapping the discrete item scores from $\{-1, 0, 1\}$ to $\{0, 0.5, 1\}$. The gap is a measurement diagnostic, not a third misinformation scale.

4.2.7 Eight-tick horizon and computational budget

The prior linear-chain study used 30 positions or hops (Maurya et al., 2025). Phase 2 limits each graph to eight nodes, eight processing ticks, and a configured maximum of eight message hops. This is an explicit computational-cost reduction: propagation can require an LLM rewrite for each reinterpretation, followed by one auditor call per scored event in continuous mode and two in dual mode. Dense and cyclic graphs can generate many events per tick, so retaining the 30-step horizon while expanding to eight topologies, 288 configurations, six articles, and dual auditing would multiply paid model calls. The eight-tick horizon therefore prioritises factorial coverage over long-tail cascade depth. Messages delivered after tick eight are not processed; the archived event records therefore contain audited input hops 0–7 and no hop-8 events.

This truncation changes the estimand. It measures early diffusion and semantic change within a short, common budget; it does not establish that a cascade has converged, and it weakens any claim of late-stage propagation. The number eight is also unrelated to Debnath et al.’s eight-word skip-gram context window (Debnath et al., 2023). The latter is a word-embedding hyperparameter, not a social-network hop protocol.

4.2.8 Raw records and derived cell summaries

Each timestamped run directory contains `metadata.json`, `state.json`, `graph_topology.json`, one `results_<articleId>.json` file per article, a `human_eval_template.csv`, and one state file per node. The raw schemas were used as follows:

metadata.json run status, fully merged configuration, model-usage counts, and per-article result pointers;

state.json propagation and audit checkpoints, completed article lists, current tick, errors, and last-update time;

graph_topology.json realised node labels, directed edges, and initial edge trusts;

nodes/node_*.json node and persona identifiers, model, relations, node parameters, inbox, action counters, and event history;

results_<articleId>.json node summaries and aggregate network, provenance, frame, and IFD metrics.

An event history record contains the tick, article, source node, hop count, action, input and output text, misinformation index, reason, provenance, timestamp, and—when auditing succeeds—the IFD score object. This event-level record, rather than the aggregate result file, is the parser’s source for Phase 2 headline scores.

For article a , let E_a^* be the set of history events with a non-null top-level misinformation index. The parsed cell headline is the event-weighted mean

$$\overline{\text{MI}}_a = \frac{1}{|E_a^*|} \sum_{e \in E_a^*} \text{MI}_e. \quad (4.7)$$

The parser also reports a mean of node-level means. It is retained as a separate quantity because it weights nodes equally rather than weighting all events equally. A cell with at most one scored event is marked dead and hatched. Configuration-level `llmCalls>0` does not prove that the particular article cell received an auditor call. Dead cells are retained in the ledger, excluded from live means, and never imputed as zero.

4.2.9 API failures and scoring coverage

Across the 288 selected directories, `llmUsage.calls` records 203,928 successful responses and `llmUsage.errors` records 64,270 parseable HTTP responses with status at least 400. These counters are not retries or duplicate reruns: the client has no retry loop and each selected directory is counted once. They are also not a request ledger, because request identifiers were not retained and concurrent dual calls can hide a sibling rejection. At least one HTTP error affected 121 runs.

Rewrite-request failure caused the engine to propagate the input text unchanged while retaining action `reinterpret`. Auditor-request failure left MI null. Of 129,664 audit-eligible final-state events, 93,470 were scored and 36,194 remained null; null events were excluded from means and k^* , and could change hatch status. The parser’s all-correct fallback for malformed auditor output is a latent fail-open risk, but the selected log sections contained no parse-fallback warnings and node files contained no recoverable scored-payload anomalies. The observed fallback count is therefore zero, not an estimated source of score bias.

Only five affected experiment names have a complete zero-error alternative run. Stochastic reruns were not silently substituted. Canonical tables, identity-output exclusion, unaffected-cell, and zero-error-run analyses are reported separately; none is a causal correction.

4.2.10 Reproducibility and limitations

The campaign records `graphRandomSeed`: 42, all merged configurations, the realised topology, node histories, checkpoints, model-usage totals, and the complete parser inputs. The graph seed is consumed by the echo-chamber and polarised builders. Exact replay must nevertheless not be claimed. In the checked source revision, the ER, small-world, and scale-free builders, as well as action sampling, call JavaScript’s unseeded `Math.random`; the ER builder

does not consume the additional seeded-options argument passed by the simulation wrapper. Language-model generation and judging can also vary across calls. Together with $N = 1$, these facts make the archived realised graphs and event logs the reproducible record of what was executed, while a fresh run is a protocol replication rather than a bitwise reproduction.

4.2.11 Campaign isolation

Phase 2 was isolated at both configuration and output level. New configurations reside in `thesisExperiment/configs/phase2/`; raw runs in `thesisExperiment/runs_phase2/`; and parsed outputs in `thesisExperiment/results_phase2/`. The Phase 1 directories `thesisExperiment/runs/`, `thesisExperiment/results/tables/`, and `thesisExperiment/configs/full/` were designated read-only and were not overwritten. Parser selection is by experiment name and retains the latest complete timestamped run with positive LLM-call usage. This separation prevents the earlier discrete 12×12 campaign, probes, and custom Debnath-network runs from silently entering the main 1,728-cell factorial analysis.

4.3 Measurement and analysis

The preceding section defined the simulation and raw event records. This section now defines the analytical units, estimands, exclusion rule, and comparisons applied to those records.

4.3.1 Analytical scope and unit of observation

The measurement analysis used four nested units: audit item, scored event, node–article MPR, and configuration–article cell. All scores are LLM-as-judge measurements produced by `gpt-4o-mini`; they are neither human annotations nor observations from Twitter.

Four nested measurement units must be distinguished:

1. an *item* is one of the five ground-truth questions;
2. a *scored event* is one agent’s audited handling of one article;
3. a *node MPR* is the mean event-level misinformation index for one node and article; and
4. an *article cell* is one configuration–article combination.

The primary cell statistic is `meanMI`, the arithmetic mean of event-level MI over scored events. The parser also records `meanNodeMPR`, the mean of the node-level means. These quantities use the same event-level instrument but weight nodes differently; they are retained as separate columns and are not substituted for one another.

4.3.2 IFD scoring and the MI–MPR hierarchy

The engine labels the auditor record *IFD*. For each event, the record contains a misinformation index (MI), correct, missing, and incorrect rates (CR, MR, and IR), and auxiliary fields. The

analysis uses MI as the event score. It does not report hop-wise CR, MR, or IR because those fields are absent from the live analysis tables.

Discrete instrument

For question j , the discrete auditor returns

$$s_j \in \{-1, 0, +1\},$$

where $+1$ denotes a correct answer, 0 a missing answer, and -1 an incorrect answer. With $m = 5$ questions, let n_C, n_M, n_I denote the corresponding counts. The implementation computes

$$\text{CR} = \frac{n_C}{m}, \quad \text{MR} = \frac{n_M}{m}, \quad \text{IR} = \frac{n_I}{m},$$

and

$$\text{MI}_d = n_M + n_I. \quad (4.8)$$

Thus $\text{MI}_d \in \{0, 1, 2, 3, 4, 5\}$. Missing and incorrect answers contribute equally to MI even though the IFD record preserves their separate rates. The dual-scoring campaign, denoted T2d, uses this discrete MI as its top-level `misinfoIndex` and hence as its headline instrument.

Continuous instrument

The continuous auditor returns one score $q_j \in [0, 1]$ per question. Its correctness rate and MI are

$$\text{CR}_c = \frac{1}{m} \sum_{j=1}^m q_j, \quad \text{MI}_c = m(1 - \text{CR}_c). \quad (4.9)$$

For the five-question articles, MI_c is a real-valued score on the same nominal 0–5 range. Scores at least 0.67 are counted as correct and scores at most 0.33 as incorrect when the continuous IFD record derives CR/MR/IR counts; intermediate scores are counted as missing. The continuous-only campaign, denoted T2c, makes one continuous auditor call per scored event and stores MI_c as the event’s headline `misinfoIndex`.

MPR aggregation

For node v , article a , and instrument $r \in \{d, c\}$, the node-level misinformation propagation rate (MPR) is

$$\text{MPR}_{v,a}^{(r)} = \frac{1}{|E_{v,a}|} \sum_{e \in E_{v,a}} \text{MI}_e^{(r)}, \quad (4.10)$$

where $E_{v,a}$ is the set of scored events for that node and article. At cell level, `meanMI` instead averages over all scored events directly:

$$\overline{\text{MI}}_{\text{cell}}^{(r)} = \frac{1}{|E_{\text{cell}}|} \sum_{e \in E_{\text{cell}}} \text{MI}_e^{(r)}. \quad (4.11)$$

The parser’s `meanNodeMPR` is $\frac{1}{|V_{\text{scored}}|} \sum_v \text{MPR}_{v,a}$. It gives each reached node equal weight, whereas `meanMI` gives each scored event equal weight. The chapter therefore uses `meanMI` as the cell headline and presents `meanNodeMPR` only as a companion summary.

4.3.3 Dual scoring and instrument separation

In a T2d event the auditor makes two separate LLM calls in parallel: one discrete call and one continuous call. The returned top-level IFD object mirrors the discrete record. A nested `ifd.dual` sidecar stores the discrete record, the continuous record, an absolute gap, and an agreement coefficient:

$$g_e = |\text{MI}_{d,e} - \text{MI}_{c,e}|, \quad (4.12)$$

$$a_e = \text{cor}\left(\frac{s_{1,e} + 1}{2}, \dots, \frac{s_{m,e} + 1}{2}; q_{1,e}, \dots, q_{m,e}\right). \quad (4.13)$$

The normalisation maps discrete item scores $-1, 0, +1$ to $0, 0.5, 1$, after which the code computes Pearson’s correlation with the continuous item scores. The stored gap is *absolute*; it is not a signed discrete-minus-continuous difference.

This design yields two campaign instruments and one within-event diagnostic:

- T2c headline: continuous MI from continuous-only events;
- T2d headline: discrete MI from dual-run events; and
- T2d sidecar: continuous MI, absolute gap, and item-score agreement on those same dual-run events.

The sidecar continuous score is not the T2c headline because it was obtained in a different campaign call on a different realised event. Likewise, the absolute gap is a scorer-disagreement diagnostic, not a third MPR. Discrete and continuous means are therefore shown in separate columns or panels, never averaged into a common headline. Cross-instrument differences are explicitly labelled as instrument contrasts and cannot identify a change in misinformation while holding the measurement process fixed.

4.3.4 Network threshold k^*

The parser forms a network mean MI at every observed tick t :

$$\overline{\text{MI}}_t = \frac{1}{|E_t|} \sum_{e \in E_t} \text{MI}_e.$$

It then defines

$$k^* = \min \left\{ t : \overline{\text{MI}}_t > 3 \ \wedge \ \overline{\text{MI}}_u > 3 \text{ for every later observed } u \right\}. \quad (4.14)$$

A temporary crossing that later returns to 3 or below does not qualify. If no crossing sustained through the last later scored observation is observed, k^* is undefined rather than zero. The observation window ends at eight ticks, so a crossing at the final scored tick is right-censored

and does not establish persistence beyond the window. T2d k^* is calculated from the discrete series; T2c k^* from the continuous series. Their rates remain separate. This threshold measures simulated semantic degradation under the study’s MI rule; it is not a Twitter ignition time or an empirical firestorm duration.

4.3.5 Hatch policy and analysis population

A cell is classified as dead when it contains at most one scored event:

$$\text{dead} \iff \text{nScored} \leq 1.$$

If the run nevertheless records positive LLM usage, the parser labels the row `hatched_dead_after_llm`. Such rows remain in the harvest and in the dead-cell ledger, but are hatched in plots and excluded from live-cell means. They are neither deleted nor imputed as $\text{MI} = 0$. Consequently, a dead heterogeneous cell is not evidence that heterogeneity prevented propagation. Missing and in-progress rows, if present, would likewise not be empirical zeros.

The official harvest contains 292 hatched-dead cells out of 1,728 (16.9%): 96 in T2c_H, 98 in T2d_H, 50 in T2c_He, and 48 in T2d_He. Of these, 232 have zero scored events and 60 have one scored event. The 60 one-event cells contain observed MI but are excluded from live means by the hatch rule. The four live populations are therefore 480, 478, 238, and 240 cells, respectively. All reported means, medians, paired-cell contrasts, and sign counts use the live subset unless a count is explicitly described as a full-harvest or dead-cell count.

4.3.6 Pre-specified comparison structure

The five comparison families in Table 4.3 were designed to make the held and varied dimensions visible. A “same topology” comparison means that summaries are aligned by one of the eight topology labels; it does not imply that independently executed T2c and T2d runs share identical event histories.

Table 4.3: Comparison families and admissible interpretation.

ID	Compared slices	Held/varied dimensions	Interpretation rule
C1	T2c_H vs. T2c_He; T2d_H vs. T2d_He	topology and instrument held; identity arrangement varied	H–He contrasts are estimated separately for each instrument.
C2	T2c_H vs. T2d_H	topology and homogeneous arm held; instrument varied	an instrument comparison, not a common-scale treatment effect.
C3	T2c_He vs. T2d_He	topology and heterogeneous arm held; instrument varied	same restriction as C2; dual sidecar is reported separately.
C4	T2c_H vs. T2d_He; T2d_H vs. T2c_He	topology held; identity arrangement and instrument both varied	two labelled, confounded contrasts; no pooled C4 headline.
C5	live cells concatenated within a labelled slice or composition subset; D-net reported separately	cell-level table union across generated topologies	descriptive pooled summaries only; no physical supergraph.

For C1, H and He means are first calculated within topology and mode. Both cell-pooled and topology-equal summaries are retained because unequal live counts can change weighting. C2 and C3 additionally use paired live cells joined on topology, persona or mix, and article. A pair enters only when both corresponding cells are live; a dead cell on either side is never replaced by zero. C4 deliberately crosses identity arrangement and instrument. Its two directions are shown separately because averaging them would manufacture a third, uninterpretable MPR.

4.3.7 Pooled table union versus a physical super-graph

In C5, “pooled” means row concatenation. Let $\mathcal{L}_{r,z,g}$ be the live article-cell rows for instrument r , identity arm z , and topology g . The all-topology pool is the disjoint table union

$$\mathcal{P}_{r,z} = \biguplus_{g=1}^8 \mathcal{L}_{r,z,g}.$$

The pooled mean weights each live article cell equally. The topology-equal companion first calculates one mean per topology and then gives each of the eight topology means weight $1/8$. Reporting both exposes weighting sensitivity.

No edges are added between runs, no node identifiers are merged, and no propagation is rerun on the union. It is therefore incorrect to interpret C5 as a graph with 64 nodes, as a union of adjacency matrices, or as a synthetic super-cascade. The D-net analysis is also not part of this pool. D-net is a separately executed 63-node, 228-directed-edge custom topology based on a hashtag co-occurrence object. The edge container is not a hydrated retweet cascade, and the object has no empirical Twitter MPR.

4.3.8 Descriptive and exploratory statistical procedures

The primary analysis is descriptive. It reports cell counts, live and dead counts, arithmetic means of `meanMI` and `meanNodeMPR`, medians of paired differences where relevant, k^* occurrence rates among live cells, dead-cell rates over all harvested cells, and absolute dual-gap and agreement means from the dual sidecar. Grouped summaries are produced by topology, article, persona or mix, identity arm, and instrument. Heat-map entries with no live cell are left missing, not displayed as zero.

The analysis reports sign counts, medians, ranges, denominators, and rank coefficients as descriptive diagnostics. It does not use p -values as evidence because the eight network conditions are fixed design strata, pooled cells are nested within runs, and no sampling distribution over independent campaign replicates exists. No causal claim, confidence interval, or population-level effect is inferred.

4.3.9 D-net structural validation

Comparable cascade-shape metrics

The validation code compares three structural quantities:

Depth the longest breadth-first-search path from the inferred root;

Breadth the maximum number of reached nodes at any depth level; and

Structural virality the mean pairwise shortest-path distance among reached nodes after treating propagation edges as undirected.

Cascade size is reported for context but excluded from the structural-similarity score. Simulated propagation trees are reconstructed from event provenance for forward and reinterpret actions. The implementation also records simulation span in ticks; the empirical hashtag object has no usable tweet clock, so ticks cannot be converted to hours.

For structural metric $j \in \{\text{depth, breadth, virality}\}$, the relative error is

$$e_j = \left| \frac{x_j^{\text{sim}}}{x_j^{\text{real}}} - 1 \right|,$$

and the aggregate structural similarity is

$$S_{\text{struct}} = \max \left(0, 1 - \frac{1}{3} \sum_j e_j \right). \quad (4.15)$$

The implementation additionally marks a metric as a scalar match when $e_j < 0.30$. For D-net, the empirical values were depth 4, breadth 33, and structural virality 2.7; the corresponding means over 16 simulated cascades were 2.5625, 40.9375, and 1.7963. Only breadth met the 30% rule, yielding $S_{\text{struct}} = 0.6885$. These are cascade-shape comparisons, not comparisons of simulated MI with an empirical Twitter MI.

KS and Jensen–Shannon comparisons

For each of depth, breadth, and structural virality, the validation module compares the empirical and simulated value arrays with an approximate two-sample Kolmogorov–Smirnov (KS) test. The KS statistic is the maximum distance between the two empirical cumulative distribution functions. A small statistic indicates closer empirical distributions; the code reports a Kolmogorov-distribution approximation to the p -value.

The same arrays are discretised into 20 common-width bins for Jensen–Shannon (JS) divergence. With base-2 logarithms, JS lies in $[0, 1]$: zero denotes identical binned distributions and one complete separation. The code labels a distributional match only when KS $p > 0.05$ and JS < 0.10 . It defines the aggregate distributional score as

$$D = \max \left(0, 1 - \frac{1}{3} \sum_j \text{JS}_j \right). \quad (4.16)$$

The D-net comparison has one empirical object and 16 simulated cascades. The observed KS statistics were 1.0000 for depth, 0.8125 for breadth, and 1.0000 for structural virality, with approximate p -values 0.1104, 0.2954, and 0.1104. JS divergence was 1.0000 for all three metrics, so no metric met the joint match rule and $D = 0$. Because $n_{\text{real}} = 1$, the KS p -values are severely underpowered and do not establish distributional equivalence. In particular, failure to reject is not evidence that the empirical and simulated distributions match.

Digital Twin Fidelity Score

The code combines structural similarity, distributional score, and a bounded content correlation into the Digital Twin Fidelity Score (DTFS):

$$\text{DTFS} = 0.40S_{\text{struct}} + 0.40D + 0.20\rho_{\text{content}}, \quad (4.17)$$

with each component clipped to $[0, 1]$. These weights, component definitions, and the 0.70 threshold are implementation choices from the repository, not externally validated criteria. DTFS is reported only to document that the implemented check failed. The implementation's validation threshold is $\text{DTFS} \geq 0.70$. D-net has $S_{\text{struct}} = 0.6885$, $D = 0$, and $\rho_{\text{content}} = 0$, giving

$$\text{DTFS} = 0.2754 < 0.70.$$

Accordingly, `isValidated` is `false`. This failure concerns the implemented structural/distributional digital-twin check. It does not turn the hashtag graph into a retweet cascade, and it does not license an empirical MPR claim: Debnath's data provide no empirical 0–5 MI/MPR against which the auditor scores could be validated.

4.3.10 Source register and reproducibility boundary

All operational definitions and numerical counts in this chapter are traceable to the following repository artefacts:

- IFD formulas, dual calls, absolute gap, and agreement: `src/Auditor.js`;
- event collection, cell means, node means, k^* , and hatch policy: `thesisExperiment/scripts/parse_phase2.js`;
- harvested counts and the $N = 1$ parse-selection rule: `thesisExperiment/results_phase2/summary.json`;
- C1–C5 joins, pooled summaries, and descriptive procedures: `thesisExperiment/analysis_full/analyze_full.py` and `thesisExperiment/analysis_full/tables/`;
- structural metric extraction: `src/ValidationMetrics.js`;
- KS, JS, structural similarity, and DTFS definitions: `src/ValidationComparison.js`; and
- D-net validation inputs and outputs: `thesisExperiment/results_phase2/debnath_compare.json`.

Phase 1 tables are not joined to these Phase 2 headlines because they use a different campaign and schema. D-net rows are not concatenated into the 1,728-cell factorial grid. These boundaries preserve the separation among instruments, campaigns, and graph objects throughout the analysis.

The Results therefore proceed in four steps: auditor-regime sensitivity (RQ1), named persona compositions (RQ2), topology-conditioned same-regime contrasts with dead-cell rates

(RQ3), and the separate D-net stress test (RQ4). All means are live-cell descriptions; dead-cell rates are reported as a distinct outcome.

5 Results

The results follow the research questions. Auditor-regime sensitivity is reported first, then named persona compositions, then bundled network-condition contrasts, then the separate D-net stress test. The D-net analysis examines whether selected descriptive patterns recur on a 63-node constructed hashtag graph; it does not test whether those patterns survive as identified effects.

5.1 Auditor-regime contrasts in live-cell mean MI

This section compares the continuous-only campaign (T2c) with the dual campaign’s discrete headline (T2d). The comparison is a whole-pipeline sensitivity, not two instruments applied to the same generated texts. T2c and T2d are separately executed stochastic campaigns. Same-event evidence is restricted to the dual sidecar.

The two instruments also apply different missing-item penalties on a nominal 0–5 range. A discrete missing item adds 1, whereas a continuous missing or neutral score of 0.5 adds 0.5. Five missing items therefore yield 5 discretely but 2.5 continuously, so a common $MI > 3$ cut is not threshold-equivalent. Dual-discrete scores exceeding continuous scores therefore combine scoring rules, separate realisations, and API coverage.

All results below are descriptive point estimates from one retained run per configuration and eight processing ticks. Dead cells ($nScored \leq 1$) are excluded from live-cell means and are never replaced by zero. “Paired” denotes a shared topology–persona/mix–article label that was live in both separately run campaigns.

5.1.1 C2: homogeneous arm, different instruments

Holding the homogeneous arm fixed, T2d is higher than T2c on all eight topologies (Table 5.1). Averaging the eight topology-specific differences gives

$$\overline{\Delta_H} = \overline{T2d_H - T2c_H} = 0.8201346884.$$

The differences range from 0.6999 on the linear chain to 0.9494 in the echo chamber. There are 398 cells live in both campaigns. Their median paired difference is 0.8125; 342 differences are positive, 26 are negative, and 30 are zero. A further 80 cells are dead only in T2c, 82 only in T2d, and 16 in both, so none of these cells enters the paired difference.

The dual-run diagnostics show that the difference also occurs when both instruments score the same event. On live dual cells with a defined cell-mean agreement, the equal-cell-weighted mean absolute gap between the discrete headline and its continuous sidecar is 0.8804 in H ($n = 478$ live cells), while mean agreement is 0.6487. Agreement is defined for only 270 of

Table 5.1: C2 and C3 live-cell mean MI by network condition. Continuous (T2c) and dual-discrete (T2d) remain separate instruments. Each Δ is dual-discrete minus continuous within the stated arm, using equal-weight live-cell means. $N = 1$; dead cells ($\text{nScored} \leq 1$) are excluded rather than set to zero.

Topology	T2c H	T2d H	Δ_H	T2c He	T2d He	Δ_{He}
Linear chain	0.9129	1.6127	0.6999	0.8638	2.0852	1.2214
Ring	0.9160	1.6402	0.7242	0.8760	2.1768	1.3008
Random ER	0.8736	1.7809	0.9073	0.6593	2.0412	1.3818
Small world	0.7215	1.5977	0.8762	0.7545	1.9096	1.1552
Scale-free	0.9205	1.7034	0.7829	0.8157	1.9924	1.1767
Echo chamber	0.8490	1.7984	0.9494	1.3775	1.9632	0.5857
Polarised	0.8356	1.7005	0.8649	1.4137	1.8830	0.4693
Hierarchical	0.8751	1.6315	0.7564	1.5242	2.5934	1.0692
Mean topology Δ			0.8201			1.0450

478 live H dual cells (56.5%) because Pearson r is undefined when either five-item vector is constant. Agreement is the Pearson correlation between the item-level continuous scores and discrete responses mapped from $\{-1, 0, +1\}$ to $\{0, 0.5, 1\}$. It is therefore a within-event scoring diagnostic, not accuracy against external labels. Likewise, the absolute dual gap is not a third MPR and the sidecar is not the separately generated T2c headline.

5.1.2 C3: heterogeneous arm, different instruments

The heterogeneous arm shows the same directional instrument separation: T2d is above T2c on all eight topologies. The topology-equal difference is

$$\overline{\Delta}_{He} = \overline{T2d_{He}} - \overline{T2c_{He}} = 1.0450048645.$$

The topology differences range from 0.4693 on the polarised graph to 1.3818 on random ER. Of the 198 cells live in both campaigns, the median paired difference is 0.97785; 169 differences are positive, 25 are negative, and 4 are zero. Cells dead on one side are again excluded: 42 are dead only in T2c, 40 only in T2d, and 8 in both.

Within the He dual runs, the equal-cell-weighted mean absolute dual gap is 1.1145 and mean agreement is 0.6174. Agreement is defined for 171 of 240 live He dual cells (71.3%). These are means of available cell-level correlations, not agreement across all live dual events. The gap is larger and agreement lower than their H summaries, but these are descriptive diagnostics; with one replicate they do not establish an arm effect. The live tables do not contain CR, MR, IR, or hop-level MI, so no corresponding contrasts are reported.

5.1.3 Differences between score levels and topology rankings

The consistent level ordering does not imply a consistent topology ranking. In H, the topology-mean rank correlation between continuous and discrete headlines is $\rho = 0.0952$; in He it is $\rho = 0.1667$ (eight topology means in each arm). These coefficients are descriptive and are not used as inferential tests.

The apparent combination—T2d above T2c everywhere, but weak agreement in rank—

Table 5.2: Same-event dual diagnostics by network condition. Gap is the equal-cell-weighted mean absolute difference between the discrete headline and continuous sidecar. Agreement is the mean of cell-level five-item Pearson correlations among cells with a defined value (270/478 live H dual cells; 171/240 live He dual cells). Neither column is a third MI or MPR.

Topology	H gap	H agreement	He gap	He agreement
Linear chain	0.9672	0.6223	1.2960	0.7196
Ring	0.8803	0.6656	1.3123	0.6404
Random ER	0.9680	0.6969	1.1335	0.6460
Small world	0.8255	0.5928	1.1049	0.6522
Scale-free	0.8463	0.6607	1.1058	0.6230
Echo chamber	0.8923	0.6306	0.8830	0.5986
Polarised	0.8553	0.6191	0.8784	0.5673
Hierarchical	0.7973	0.6906	1.1354	0.2947
Live-cell summary	0.8804	0.6487	1.1145	0.6174

follows from two distinct comparisons. The first asks whether the discrete level exceeds the continuous level within each topology. The second asks whether the size ordering among topologies is preserved. It is not: the instrument increment is topology-dependent rather than a constant offset. For example, the H increment is 0.6999 on the linear chain but 0.9494 in the echo chamber; in He it is 0.4693 on the polarised graph but 1.3818 on random ER. The instruments also use different response representations—hard missing/incorrect counts versus graded scores—and the T2c and T2d campaigns make separate auditor calls. Thus borderline item judgements can change both scale spacing and rank.

These observations explain why “discrete is higher” is an instrument result, not evidence that discrete scoring recovered more true misinformation. Continuous scoring is not a ground-truth benchmark for discrete scoring, and discrete scoring is not a ground-truth correction of continuous scoring. The same-event gaps and non-unit agreement document their divergence; they do not adjudicate which instrument is more valid.

5.1.4 C4: cross-mode, two-factor contrast

C4 changes arm and instrument simultaneously. Comparing T2d He with T2c H gives a positive mean topology difference,

$$\overline{T2d_{He}} - \overline{T2c_H} = 1.2175517040,$$

whereas comparing T2c He with T2d H gives

$$\overline{T2c_{He}} - \overline{T2d_H} = -0.6475878488.$$

The sign reverses on every topology when the instrument is swapped against the arm (Table 5.3). The sign reversal is consistent with the large instrument-level differences observed in C2 and C3. It prevents either cross-mode difference from being interpreted as an H–He effect: each combines identity composition and scoring mode, and neither factor is held fixed.

C4 is reported only as a labelled sensitivity contrast. Averaging its two deltas would cancel instrument-specific differences and yield an uninterpretable quantity; hence the analysis

5 Results

Table 5.3: C4 cross-mode topology means and exact contrast direction. The two deltas change arm and instrument together and are not averaged into a headline.

Topology	T2c H	T2d He	$\Delta(\text{T2d}_{\text{He}} - \text{T2c}_{\text{H}})$	T2d H	T2c He	$\Delta(\text{T2c}_{\text{He}} - \text{T2d}_{\text{H}})$
Linear chain	0.9129	2.0852	1.1724	1.6127	0.8638	-0.7489
Ring	0.9160	2.1768	1.2608	1.6402	0.8760	-0.7642
Random ER	0.8736	2.0412	1.1675	1.7809	0.6593	-1.1216
Small world	0.7215	1.9096	1.1881	1.5977	0.7545	-0.8432
Scale-free	0.9205	1.9924	1.0719	1.7034	0.8157	-0.8877
Echo chamber	0.8490	1.9632	1.1141	1.7984	1.3775	-0.4209
Polarised	0.8356	1.8830	1.0474	1.7005	1.4137	-0.2868
Hierarchical	0.8751	2.5934	1.7182	1.6315	1.5242	-0.1074
Mean topology Δ			1.2176			-0.6476

reports no pooled C4 mean.

5.2 Persona composition, article identity, and within-window persistence

This section separates the homogeneous-persona (**H**) and heterogeneous-mix (**He**) arms before interpreting them. The labels describe how identities were assigned to the eight nodes: every node received the same persona in H, whereas He used a documented mixture of eight persona identifiers. They do not denote low- and high-diversity treatments with otherwise matched content. In particular, the H arm pools conspiracy personas with climate-action and science/environment personas, while the He arm pools conspiracy-heavy, conspiracy-sparse, and conspiracy-free compositions. Consequently, an aggregate H–He difference combines composition with heterogeneity and cannot by itself identify an effect of diversity.

5.2.1 Homogeneous persona families

Figures 5.1 to 5.3 cross the twelve H personas with the six articles. They show event-weighted live-cell mean MI, maximum event-level Misinformation Index, and within-window k^* occurrence. The columns are not replicates: each cell is one executed configuration–article observation, and cells with at most one scored event are excluded rather than entered as zero.

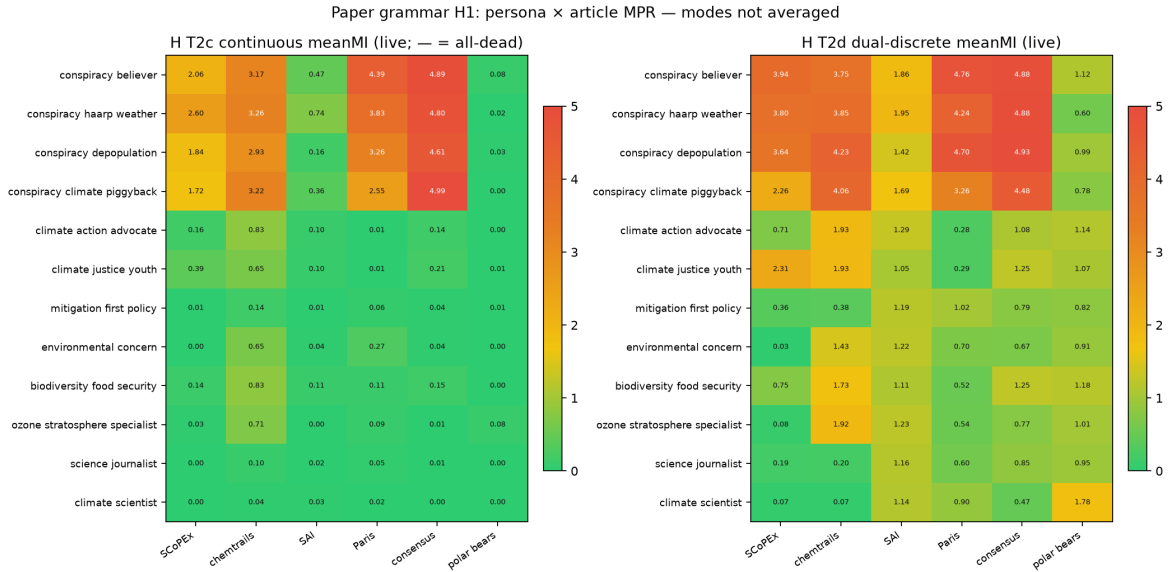


Figure 5.1: Homogeneous persona-by-article live-cell mean MI (event-weighted cell mean) under continuous (T2c) and dual-discrete (T2d) instruments. Each cell is one configuration–article observation, not an independent replicate. Hatched or missing cells have at most one scored event and are excluded from live means.

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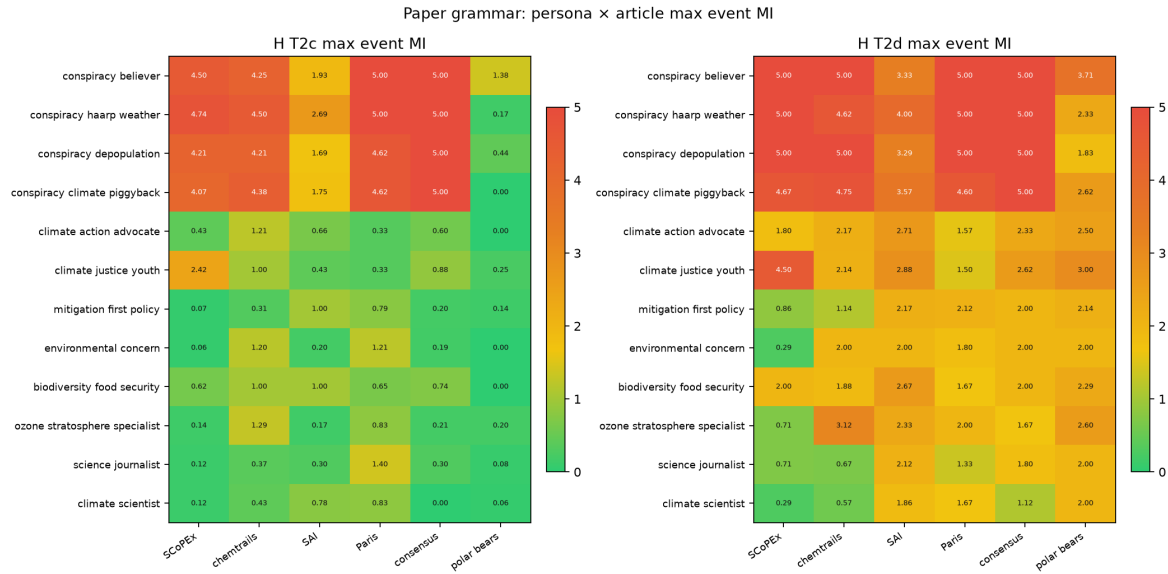


Figure 5.2: Homogeneous persona-by-article maximum event-level Misinformation Index among scored events. Continuous and dual-discrete panels remain separate. A high maximum is not a within-window sustained crossing.

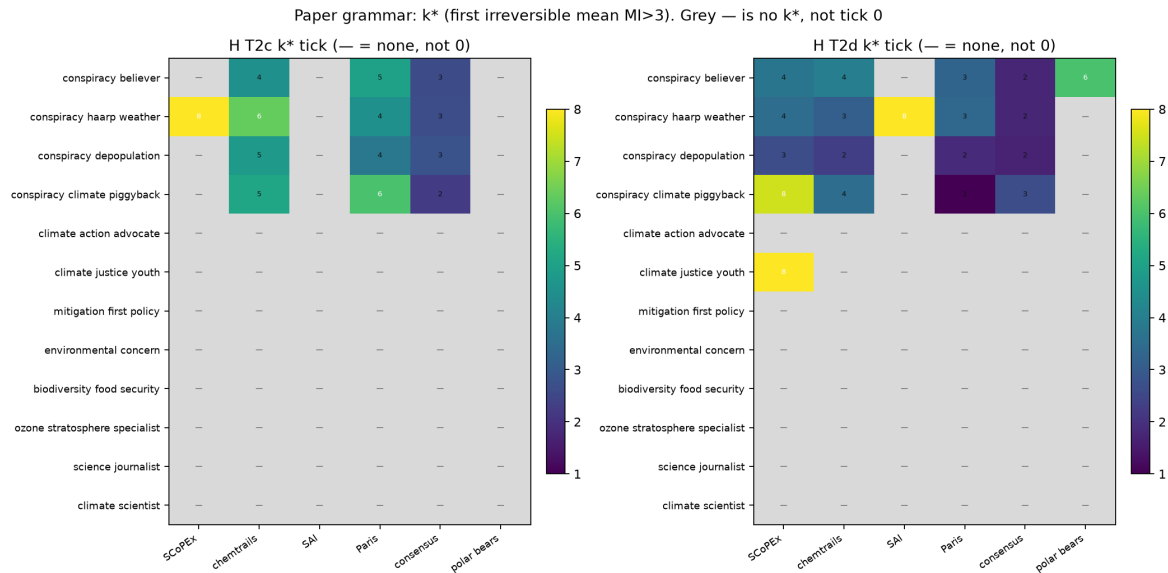


Figure 5.3: Homogeneous persona-by-article occurrence of k^* , the first scored tick above MI 3 sustained through the last later scored observation within eight ticks. Grey cells indicate no such crossing in the observed window.

Table 5.4 reports the twelve homogeneous personas. Across live H cells, the four conspiracy personas have continuous live-cell mean MI 2.2194 (168 live cells) and dual-discrete live-cell mean MI 3.1691 (159 live cells). Their k^* rates are 40.5% and 59.7%, respectively. Climate-action personas are much lower: continuous live-cell mean MI 0.1729 (117 live) and dual-discrete live-cell mean MI 1.0488 (126 live), with k^* rates 0.0% and 0.8%. The science/environment family is lowest on continuous live-cell mean MI, 0.1118 (195 live), but its dual-discrete live-cell mean MI, 0.8687 (193 live), is not zero; its k^* rate is 0.0% in both instruments. Thus the discrete instrument can assign nonzero scores to science/environment personas without those runs remaining above the k^* threshold through the last later scored observation. Figure 5.13 supports reading this as instrument disagreement: among H dual cells with a defined agreement mean the equal-cell-weighted gap is 0.8804, and mean agreement is 0.6487.

The individual rows qualify the family averages. On continuous live-cell mean MI, `conspiracy_believer` (2.4842) and `conspiracy_haarp_weather` (2.4829) are nearly tied, whereas `conspiracy_depopulation` (1.9535) and `conspiracy_climate_piggyback` (1.9570) form a lower pair. The dual-discrete ordering changes: `conspiracy_haarp_weather` is highest at 3.3571, followed by `conspiracy_believer` at 3.3522, `conspiracy_depopulation` at 3.2823, and `conspiracy_climate_piggyback` at 2.6838. Their dual k^* rates span 42.5%–70.5%; the corresponding continuous range is 31.7%–47.7%.

Among the climate-action personas, `climate_justice_youth` is highest on both instruments (continuous 0.2454, dual 1.3072); it is also the only non-conspiracy H persona with any observed k^* , a 2.2% dual rate (one of 46 live cells). `mitigation_first_policy` is lowest in that family (continuous 0.0464, dual 0.7571). Within science/environment, `biodiversity_food_security` is highest (continuous 0.2115, dual 1.1470), while `climate_scientist` has the lowest continuous live-cell mean MI (0.0130) and `science_journalist` the lowest dual-discrete live-cell mean MI (0.6745). These differences are visible by article in Figures 5.1 and 5.3 and by network condition in Figure 5.6. Figure 5.2 is also important methodologically: maximum event MI can be high when live-cell mean MI is modest, so a transient peak should not be reported as a sustained firestorm.

5.2.2 Heterogeneous compositions

Figures 5.4 and 5.5 provide the corresponding mix-by-article heatmaps for live-cell mean MI and maximum event MI, and Figure 5.7 crosses each mix with network condition.

5 Results

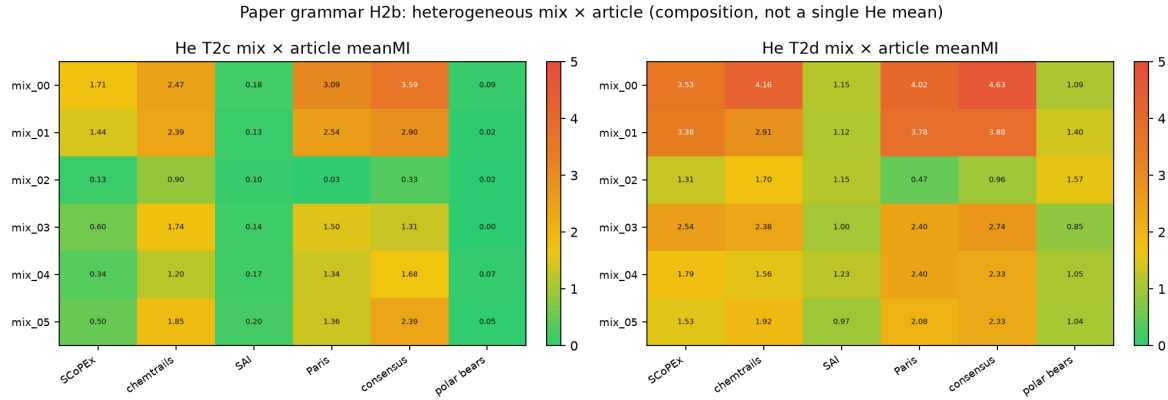


Figure 5.4: Heterogeneous mix-by-article live-cell mean MI under continuous and dual-discrete instruments. Live cells only; hatched cells have at most one scored event.

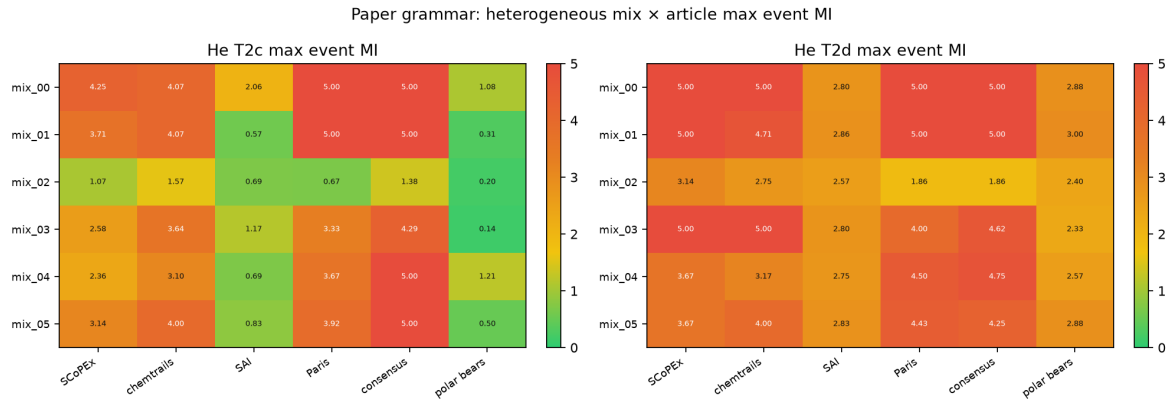


Figure 5.5: Heterogeneous mix-by-article maximum event-level Misinformation Index. Continuous and dual-discrete panels remain separate.

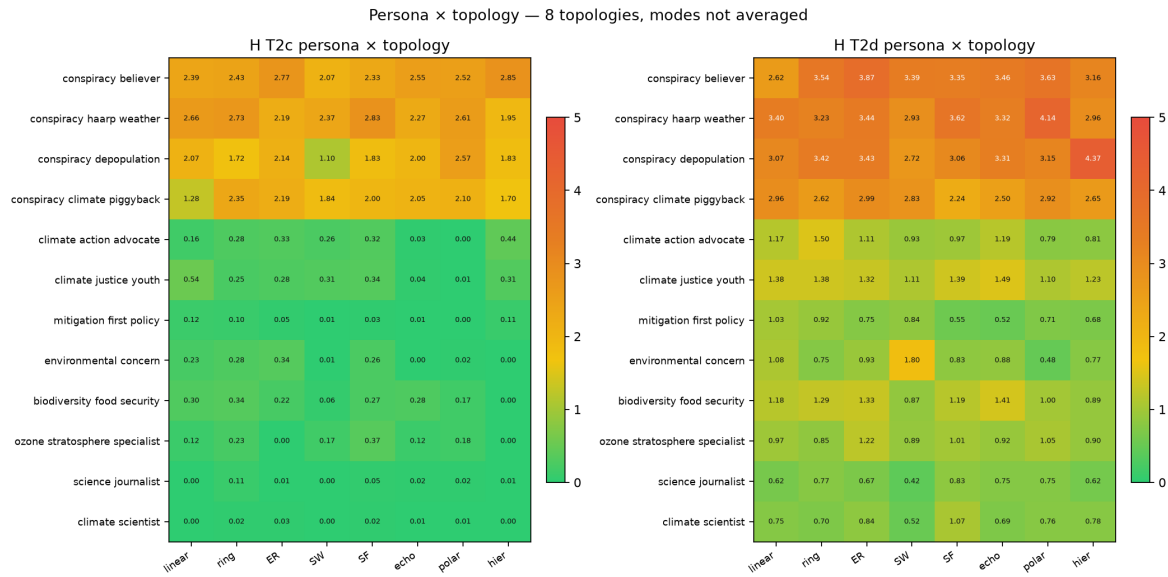


Figure 5.6: Homogeneous persona-by-network-condition live-cell mean MI. The linear chain and ring also differ in action probabilities and trust threshold, so the eight-condition ranking is not a topology-only contrast.

Table 5.4: Homogeneous personas, collapsed across eight network conditions and six articles. Live-cell mean MI ($\overline{MI}$) is the equal-weight mean of event-weighted cell means. Subscripts c and d denote continuous (T2c) and dual-discrete (T2d). n is the live-cell count; Dead c/d counts cells with at most one scored event. k^* is an eight-tick sustained crossing among live cells. Each persona has 48 expected cells per instrument.

Persona	Family	n_c	$\overline{MI}_c$	k_c^*	n_d	$\overline{MI}_d$	k_d^*	Dead c/d
conspiracy_believer	conspiracy	44	2.4842	47.7%	38	3.3522	65.8%	4/10
conspiracy_harp_weather	conspiracy	40	2.4829	47.5%	44	3.3571	70.5%	8/4
conspiracy_depoulation	conspiracy	43	1.9535	34.9%	37	3.2823	59.5%	5/11
conspiracy_climate_piggyback	conspiracy	41	1.9570	31.7%	40	2.6838	42.5%	7/8
climate_action_advocate	climate action	37	0.2258	0.0%	37	1.0665	0.0%	11/11
climate_justice_youth	climate action	41	0.2454	0.0%	46	1.3072	2.2%	7/2
mitigation_first_policy	climate action	39	0.0464	0.0%	43	0.7571	0.0%	9/5
environmental_concern	science/env.	40	0.1423	0.0%	37	0.8359	0.0%	8/11
biodiversity_food_security	science/env.	39	0.2115	0.0%	37	1.1470	0.0%	9/11
ozone_stratosphere_specialist	science/env.	38	0.1627	0.0%	37	0.9639	0.0%	10/11
science_journalist	science/env.	37	0.0310	0.0%	39	0.6745	0.0%	11/9
climate_scientist	science/env.	41	0.0130	0.0%	43	0.7517	0.0%	7/5

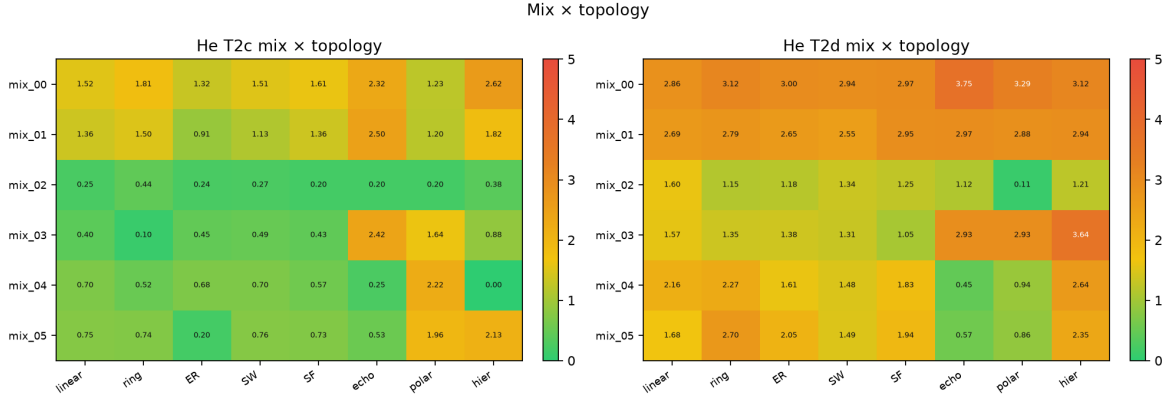


Figure 5.7: Heterogeneous mix-by-network-condition live-cell mean MI. Echo-chamber, polarised, and hierarchical mixes generally do not realise the declared eight-seat order.

Table 5.5 reports the six heterogeneous mixes. The conspiracy content of the mix tracks the principal live-cell mean-MI gradient more closely than the generic He label. The most direct contrast is between **mix_00** (four conspiracy plus four climate/environment personas) and **mix_02** (zero conspiracy plus eight science/climate personas). **mix_00** reaches continuous live-cell mean MI 1.7712 and dual-discrete live-cell mean MI 3.1321, compared with 0.2645 and 1.1882 for **mix_02**. Its k^* rates are 12.8% continuous and 57.5% dual, whereas **mix_02** is 0.0% on both.

mix_01, with two conspiracy personas, remains high (continuous 1.4770, dual 2.7795) and has the highest dual k^* rate, 58.5%. A single conspiracy persona in **mix_03** is associated with live-cell mean MI 0.8945 continuous and 2.0410 dual, with k^* rates 10.3% and 33.3%. The two conspiracy-adjacent compositions do not produce a monotone dose response: **mix_04** is lower than **mix_05** continuously (0.7021 versus 1.0134), but slightly higher in dual-discrete scoring (1.6896 versus 1.6713). Nor is **mix_00** uniformly above **mix_01**: its MPR is higher, but its dual k^* rate is lower by 1.0 percentage point. The defensible result is therefore compositional—conspiracy-heavy mixes generally score above the conspiracy-free mix—rather than a universal claim that heterogeneous networks amplify or suppress a cascade.

5 Results

Table 5.5: Heterogeneous mixes collapsed across network conditions and articles. Live-cell mean MI ($\overline{MI}$) is the equal-weight mean of event-weighted cell means. Subscripts c and d denote continuous (T2c) and dual-discrete (T2d). n is the live-cell count; Dead c/d counts cells with at most one scored event. k^* is an eight-tick sustained crossing among live cells. Each mix has 48 expected cells per instrument. Echo-chamber, polarised, and hierarchical realisations generally do not instantiate the declared eight-seat order.

Mix	Composition	n_c	$\overline{MI}_c$	k_c^*	n_d	$\overline{MI}_d$	k_d^*	Dead c/d
mix_00	4 conspiracy + 4 climate/env.	39	1.7712	12.8%	40	3.1321	57.5%	9/8
mix_01	2 conspiracy + 6 climate/env.	41	1.4770	9.8%	41	2.7795	58.5%	7/7
mix_02	0 conspiracy + 8 science/climate	41	0.2645	0.0%	41	1.1882	0.0%	7/7
mix_03	1 conspiracy + 7 other	39	0.8945	10.3%	36	2.0410	33.3%	9/12
mix_04	2 conspiracy-adj. + 6 other	38	0.7021	5.3%	39	1.6896	20.5%	10/9
mix_05	2 conspiracy-adj. + 6 other	40	1.0134	7.5%	43	1.6713	23.3%	8/5

5.2.3 Article-specific and topology-conditioned differences

Article effects cut across persona composition. Figures 5.9 and 5.10 collapse the six articles separately for H and He, while Figure 5.8 shows the network-condition-by-article pattern. The exact article means are:

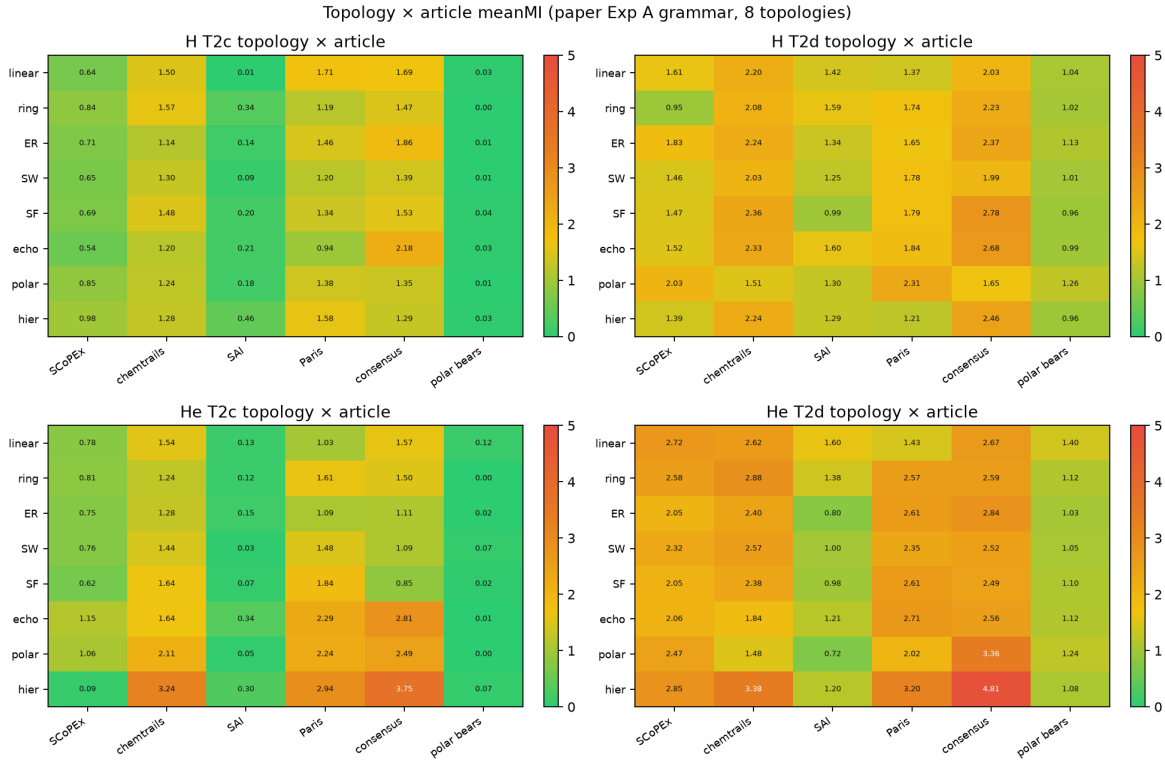


Figure 5.8: Network-condition-by-article live-cell mean MI under both instruments. Values exclude cells with at most one scored event.

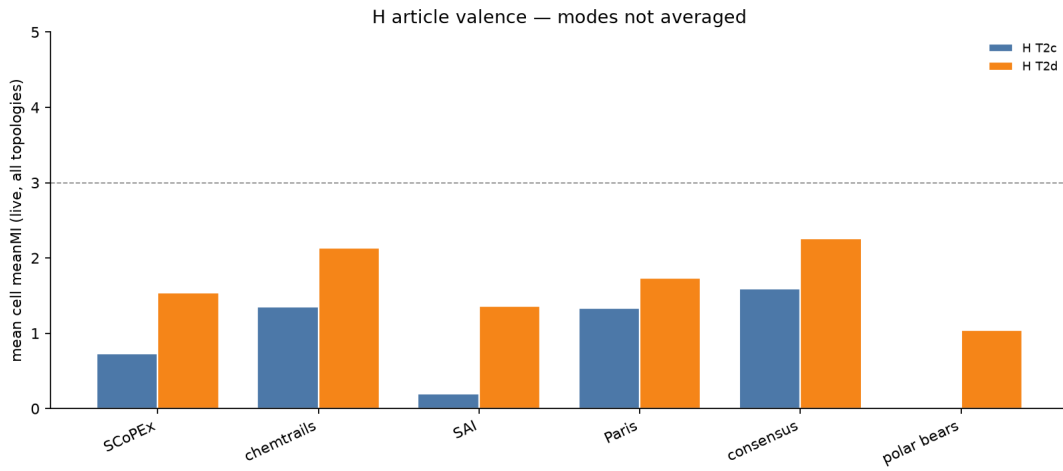


Figure 5.9: Homogeneous-arm article-level live-cell mean MI. Article identity bundles topic, wording, ground-truth questions, and persona fit; the figure does not isolate valence.

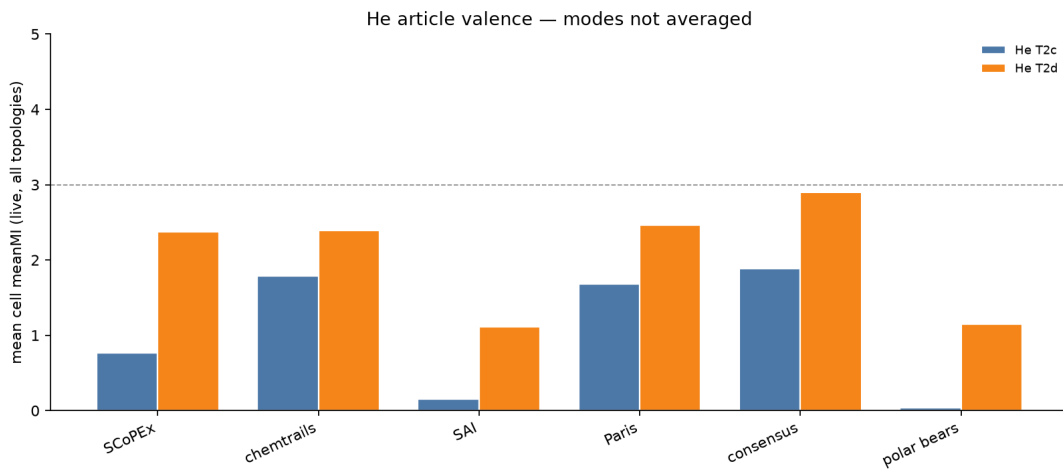


Figure 5.10: Heterogeneous-arm article-level live-cell mean MI, using the same 0–5 axis convention as the homogeneous panel. Instruments remain separate.

`climate_consensus` is highest in all four arm-instrument combinations: H continuous 1.5934, H dual 2.2610, He continuous 1.8896, and He dual 2.8991. `polar_bears` is lowest in all four (0.0185, 1.0399, 0.0415, and 1.1444, respectively). The chemtrails-Gates article is consistently above SCoPEX: 1.3541 versus 0.7322 in H continuous, 2.1383 versus 1.5390 in H dual, 1.7852 versus 0.7694 in He continuous, and 2.3885 versus 2.3718 in He dual. Because the study did not manipulate valence independently, these article differences jointly reflect topic, framing, and compatibility with persona prompts. Figures 5.1 and 5.4 show that the pattern is an interaction, not an article-only treatment.

5.2.4 k^* and dead-cell sensitivity

Figure 5.12 summarises within-window sustained crossings over eight ticks. Collapsed H k^* rates are 14.2% continuous and 20.1% dual; collapsed He rates are 7.6% and 32.1%. The reversed continuous/dual H-He ordering is further evidence against assigning a single causal meaning to the arm label. It also shows why k^* should not be inferred from live-cell mean MI alone: the dual He arm has both a much higher crossing rate and a different scoring scale. With only eight ticks, “no k^* ” means no observed within-window sustained crossing, not proof that a later crossing is impossible.

Figure 5.11 reports missingness after the executed campaign. Of 1,728 expected thesis cells, 292 are dead (`nScored` ≤ 1): 232 have zero scored events and 60 have one. They remain hatched and are never converted to zeros. Configuration-level LLM usage does not prove that a given article cell received a successful auditor call. Client invocations also recorded 64,270 HTTP-error responses across 121 of 288 selected runs, and 36,194 audit-eligible events remained null. Hatch counts therefore mix sparse propagation with scoring unavailability.

By arm and instrument, dead rates are 16.7% for H continuous (480/576 live), 17.4% for He continuous (238/288 live), 17.0% for H dual (478/576 live), and 16.7% for He dual (240/288 live). Missingness is not uniform by topology. The largest rate is polarised He continuous at 30.6% (25/36 live), followed by hierarchical H continuous and hierarchical He continuous, both 27.8% (52/72 and 26/36 live), and polarised He dual at 27.8% (26/36 live). At the low end, linear-chain H dual is 5.6% dead (68/72 live), and ring He is 8.3% dead in both instruments (33/36 live).



Figure 5.11: Share of hatched-dead cells ($n_{\text{Scored}} \leq 1$) by network condition and instrument. Dead cells remain in the ledger and are never imputed as zero. The hatch mixes sparse propagation with auditor/API unavailability.



Figure 5.12: Share of live cells with a within-window k^* under each instrument. The cut $MI > 3$ is not equivalent across discrete and continuous scoring.

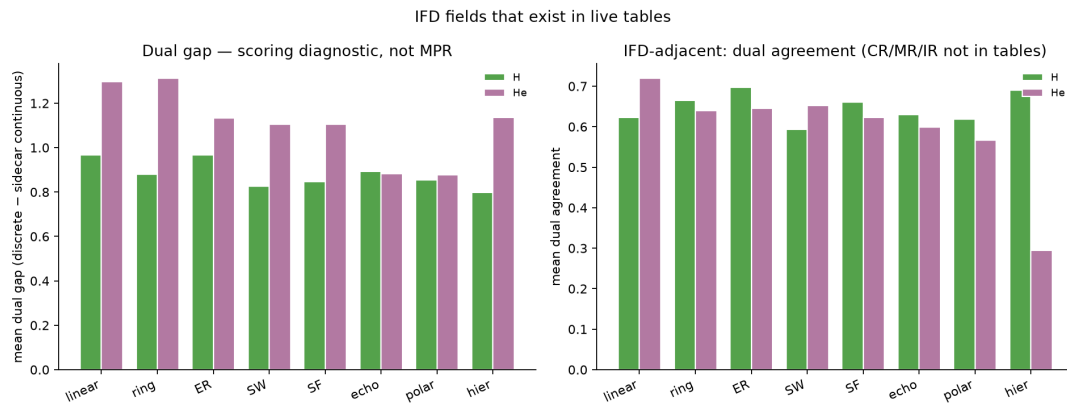


Figure 5.13: Same-event dual diagnostics: equal-cell-weighted mean absolute gap and Pearson agreement where defined. Agreement is missing when either five-item vector is constant (H: 270/478 live dual cells; He: 171/240).

Table 5.6: Article-level live-cell mean MI by arm and instrument. Values are equal-cell-weighted means of event-weighted cell `meanMI`; dead cells are excluded and the two instruments are not averaged.

Article	H continuous	H dual	He continuous	He dual
<code>scopex_2017</code>	0.7322	1.5390	0.7694	2.3718
<code>chemtrails_gates_2018_2021</code>	1.3541	2.1383	1.7852	2.3885
<code>sai_geoengineering</code>	0.1974	1.3576	0.1523	1.1160
<code>paris_agreement</code>	1.3357	1.7321	1.6821	2.4591
<code>climate_consensus</code>	1.5934	2.2610	1.8896	2.8991
<code>polar_bears</code>	0.0185	1.0399	0.0415	1.1444

Accordingly, the live-cell means in Figures 5.1, 5.4 and 5.6 to 5.10 describe surviving scored cascades. A pale heatmap cell means low live-cell mean MI, while a hatched cell supplies no live mean. Neither dead He cells nor the 0.0% k^* rate of `mix_02` demonstrates that heterogeneous or expert-inclusive networks are immune. Across both instruments, conspiracy-heavy mixes generally scored above the conspiracy-free mix; the ordering among intermediate mixes was not monotonic.

5.3 Topology results

This section describes outcomes across eight named network conditions and comparison C1, which holds the scoring mode fixed while comparing homogeneous (H) and heterogeneous (He) assignments. The linear-chain and ring conditions also differ in action probabilities and trust threshold, so the full eight-condition ranking is not attributable to topology alone. Comparisons among the remaining six graph-policy conditions still describe one realised graph per configuration. Configurations recorded `graphRandomSeed`: 42, but only selected generators consumed it. The differences below are therefore descriptive properties of bundled network conditions. They do not identify a causal effect of topology or identity composition.

There were 72 candidate H cells per topology and mode (12 personas $\times$ 6 articles) and 36 candidate He cells (6 mixes $\times$ 6 articles). A cell was classified as dead when `nScored` $\leq$ 1 after the LLM calls. Dead cells were retained and hatched in the grids, but were neither replaced by zero nor included in the means. Thus, the denominators in Tables 5.7 and 5.8 vary by topology. Across the full 1,728-cell harvest, 292 cells were dead (rather than the preliminary brief’s 290).

5.3.1 Continuous instrument

Table 5.7 reports the continuous-only (T2c) headline separately from the dual-discrete results. Among H cells, the topology means ranged from 0.7215 for small world (56/72 live) to 0.9205 for scale-free (60/72 live). Among He cells, they ranged from 0.6593 for random ER (29/36 live) to 1.5242 for hierarchical (26/36 live). These extrema describe live cells only; for example, hierarchical He had 10 dead cells and polarised He had 11.

The continuous C1 sign was not uniform. He exceeded H on four of eight topologies (small world, echo chamber, polarised, and hierarchical), whereas H exceeded He on linear chain, ring, random ER, and scale-free. Giving each topology equal weight, the mean $\Delta(\text{He} - \text{H})$ was

Table 5.7: Continuous (T2c) live-cell mean `meanMI` by topology and C1 same-mode difference. “Live” gives the scored numerator and full-cell denominator; dead is the excluded-from-mean count. $\Delta = \text{He} - \text{H}$. $N = 1$.

Topology	H live	H dead	H mean	He live	He dead	He mean	Δ
linear chain	62/72	10	0.9129	32/36	4	0.8638	−0.0491
ring	65/72	7	0.9160	33/36	3	0.8760	−0.0400
random ER	57/72	15	0.8736	29/36	7	0.6593	−0.2143
small world	56/72	16	0.7215	31/36	5	0.7545	0.0330
scale-free	60/72	12	0.9205	30/36	6	0.8157	−0.1048
echo chamber	63/72	9	0.8490	32/36	4	1.3775	0.5284
polarised	65/72	7	0.8356	25/36	11	1.4137	0.5780
hierarchical	52/72	20	0.8751	26/36	10	1.5242	0.6490

Table 5.8: Dual-discrete (T2d) live-cell mean `meanMI` by topology and C1 same-mode difference. “Live” gives the scored numerator and full-cell denominator; dead is the excluded-from-mean count. $\Delta = \text{He} - \text{H}$. $N = 1$.

Topology	H live	H dead	H mean	He live	He dead	He mean	Δ
linear chain	68/72	4	1.6127	33/36	3	2.0852	0.4725
ring	61/72	11	1.6402	33/36	3	2.1768	0.5366
random ER	56/72	16	1.7809	29/36	7	2.0412	0.2602
small world	59/72	13	1.5977	32/36	4	1.9096	0.3120
scale-free	57/72	15	1.7034	29/36	7	1.9924	0.2890
echo chamber	61/72	11	1.7984	30/36	6	1.9632	0.1647
polarised	57/72	15	1.7005	26/36	10	1.8830	0.1825
hierarchical	59/72	13	1.6315	28/36	8	2.5934	0.9618

0.1725. Pooling live cells instead gave H 0.8643 ($n = 480/576$ live) and He 1.0192 ($n = 238/288$ live), a difference of 0.1549. The two summaries differ because live-cell counts vary; neither treats the 146 dead continuous cells as zeros.

The near-balanced signs are visible in the continuous panel of Figure 5.15. Restricting the same contrast to zero-HTTP-error runs changes the pooled continuous He–H difference from +0.1549 to −0.0295; that direction is therefore not robust. Sign counts of the eight topology-level differences are four positive and four negative, with median delta −0.0035. These eight means are fixed design strata, not independent experimental replicates.

5.3.2 Dual-discrete instrument

Table 5.8 reports the discrete headline from dual runs (T2d). It is a different instrument from T2c and is not averaged with the continuous headline. H means ranged from 1.5977 for small world (59/72 live) to 1.7984 for echo chamber (61/72 live). He means ranged from 1.8830 for polarised (26/36 live) to 2.5934 for hierarchical (28/36 live).

He exceeded H on all eight dual-discrete topology means. The topology-equal mean difference was 0.3974. Pooling live cells gave H 1.6814 ($n = 478/576$ live) and He 2.0800 ($n = 240/288$ live), a difference of 0.3986; the 146 dead dual cells were not imputed. In zero-HTTP-error runs the pooled dual-discrete difference remained positive (0.4695). The largest topology-level gap was hierarchical (0.9618), and the smallest was echo chamber (0.1647). Sign counts of the eight topology-level differences are eight positive and zero negative, with median delta 0.3005. Because $N = 1$, this is not evidence of a general identity-mix effect. In particular, collapsed H combines conspiracy personas with low-scoring scientist personas, while collapsed He combines conspiracy-heavy and conspiracy-free mixes. The observed ordering therefore

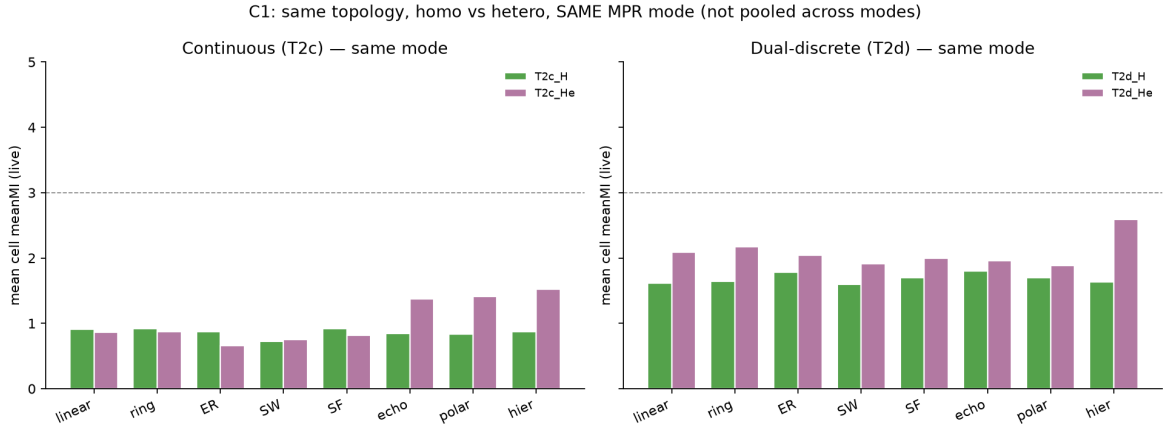


Figure 5.14: C1 same-mode H and He live-cell mean MI for all eight network conditions. Continuous and dual-discrete panels remain separate; dead cells are excluded from means rather than set to zero. The dashed line at 3, if present, is the event-level k^* cut and is not a topology-level criterion. $N = 1$.

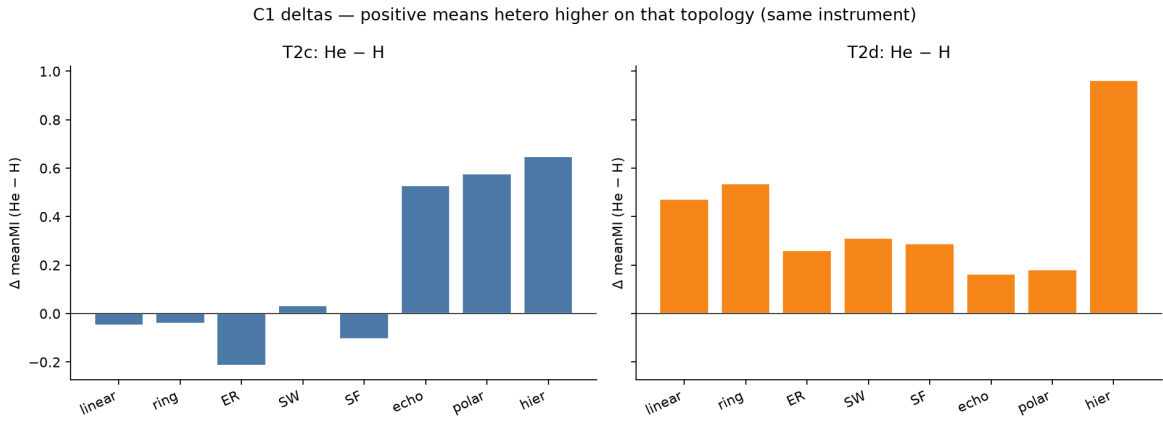


Figure 5.15: C1 topology-level $\Delta(\text{He} - \text{H})$. Continuous has four positive and four negative differences; dual-discrete has eight positive differences. These are descriptive one-seed contrasts.

does not support a causal claim that heterogeneity increases or buffers firestorms.

The article-resolved network-condition pattern is shown in Figure 5.8; persona and mix views are in Figures 5.6 and 5.7. Dead-cell denominators are visualised in Figure 5.11. Those displays follow the same live-only and instrument-separation rules used in the tables.

5.4 Results on the 63-node D-net

5.4.1 Graph and persona placement

The D-net experiments used the custom graph in `thesisExperiment/data/derived/debnath_hashtag_cascade.json`. It contains 63 nodes and 228 directed edges, collapsing to 171 unique undirected pairs. The nodes are accounts keyed by published hashtags and the edges encode hashtag co-occurrence and same-cluster links. Although the edges occupy the importer’s `retweets[]` field, they are not observed retweets: Twitter/X hydration failed, and the object is not a retweet cascade. Both D-net persona conditions used exactly this topology and the same seed node, `user_chemtrails_hub`.

The heterogeneous placement retained the reconstructed Debnath type mix. Of the 63 placements, 56 used the exact reconstructed persona identifier. The remaining seven used a coarser persona identifier while retaining the mapped Debnath type:

- `user_depopulation_amp`: `conspiracy_depopulation` $\rightarrow$ `conspiracy_believer`;
- `user_climateaction_piggyback_amp` and `user_climateaction_piggyback_peri_1`: `conspiracy_climate_piggyback` $\rightarrow$ `conspiracy_believer`;
- `user_biodiversity_hub`, `user_biodiversity_peri_1`, and `user_foodsecurity_amp`: `biodiversity_food_security` $\rightarrow$ `environmental_concern`; and
- `user_mitigation_amp`: `mitigation_first_policy` $\rightarrow$ `climate_action_advocate`.

Thus the heterogeneous arm placed 28 conspiracy-family, 19 climate-action, 13 environmental, and three expert personas. The homogeneous arm instead assigned `conspiracy_believer` to all 63 nodes. The assigned reconstruction identities were 28 conspiracy, 19 climate-action, 14 environmental-concern, and two other nodes; this difference in counting reflects the coarser simulation placement, including the type-only fallback for `user_mitigation_amp`, rather than a change to the graph.

5.4.2 Four D-net arms

Four completed configurations crossed continuous versus dual-discrete auditing with homogeneous conspiracy versus heterogeneous Debnath-mapped personas. Each configuration propagated two seed articles, producing the eight article-level results in Table 5.9. These are auditor scores from the simulation, not measurements made on Twitter.

Table 5.9: Article-level auditor scores in the four D-net configurations. The continuous sidecar from a dual run is shown only as a diagnostic and is not interchangeable with the continuous-run headline.

Configuration	Article	Headline score	n scored	Dual sidecar
Dnet_c_H_conspiracy	SCoPEX	1.7349	970	—
Dnet_c_H_conspiracy	chemtrails-Gates	3.3378	984	—
Dnet_c_He_mixed	SCoPEX	0.9983	591	—
Dnet_c_He_mixed	chemtrails-Gates	2.5068	1,035	—
Dnet_d_H_conspiracy	SCoPEX	4.2537	1,076	2.2495
Dnet_d_H_conspiracy	chemtrails-Gates	3.7303	749	3.1251
Dnet_d_He_mixed	SCoPEX	2.6880	407	1.1302
Dnet_d_He_mixed	chemtrails-Gates	1.5297	1,027	1.1497

Within each article and instrument, homogeneous conspiracy scored above the mixed placement: 1.7349 versus 0.9983 and 3.3378 versus 2.5068 under the continuous auditor, and 4.2537 versus 2.6880 and 3.7303 versus 1.5297 under the dual-discrete auditor. Article ordering depended on the instrument. Chemtrails-Gates exceeded SCoPEX in both continuous arms, whereas SCoPEX exceeded chemtrails-Gates in both dual-discrete arms.

5.4.3 Do the small-topology findings hold?

One directional pattern recurs in D-net: the conspiracy-only assignment scores above the mixed assignment for both articles under both auditor regimes. This recurrence is consistent with the small-grid finding that conspiracy-family homogeneous cells were the high-score family on both instruments (3.1691 dual-discrete and 2.2194 continuous). It is not a replication, because graph size, assignment definitions, event counts, and article coverage differ.

The collapsed H/He result does not transfer as an H/He law. Across the eight small topologies, collapsed He exceeded collapsed H by 0.1549 under continuous scoring and by 0.3986 under dual-discrete scoring, although the continuous He>H ordering held on only four of the eight individual topologies. On D-net, H exceeded He in all four like-for-like article and instrument comparisons. This reversal is consistent with the different composition of the labels: small-topology H pooled conspiracy personas with scientists, while D-net H means conspiracy-only.

Nor does the small-grid instrument ordering hold without exception. The eight-topology aggregates placed dual-discrete above continuous on every topology. On D-net, the dual headline exceeded the corresponding continuous-run headline for homogeneous SCoPEX, homogeneous chemtrails–Gates, and mixed SCoPEX, but not for mixed chemtrails–Gates (1.5297 dual-discrete versus 2.5068 continuous). Because these are different auditors, even the three concordant comparisons are instrument differences rather than estimates in a shared unit. D-net has only one fixed 63-node topology, so it cannot replicate or rank the eight small topology treatments.

5.4.4 Auditor MI is not empirical Twitter MPR

Debnath provides no empirical 0–5 misinformation-propagation-rate (MPR) outcome for this graph. The source tweet identifiers were not hydrated, the graph has no empirical tweet clock, and no tweet-level content score can be paired with a simulated event. Accordingly, the values in Table 5.9 are LLM-as-judge auditor misinformation indices (MI) inside the simulation. They must not be described as Twitter MPR, as Twitter MI, or as a match to observed misinformation. The dual-discrete headline, its continuous sidecar, and the continuous-only headline are also separate instruments and are not averaged.

5.4.5 Structural comparison and validation diagnostics

The imported hashtag co-occurrence graph has a global transitivity of 0.3146, mean local clustering 0.5153, and mean unique degree 5.4286. The `user_chemtrails_hub` node has 33 unique neighbours. Using unweighted unique-undirected pairs, the mixed-label conspiracy cut has Newman–Girvan modularity 0.4297; the homogeneous one-community partition has modularity 0. Mixed-label identity homophily is 0.8377 in the simulation; homogeneous identity homophily is 1.0 by construction, so the latter is not evidence of an emergent echo chamber.

For the rooted cascade-shape comparison, the constructed hashtag object had depth 4, breadth 33, reached size 61, and structural virality 2.7. Across 16 simulated D-net cascades,

the corresponding means were 2.5625, 40.9375, 53.5, and 1.7963. Breadth was the only one of the three validation metrics (depth, breadth, and structural virality) within the pre-specified 30% band; the resulting structural-similarity score was 0.6885.

The distributional diagnostics do not validate a digital twin. With one constructed reference graph and 16 simulated cascades, the Kolmogorov–Smirnov distances were $D = 1.0000$ for depth, $D = 0.8125$ for breadth, and $D = 1.0000$ for structural virality. Jensen–Shannon divergence was 1.0000 for all three metrics, and all three distribution-match flags were false. Because $n_{\text{real}} = 1$, the KS reference distribution is not a campaign-level sampling distribution; those distances are reported only as implementation diagnostics. Finally, DTFS was 0.2754, below the 0.70 software threshold (`isValidated=false`). Its weights 0.40/0.40/0.20, the 0.70 cut, and the content-correlation component fixed at zero because no empirical Twitter MI exists are repository choices, not validated fidelity criteria. The structural evidence therefore supports a comparison of cascade shape on a designed reference graph, not empirical misinformation fidelity.

5.5 Applying the Pfeffer firestorm framework

5.5.1 The original framework and the thesis remapping

Pfeffer et al. (2014) define an online firestorm as a sudden discharge of large quantities of negative word-of-mouth and complaint messages directed at a person, company, or group. The messages are predominantly opinion rather than fact and are therefore highly affective. The article is a qualitative–synthetic account based on observed cases; it does not present an agent-based model, a tipping-point estimator, or a climate-discourse experiment.

The Outlook of Pfeffer et al. (2014) identifies seven interrelated factors:

1. *speed and volume of communication*;
2. *binary choices*;
3. *network clusters*;
4. *unrestrained information flow*;
5. *lack of diversity*;
6. *cross-media dynamics*; and
7. *network-triggered decision processes*.

These are the original factors used here. In particular, valence and surprise are not numbered Outlook factors. Valence is motivated by the affective character in the paper’s definition of a firestorm, while surprise is a thesis operationalisation of the novelty, speed, and volume mechanism in factor 1.

The thesis vocabulary—valence, surprise, identity, clustering, echo, temporal, and cross-media—is consequently a computational remapping, not a renaming of Pfeffer et al.’s list (Table 5.10). The first six terms describe variables or observables that can at least partly

Table 5.10: Original Outlook factors and their treatment in the thesis. The right-hand column is an operational mapping, not Pfeffer et al.’s terminology.

#	Pfeffer et al. (2014)	Thesis treatment
1	Speed and volume of communication	Surprise (shock versus drip) and temporal horizon; surprise is held at drip and empirical timing is unavailable.
2	Binary choices	No dedicated observable or sweep; simulated actions are ternary.
3	Network clusters	Clustering and part of the echo measurement.
4	Unrestrained information flow	Partly represented by echo and hub reach, but not the hundreds or thousands of weak ties discussed in the paper.
5	Lack of diversity	Identity composition and part of echo/homophily; homogeneous versus mixed belief profiles are compared.
6	Cross-media dynamics	Held; neither the empirical reconstruction nor the engine contains a legacy-media or newsroom layer.
7	Network-triggered decision processes	Related to the temporal ledger row, but the four adoption stages are not logged or estimated.

be represented in the simulation; cross-media is included as a seventh ledger row but is held because the model has no legacy-media layer. Conversely, Pfeffer et al.’s binary-choice factor has no dedicated thesis row. The agents choose among **forward**, **reinterpret**, and **drop**, so the engine is ternary rather than an implementation of the paper’s like/share/retweet either-or mechanism.

5.5.2 Order of application: constructed structure before simulation

The application proceeds in two stages. First, structural observables are calculated on a 63-node hashtag co-occurrence fallback derived from published discourse categories. Because tweet hydration failed, this object is not an observed user graph, retweet cascade, or temporal diffusion record. Second, D-net imports that fixed fallback structure and simulates propagation on it. The simulation therefore tests behaviour on a documented stand-in; it does not reconstruct or validate the original Twitter network.

The application proceeds in two distinct stages. First, structural observables are calculated on the empirical fallback derived from Debnath et al. (2023). This object is a hashtag co-occurrence graph with 63 nodes, 228 directed edges, and 171 unique undirected pairs. It is not a retweet, reply, follower, or **conversation_id** cascade. The 814,924 tweet identifiers in the associated dataset were not hydrated, and the graph contains no usable tweet clock. Its labels and edges can therefore support a bounded comparison of hashtag structure, but not a reconstruction of user-to-user diffusion.

Second, D-net imports the same 63-node, 228-edge topology and runs agents on it. This ordering matters: transitivity, local clustering, hub degree, and the mixed-label conspiracy cut already exist before any simulated message is generated. D-net copies this structure; it does not produce or validate it. The post-simulation evidence consists of content-auditor scores, realised actions and reach on that fixed graph, together with label-dependent quantities for the homogeneous and mixed assignments.

Empirical hashtag structure. The graph has global transitivity 0.315, mean local clustering 0.515, and mean unique degree 5.43. The `#chemtrails` hub has 33 unique neighbours. On the 171 unique-undirected pairs, the six assigned graph clusters have $Q = 0.5489$, the four assigned discourse identities have $Q = 0.5322$, and the binary conspiracy cut has $Q = 0.4297$. These label-partition statistics are distinct from clustering coefficients. Identity edge homophily is 0.873; belief-profile-family homophily is 0.838. The identity composition is 28 conspiracy, 19 climate-action, 14 environmental, and two expert/other nodes. Thus conspiracy discourse is a plurality ($28/63 = 0.444$), not a monoculture.

D-net after simulation. The mixed D-net condition retains 28 conspiracy nodes and the empirical type-level composition, with 19 climate-action, 13 environmental, and three expert nodes after belief-profile assignment. The homogeneous condition instead assigns all 63 seats the `conspiracy_believer` profile. Because both conditions use the same edge set, transitivity, local clustering, and hub degree remain 0.315, 0.515, and 33. Mixed D-net identity homophily is 0.838, whereas the homogeneous value is 1.0 by construction. The latter is not evidence that an echo chamber emerged. It is the mechanical result of giving every node the same identity. Mixed belief-profile families have $Q = 0.4908$, and the mixed binary conspiracy cut has $Q = 0.4297$. Under homogeneous labels there is only one community, for which $Q = 0$.

Across all eight article-level D-net cells (four configurations), 61 of 63 nodes receive a message within the held eight-tick horizon. Reaching 61/63 is saturation of the seeded connected component under the engine; the two unreached seats form a separate expert component. This does not establish that the Twitter discourse spread to 61 users in eight equivalent units of time. Simulation ticks have no defensible conversion to hours, and the eight-hop budget is a computationally compressed horizon.

5.5.3 What the application can and cannot reveal

Table 5.11 summarises what the application can and cannot reveal.

The valence result requires particular care. In the continuous runs, mean auditor MI for SCoPEX is 1.735 in homogeneous D-net and 0.998 in mixed D-net; for the chemtrails seed it is 3.338 and 2.507, respectively. These values show content degradation under a particular auditor and graph intervention. They do not measure indignation or toxicity. The dual-discrete instrument is a separate scale and does not preserve the same article ordering in every cell, so the two instruments are not averaged.

Likewise, the network-mean threshold k^* , where used, is a thesis rule based on simulated MI. It is neither a statistic introduced by Pfeffer et al. (2014) nor an empirical estimate from Debnath et al. (2023). Structural cascade comparison also does not repair this gap: the available comparison has one empirical graph, and its reported digital-twin fit score is 0.2754, below the 0.70 validation threshold. The D-net application should therefore be read as a controlled, theory-mapped simulation on a constructed hashtag stand-in, not as a validated digital twin of the geoengineering conversation.

Table 5.11: Interpretive boundary of the Pfeffer application. Pre-simulation values refer to the reconstructed hashtag graph; simulated values refer to D-net after a run. Homogeneous one-community modularity is $Q = 0$.

Observable	What the method reveals	What the method cannot reveal
Clustering	Before simulation, the hashtag construction contains substantial closure: transitivity 0.315, local clustering 0.515, and a hub of degree 33. The mixed-label conspiracy cut has $Q = 0.4297$.	These values are not retweet triadic closure or emergent D-net clusters. The graph is a documented co-occurrence construction, not a Watts–Strogatz fit, and previously reported Debnath eigenvector centralities were not re-estimated here.
Echo	Empirical identity homophily 0.873 and belief-profile-family homophily 0.838 show that most represented ties remain within discourse blocks. Mixed D-net retains comparable block echo (0.838).	The measure does not identify algorithmic ranking, weak-tie exposure, or an emergent chamber. Homogeneous D-net homophily of 1.0 is tautological and cannot measure echo formation.
Identity	The comparison separates a mixed assigned-label composition (conspiracy share 0.444) from the homogeneous intervention (share 1.0). It therefore isolates identity composition as a model input on a common topology.	The labels are theory-grounded reductions, not HDBSCAN communities recovered from hydrated tweets, and they do not establish individual users' identities or follower-network homophily.
Valence proxy	The empirical side supplies only the paper-reported toxicity mean (approximately 0.17) and a hashtag prior proxy (0.1097); the simulated side supplies an LLM-auditor misinformation index. On the continuous instrument, the chemtrails seed has higher mean MI than the SCoPEX seed in both homogeneous and mixed D-net.	Toxicity, a hashtag prior, and auditor MI are different constructs and scales. Debnath provides no empirical MPR. The comparison cannot be described as Twitter MI, an empirical-to-simulated MPR match, or validation of a firestorm.
Surprise	The ledger records the seeded regime: a single <code>user_chemtrails_hub</code> provides a single-node, single-time seed. The paper-reported SCoPEX volume spike motivates the construct.	Surprise is held, not tested. There is no all-node shock condition and no reconstructed empirical volume series, novelty half-life, or attention decay.
Temporal	D-net records engine progression for eight ticks and shows 61/63-node reach within that budget.	Empirical timing is unavailable because the hashtag fallback has no tweet clock. Ticks cannot be translated into minutes or hours, and the method cannot estimate inter-arrival times, half-life, or Pfeffer et al.'s knowledge–persuasion–propagation–affirmation sequence.
Cross-media	The ledger makes the omission explicit rather than attributing it to a topology.	Cross-media dynamics are held on both sides. There is no social-to-legacy- media-to-social loop, newsroom agent, external-volume estimate, or basis for treating a hierarchical graph as cross-media amplification.

5.5.4 Interpretation

The application supports three bounded conclusions. First, clustering and block homophily are properties of the reconstructed hashtag graph before simulation; they are consistent with the structural mechanisms in Pfeffer et al.'s factors 3 and 5, but do not prove a user-level firestorm. Second, holding topology fixed while changing the belief-profile composition distinguishes mixed from homogeneous D-net outcomes and avoids mistaking an all-identical assignment for emergent echo. Third, content auditing can describe simulated factual degradation after propagation, but it cannot supply the missing empirical temporal, binary-choice, weak-tie, or cross-media processes. The framework is thus applied as an explicit mapping and limitation ledger, not as a claim that all seven original factors were experimentally realised.

6 Discussion

This chapter interprets the Phase 2 results in relation to the research questions and the literature. The distinction between evidence, interpretation, and implication is maintained throughout. This distinction is necessary because the campaign comprises one run per configuration ($N = 1$), one retained realisation per configuration, an eight-tick horizon, and 292 dead cells among 1,728 configured cells. The reported quantities are therefore descriptive properties of the present LLM simulation. They are neither estimates of population effects nor measurements of misinformation on Twitter.

6.1 Synthesis of the research questions

6.1.1 RQ1: What does the instrument comparison establish?

The instrument comparison constrains the interpretation of all substantive results. Dual-discrete scores exceeded continuous scores on every network-condition aggregate in both arms, while topology ranks aligned weakly. Same-event dual diagnostics show the same directional gap, but agreement is defined for only 270 of 478 live H dual cells and 171 of 240 live He dual cells. Because missing items receive different penalties, a common $MI > 3$ cut is not equivalent across instruments. T2c and T2d are also separate realisations, so campaign-level T2d–T2c contrasts are whole-pipeline sensitivity rather than a clean instrument effect. The campaign therefore establishes that severity levels and network-condition ordering depend on the auditor regime. It does not separately answer an instrument effect on threshold occurrence from a dedicated topology-resolved k^* table.

6.1.2 RQ2: What is the role of identity composition?

Evidence. When all eight generated topologies were collapsed within one instrument, the heterogeneous arm exceeded the homogeneous arm: by 0.1549 in continuous MI (H 0.8643, He 1.0192) and by 0.3986 in dual-discrete MI (H 1.6814, He 2.0800). The continuous direction was nevertheless split: He exceeded H on four of eight topologies, whereas H exceeded He on the other four. The dual-discrete direction was $He > H$ on all eight topologies. Composition-specific results were substantially more differentiated than these arm labels. Homogeneous conspiracy-family cells reached 2.2194 continuous and 3.1691 dual-discrete, compared with 0.1118 and 0.8687 for homogeneous science/environment personas. Among heterogeneous mixes, `mix_00`, containing four conspiracy personas, scored 1.7712 continuous and 3.1321 dual-discrete; conspiracy-free `mix_02` scored 0.2645 and 1.1882. Restricting the collapsed comparison to zero-HTTP-error runs reverses the continuous difference (-0.0295 rather than $+0.1549$) while the dual-discrete difference remains positive ($+0.4695$). The continuous

collapsed-arm direction is therefore not robust to API availability.

Interpretation. The collapsed H/He contrast is not an estimate of a general effect of diversity. The homogeneous arm contains both conspiracy-only and scientist-only societies, while the heterogeneous arm contains mixes with different conspiracy shares and different persona identities. Averaging within these broad labels changes the distribution of substantive roles. Consequently, the aggregate He>H difference is consistent with a strong composition gradient: conspiracy-conditioned cells score high, whereas science-conditioned cells score low. It does not establish that heterogeneity intensifies misinformation, just as the low value of `mix_02` does not establish that diversity prevents it.

Implication. Future comparisons should hold the proportion and identity of conspiracy-conditioned seats constant while varying only within-network diversity or placement. Network means should be accompanied by persona-conditioned MPR and exposure-matched comparisons. This would separate a genuine interaction between heterogeneous neighbours from the mechanical dilution or enrichment produced by changing the number of high-scoring personas.

6.1.3 RQ3: How does network topology affect misinformation propagation?

Evidence. Topology changed the live-cell scores, the frequency of within-window k^* , and the probability that a cascade remained scorable. However, no topology occupied the highest or lowest position consistently across identity composition and scoring instrument. In homogeneous continuous runs, scale-free networks had the highest topology mean (0.9205), whereas small-world networks had the lowest (0.7215). In homogeneous dual-discrete runs, echo chambers had the highest mean (1.7984), again with small-world networks lowest (1.5977). In heterogeneous runs, the hierarchical topology had the highest mean under both instruments (1.5242 continuous and 2.5934 dual-discrete). These latter estimates must be read together with the hierarchical dead-cell rates of 27.8% and 22.2%, respectively. Polarised heterogeneous continuous runs likewise combined a high live-cell mean (1.4137) with a 30.6% dead-cell rate.

Interpretation. The results do not support a topology-invariant ranking. Topology appears to condition which messages are observed and scored, but it does so jointly with persona placement, article content, and cascade survival. In particular, live-only means on topologies with many dead cells describe the surviving cascades rather than the generator as a whole. The scale-free versus random comparison is primarily a hub and degree-sequence contrast; it is not a controlled comparison of clustering coefficients. Similarly, the hierarchical generator represents authority-oriented transmission and should not be interpreted as a model of cross-media dynamics.

Implication. Topology should be treated as a conditioning structure, not as a sufficient cause of factual degradation. Subsequent work should use repeated graph seeds, report cascade survival as an outcome, and match graph properties when isolating a specific mechanism such

as clustering, centralisation, or community separation. On the present evidence, selecting a single ‘most harmful’ topology would overstate what the design identifies.

6.1.4 Measurement implications across RQ2 and RQ3

Evidence. Dual-discrete scores exceeded continuous scores on all eight topologies in both arms. The topology-mean difference was 0.8201 for homogeneous and 1.0450 for heterogeneous runs. Among cells live under both modes, the median paired difference was 0.8125 for H ($n = 398$) and 0.9778 for He ($n = 198$). Instrument choice also affected the incidence of within-window k^* : the collapsed He rates were 32.1% under dual-discrete scoring and 7.6% under continuous scoring. Moreover, topology rankings agreed only weakly across modes (Spearman $\rho = 0.095$ for H and $\rho = 0.167$ for He). These coefficients describe this harvest’s rank order; they are not population inferences.

Interpretation. The two modes are not interchangeable measurements on a common numerical scale. The discrete instrument awards categorical penalties for omitted or contradicted answers, whereas the continuous instrument resolves degradation differently. The systematic level difference and weak rank agreement show that instrument choice can affect both absolute severity and comparative claims about topology. The continuous sidecar generated during a dual run is useful for diagnosing disagreement, but it is not a third independent MPR.

Implication. Substantive conclusions should be reported separately for each instrument. Averaging the two modes would create a quantity without a defined auditor. The disagreement also motivates external validation: a preregistered sample of rewrites should be assessed by human coders, and calibration should examine whether each instrument preserves article- and persona-level ordering rather than merely correlating at the event level. Until such validation is available, k^* is instrument-relative.

6.1.5 RQ4: Does the D-net analysis generalise the generated-topology findings?

Evidence. D-net used a 63-node custom graph reconstructed from hashtag co-occurrence, not a retweet cascade. Its homogeneous condition assigned `conspiracy_believer` to all 63 seats; its mixed condition retained a conspiracy share of 28/63. On the identical edge set, homogeneous conspiracy exceeded the mixed condition for both articles and both instruments: continuous SCoPEx 1.7349 versus 0.9983, continuous chemtrails–Gates 3.3378 versus 2.5068, dual-discrete SCoPEx 4.2537 versus 2.6880, and dual-discrete chemtrails–Gates 3.7303 versus 1.5297. Structural comparison was less favourable. Breadth fell within the specified 30% band, whereas depth and structural virality did not; DTFS was 0.2754 and failed the 0.70 validation threshold. Tests comparing one empirical graph with 16 simulated cascades were underpowered.

Interpretation. D-net provides a useful stress test of the composition interpretation because the topology is held fixed while persona labels change. It does not provide out-of-sample

validation of the eight-node generators, and it is not a digital twin of Debnath et al.’s Twitter corpus. The graph imports co-occurrence structure and then simulates message propagation on that structure; it neither reconstructs empirical retweet paths nor reproduces an empirical misinformation score. The D-net result therefore supports the narrow statement that a conspiracy-only assignment scored above the retained mixed assignment on this larger custom graph.

Implication. The qualitative composition pattern recurs in D-net, but numerical effect sizes and cascade-shape claims do not. Stronger generalisation would require hydrated interaction data, common community and centrality estimators, temporal information, multiple reconstructed cascades, and repeated simulation seeds. Simulated auditor MI must remain separate from Debnath et al.’s toxicity and network measures (Debnath et al., 2023; Ruths & Pfeffer, 2014).

6.2 Reconciling the collapsed and D-net orderings

Evidence. The generated-topology collapse gives $H_e > H$. In D-net, homogeneous conspiracy scores exceed the mixed assignment. These comparisons use different definitions of H. Across the eight-node grid, H denotes twelve separate homogeneous persona societies, including four conspiracy personas as well as climate-action, environmental, journalist, and scientist personas. D-net H denotes one specific treatment: all 63 nodes are assigned the `conspiracy_believer` persona. D-net H_e preserves a majority non-conspiracy composition. Restricting the generated-topology H cells to the conspiracy family already reverses the broad collapse: conspiracy H is well above overall H_e on both instruments.

Interpretation. The two orderings address different comparisons because the H condition has different compositions in the generated-topology and D-net analyses. The label H encodes within-network uniformity, but it does not encode which identity is uniform. Collapsing across homogeneous scientist and homogeneous conspiracy societies answers a different question from comparing a conspiracy monoculture with a mixed society. D-net changes both diversity and the prevalence of the high-scoring persona when moving from H to H_e . Its ordering is consequently consistent with composition, but cannot distinguish composition from a causal benefit of heterogeneity.

Implication. H and H_e should not be used as portable substantive treatments without an attached composition descriptor. A defensible notation would distinguish, for example, H-conspiracy, H-science, H_e -44%-conspiracy, and H_e -0%-conspiracy. A matched factorial design could then vary diversity while holding conspiracy share, seed position, and topology fixed. This reformulation is more informative than asking whether H or H_e is ‘higher’ in the abstract.

6.3 Relation to the prior arXiv preprint and adjacent literature

6.3.1 The prior auditor–node study as prior method, not a benchmark result

Evidence from the literature. Maurya et al. (2025) introduced the auditor–node framework, MI and MPR, and the severity taxonomy used by the present system. Its published evaluation concerned 30-hop linear branches, a different article collection, a larger persona inventory, ten auditor questions, and a different model configuration. It reported that identity- and ideology-oriented homogeneous branches could accelerate degradation and that experts could stabilise it; it also reported propaganda-tier outcomes in approximately 85% of heterogeneous branch–domain pairs. Thus, the prior preprint itself does not establish a universal rule that heterogeneous societies must score below homogeneous societies.

Interpretation of the present comparison. The present study transfers the instrument to climate and geoengineering content and extends the setting from linear branches to generated and custom graphs. The non-zero, persona-sensitive scores indicate that the instrument remains responsive in this domain. The observed collapsed He>H ordering is not a failed replication of a universal prior-study H>He hypothesis, because no such universal hypothesis follows from the prior heterogeneous result. Conversely, D-net H-conspiracy>He-mixed cannot be presented as a replication: identity inventories, graph objects, hop budgets, auditor questions, and model settings differ, and the present campaign has only one replicate.

Implication. The defensible contribution is a boundary analysis of the prior method, not a numerical reproduction of the prior preprint. A direct conceptual replication would require matched linear chains, an expert-inclusive heterogeneous arm, the same climate articles in both arms, a sufficiently broad article set, and node-type-conditioned outcomes. The eight-hop graph campaign can motivate that test but cannot substitute for it.

6.3.2 Topology and diffusion literature

Evidence from the literature. Classical diffusion models establish that network structure affects reach and activation opportunities (Kempe et al., 2003). Small-world and preferential-attachment models formalise different structural properties (Barabási & Albert, 1999; Watts & Strogatz, 1998). Empirical work further shows that cascade depth and breadth are not captured by size alone (Goel et al., 2016), and that false news can diffuse farther and faster than true news in observational data (Vosoughi et al., 2018).

Interpretation of the present comparison. The present results are compatible with the general proposition that structure conditions diffusion, but they do not re-estimate these empirical findings. Here the state is mutable text, forwarding is mediated by an LLM, and live-cell MI is an auditor output. Topology effects are also entangled with cascade death and composition. The data therefore add a content-transformation perspective to network

diffusion; they do not show that one canonical graph class is intrinsically most susceptible to misinformation.

Implication. Network controls in LLM societies should include both structural summaries and content outcomes. Matching degree, density, seed centrality, and clustering would permit more specific tests than topology names alone.

6.4 Pfeffer’s firestorm theory: insights and boundaries

Evidence. Pfeffer et al. (2014) identify seven interrelated factors: speed and volume of communication, binary choices, network clusters, unrestrained information flow, lack of diversity, cross-media dynamics, and network-triggered decision processes. The present six-knob vocabulary is a computational remapping rather than this original list. The campaign varied network structure and identity composition and compared article frames. It held surprise as a single-node drip, fixed the horizon at eight ticks, omitted a cross-media layer, and used ternary actions—forward, reinterpret, and drop—rather than binary platform actions. The constructed D-net hashtag graph had transitivity 0.315, mean local clustering 0.515, and mixed-label homophily close to 0.84, but these structures were pre-wired rather than generated by the simulation.

Interpretation. Two Pfeffer insights receive limited support as useful design principles. First, network clusters and identity structure cannot be reduced to a single topology label: the high-scoring composition can be concentrated or diluted within the same graph. Second, lack of diversity is not adequately tested by the binary H/He label unless identity prevalence is controlled. The D-net comparison illustrates both points particularly clearly. By contrast, this campaign does not test speed and volume as a shock manipulation, unrestrained information flow at platform scale, binary-choice dynamics, cross-media amplification, or the four stages of network-triggered adoption. Nor does k^* originate in Pfeffer’s framework; it is a thesis-specific auditor threshold.

Implication. In this thesis, Pfeffer’s framework serves primarily to identify what the simulation represents and omits. It should not be presented as a seven-factor confirmation. A fuller test would add shock versus drip seeding, a binary-action ablation, a legacy-media or exogenous-broadcast layer, and explicit logging of repeated exposure and affirmation. Such extensions would move the framework from retrospective mapping towards identified mechanism tests.

6.5 Alternative explanations and validity of the interpretation

Several alternative explanations remain plausible. Persona prompts may induce role compliance rather than persistent beliefs; high conspiracy scores may therefore partly measure how faithfully the model follows an instruction to use conspiratorial language. The action policy

gives reinterpretation a substantial role, which creates opportunities for factual mutation by design. The same model family participates in generation and automated assessment, and the five author-defined questions constrain what counts as degradation. Article differences combine topic, wording, affective charge, and prior fit with persona prompts, so they are not a clean manipulation of valence. Finally, excluding dead cells from means avoids coding them as factual zeros, but produces survivor-conditioned estimates.

These limitations do not make the campaign uninformative. They narrow the appropriate claim. The experiment demonstrates that, under the documented prompts and graph protocols, auditor scores vary with persona composition, topology, and scoring mode. It does not establish human susceptibility, platform-level firestorm dynamics, or causal effects in an empirical population. Human evaluation, model ablations, repeated seeds, and a fail-closed auditor are therefore validation requirements rather than optional refinements (Park et al., 2023; Zheng et al., 2023).

6.6 Overall implications

For simulation methodology, the principal implication is that treatment labels must preserve their substantive content. ‘Homogeneous’ and ‘heterogeneous’ are insufficient without persona shares, placement, and node-conditioned outcomes. Likewise, topology names should be accompanied by realised graph statistics and cascade-survival rates.

For measurement, discrete and continuous MI should be treated as separate operationalisations. Agreement in a direction across both instruments is stronger descriptive evidence than a result present under only one, but neither becomes ground truth through convergence. Divergence is itself a result about the sensitivity of the conclusion to the auditor.

For empirical generalisation, D-net is an intermediate case: larger and structurally grounded in a hashtag graph, yet still neither a retweet cascade nor a temporal reconstruction. Its contribution is to show that the composition interpretation survives a change in graph size and source under the same simulator. It does not validate the simulator against Twitter.

Taken together, the research questions yield a restrained conclusion. Topology conditions propagation but has no stable winner; identity composition is more informative than the H/He label; severity and k^* are instrument-dependent; and D-net supports only qualitative, not numerical, generalisation. Pfeffer’s framework explains why these dimensions should be considered jointly, while also making visible the mechanisms that the current campaign leaves untested.

7 Validity, Ethics, and Reproducibility

The Phase 2 results describe one configured simulation campaign. They are not population estimates and not a validated digital twin of online climate discourse. The harvest is therefore diagnostic: it shows which patterns appeared under the specified graph generators, personas, articles, language model, and scoring instruments, and where those patterns are sensitive to design choices and API availability.

7.1 Internal and statistical validity

One replicate. The parser retained one run per configuration ($N = 1$). Configurations recorded `graphRandomSeed:` 42, but only the echo-chamber and polarised builders consumed it; ER, small-world, scale-free, and action sampling used unseeded `Math.random`. Article cells within a configuration share the realised graph, persona assignment, protocol, and hosted-model context, so they are not independent campaign replicates. Inboxes and per-article histories were cleared between articles, and post-hoc trust updates could not alter already completed trajectories. The reported means and rank orders are descriptive comparisons of this harvest; they do not provide confidence intervals, population-level significance tests, or causal estimates of a topology or identity-mix effect. In particular, a small p -value computed over pooled cells would not repair the absence of independent simulation replicates.

Mitigation. Results are separated by scoring mode and accompanied by live-cell counts and dead-cell rates. Claims are phrased as observations ‘in this harvest’, and the analysis avoids interpreting event counts as sample size. Future work should run independently seeded graph and model replicates, predefine topology-level contrasts, and estimate uncertainty with a multilevel or cluster-aware analysis whose experimental unit is the independent run.

Eight-hop horizon. The campaign stops after eight ticks/hops, a cost compression relative to the prior linear-chain length of 30. This horizon is neither an empirical firestorm duration nor Debnath et al.’s protocol: their ‘eight’ is a skip-gram context window of eight words. The short horizon right-censors propagation and weakens the interpretation of k^* , which is defined as the first observed tick above the threshold with no later observed recovery. A cascade that appears sustained within the window at tick eight may recover at tick nine.

Mitigation. Tick counts are not translated into clock time, and k^* is reported as an eight-tick simulation outcome. Future work should conduct a horizon sensitivity analysis (including the 30-step reference), use survival or censoring-aware summaries, and calibrate a time scale only if timestamped empirical cascades become available.

Model dependence and stochastic services. Both the agents and the auditor use `gpt-4o-mini`. The findings may therefore reflect this model’s instruction following, safety tuning, linguistic priors, and version at the time of execution. A hosted model can change without a corresponding repository change, and API sampling may not be perfectly reproducible even when the graph seed is fixed. High misinformation scores for conspiracy personas are compatible with prompt obedience; they are not evidence that a field community would behave in the same way.

Mitigation. The model name, configuration, graph seed, raw run directories, parser rules, and harvest timestamp are retained. Future work should pin provider model snapshots where possible, record sampling and API metadata, repeat runs across several model families and temperatures, and report model-by-condition interactions rather than presenting one model as a generic human proxy.

API failure and fallback process. A forensic audit of the 288 selected runs found 203,928 successful responses and 64,270 HTTP-error responses in the client counters; 121 runs recorded at least one error. The client made no automatic retries. Errors were strongly time-clustered: 57,417 occurred among runs started in the 04:00 UTC hour. Failed rewrites propagated the unchanged input while retaining the `reinterpret` action; failed audits left 36,194 eligible events with a null score. Null scores were excluded from MI, MPR, and k^* , but reduced `nScored` and could cause hatching. The malformed-response path would instead fail open to all-correct scores (MI = 0); no such parse warning was found in the selected archived log sections. Raw responses and request identifiers were not retained, so provider-side uniqueness and exact event linkage cannot be reconstructed.

The same-mode He–H sensitivity is consequential. The canonical continuous delta is +0.1549, but becomes −0.0295 when restricted to runs with zero recorded HTTP errors. The dual-discrete delta remains positive (+0.3986 canonical; +0.4695 in zero-error runs). This is a conditioned sensitivity analysis, not a corrected causal estimate, because API availability was associated with time and condition. The continuous collapsed H–He direction is therefore not robust to this restriction, while the dual-discrete direction is stable across the audited checks. Complete forensic tables are retained separately; raw run results were not rewritten.

Mitigation. These failures are reported as an outcome-relevant execution process rather than folded into cascade death. Future runs should use bounded backoff, fail-closed schema validation, per-request identifiers, stage and article labels, immutable raw-response hashes, and explicit fallback flags on events. Propagation sparsity, rewrite failure, audit failure, and parse failure should be separate denominators.

Auditor circularity. The generator and judge use the same `gpt-4o-mini` endpoint and client defaults, so dual scoring tests two prompts of one model rather than two independent instruments. High conspiracy-persona scores may primarily measure role-prompt compliance. The campaign has no human or cross-family criterion for accuracy, realism, or calibration. The same model family generates the messages and judges them against five author-curated questions. Shared model priors can create correlated generator and judge errors, while the

selected questions determine what counts as missing or incorrect. Discrete and continuous scores are also different instruments: the dual-run discrete headline, its continuous sidecar, and the continuous-only campaign are not interchangeable. Their disagreement shows that misinformation prevalence is partly measurement-dependent.

Mitigation. The two headlines are never averaged, dual gap is treated as an instrument diagnostic rather than a third MPR, and the question key is retained for inspection. Future validation should blind independent human coders to condition, measure inter-rater reliability, adjudicate disagreements, test alternative question sets, and use judge models from model families not used to generate the content.

Hatched-cell policy. Of 1,728 thesis cells, 292 are hatched because $nScored \leq 1$ (232 with zero scored events and 60 with one). Positive configuration-level LLM usage does not prove that the article cell itself received a successful auditor call. They are kept in the ledger but excluded from live-only means; they are neither zeros nor evidence that a mix prevented misinformation. If death depends on topology, persona, or scoring mode, complete-case means describe surviving cascades and may be selection-biased. This concern is material for slices such as hierarchical and polarised heterogeneous runs, which have comparatively high dead rates.

Mitigation. Every live mean is paired with its denominator and hatch rate, and no dead cell is imputed as zero. Future work should diagnose the mechanism producing unscored cells, predefine a minimum-information rule, rerun failed cells under the same protocol, and report sensitivity bounds or selection models rather than relying only on complete cases.

7.2 Construct and external validity

Descriptive comparisons, not interventions. The grid changes several consequential features at once. Homogeneous and heterogeneous labels encode different persona compositions, while generators also differ in degree sequence, reachability, and cluster assignment. Collapsed H–He differences therefore do not isolate ‘diversity’, and topology rank differences do not identify a causal network effect. The pooled tables concatenate cells; they do not form a physical super-graph.

Mitigation. Same-mode contrasts are kept separate, composition is shown by persona family and mix, and D-net is reported outside the eight-node pool. Future experiments should use matched compositions, graph randomisations that preserve relevant degree sequences, factorial ablations, and preregistered estimands for identity, topology, and their interaction.

Persona reduction. The personas are theory-faithful prompt reductions of discourse types, not individuals, survey respondents, or HDBSCAN centroids reconstructed from the full Debnath corpus. Rich and changing identities are compressed into short, stable instructions. The simulation therefore risks stereotyping conspiracy-oriented, activist, scientific, or environmental participants and may overstate within-group consistency. A model’s performance of a persona must not be interpreted as a psychological diagnosis or a moral ordering of real

people.

Mitigation. Results are attributed to prompt configurations rather than demographic groups, and the auditor’s scientific key evaluates claims rather than the worth of a persona. Future work should co-design personas with domain experts and affected communities, represent uncertainty and within-group variation, test multiple prompt formulations, and audit outputs for caricature and disparate treatment.

Hashtag graph, hydration failure, and absent empirical MPR. The 63-node D-net object is a hashtag co-occurrence reconstruction with 228 directed importer edges; it is explicitly marked `notARetweetCascade=true`. The edge container must not be read as observed retweets, replies, or follower ties. The study did not hydrate the reported 814,924 tweet identifiers. The accessible OSF identifiers were stored in lossy scientific notation, the Mendeley identifier dump was unavailable to the study, and the researchers had no Twitter/X bearer token. The fallback therefore cannot reproduce user-level diffusion or temporal ordering. Debnath et al. did not report the present 0–5 misinformation index or MPR, so there is no empirical Twitter MPR against which simulated MPR can be matched.

Mitigation. The graph is labelled as a structural hashtag fallback, tweet IDs are not invented, and simulated auditor scores are never described as observed Twitter misinformation. Future work requires a lawful, lossless identifier source and approved API access, followed by an explicit separation of hashtag co-occurrence, retweet, reply, mention, and follower networks. Any empirical content outcome would also require a separately validated annotation protocol rather than retrofitting simulated MPR.

Distributional validation with $n_{\text{real}} = 1$. The KS and Jensen–Shannon comparisons use one empirical graph against simulated cascades. With $n_{\text{real}} = 1$, a two-sample KS result has little power and does not estimate natural variation among empirical firestorms. The DTFS of 0.2754 falls below the configured validation threshold of 0.70. Individual metric matches do not alter this failed implementation diagnostic. Breadth similarity alone does not establish a digital twin.

Mitigation. These quantities are presented as diagnostics and the failed threshold is retained. Future work should collect multiple comparable empirical cascades, define metrics and tolerances before looking at simulation outputs, use uncertainty-aware posterior predictive checks, and validate structure, timing, content, and intervention response separately.

7.3 Reproducibility and resource constraints

The repository retains configs, model names, seeds, raw outputs, parser code, derived tables, hatch rules, and the official harvest timestamp. These artefacts support computational traceability: a reader can follow a reported number to a cell and inspect the transformation. They do not guarantee exact reproduction because the external model service, model weights, API defaults, and provider-side moderation can change. Re-running the full campaign also incurs substantial monetary, energy, and time costs, and indiscriminate reruns could overwrite

or silently diverge from the recorded harvest.

Mitigation. Phase 2 outputs are isolated from Phase 1, completed runs are preserved, secrets are excluded, and analysis can be regenerated without new LLM calls. A reproducibility release should add immutable checksums, software and package lockfiles, model snapshot identifiers where available, machine-readable provenance, exact cost/token totals, and a small deterministic fixture for testing parsers without paid calls. Replication plans should budget API calls in advance and prioritise statistically informative replicates over a larger but unreplicated grid.

7.4 Privacy, research ethics, and dual use

API privacy and cost. Sending content to hosted APIs transfers data to a third party and may create retention, jurisdiction, and terms-of-service obligations. Although the present campaign uses curated articles and synthetic personas, a future empirical collection could process usernames, tweet text, profile fields, or subsequently deleted content. Public availability does not remove contextual privacy or justify redistributing personal data. API access is also financially unequal and can make nominally reproducible research inaccessible to groups without equivalent budgets.

Mitigation. The failed hydration workflow discarded sampled tweet text and user fields after aggregate hashtag extraction, did not fabricate IDs, and did not expose credentials. Any future empirical collection should undergo institutional ethics review, follow platform terms and data-minimisation principles, store identifiers separately with access controls, honour deletion where feasible, publish aggregates instead of raw personal data, document retention, and disclose expected and realised API costs.

Dual-use risk. A simulator that identifies combinations of topology, identity prompts, and message framing associated with high propagation or distortion could support defensive stress testing, but it could also be used to optimise propaganda, harassment, astroturfing, or evasion of moderation. Detailed persona prompts and high-performing cascade recipes can lower the cost of such misuse.

Mitigation. The thesis should report scientific findings without packaging operational targeting instructions, avoid claims about vulnerable real-world groups, and release sensitive artefacts proportionately. Future work should include a formal dual-use review, threat modelling, access tiers for potentially actionable configurations, abuse monitoring, and evaluation of defensive applications such as fact-check timing without deploying content to live platforms.

No human evaluation. Human-evaluation templates exist in run directories, but no completed, ethics-approved human study was conducted. There is therefore no evidence that people find the rewrites realistic, that experts agree with the auditor, or that the personas faithfully represent lived discourse. The absence of human evaluation is a criterion-validity limitation as well as an ethical safeguard: incomplete templates must not be presented as participant data. The computational campaign used curated public texts and synthetic perso-

nas. No formal ethics-determination, review, exemption, or approval artefact for this master's thesis is present in the repository. This sentence records that absence. It does not claim that review was not required, that an exemption was granted, or that the work has been institutionally cleared. **[USER INPUT REQUIRED: confirm with the responsible TUM school or ethics body whether review, exemption, or a not-required determination applies, and deposit the written artefact]** Licensing of OSF/Mendeley identifier access remains a submission-time documentation requirement.

Mitigation and future work. A follow-up study should obtain ethics approval before recruitment, use informed consent and fair compensation, minimise exposure to harmful misinformation, provide withdrawal and debriefing procedures, and recruit both subject-matter experts and lay raters for distinct questions. The protocol should preregister realism and factuality rubrics, blind raters to condition, measure inter-rater agreement, and compare human, independent-model, and current-auditor scores. Human review should complement, not merely rubber-stamp, automated scoring.

7.5 Priority future work

The next study should prioritise a smaller, preregistered validation design over a larger unreplicated grid. Such a study should use independent seeds, matched compositions, longer horizons, cross-model generation and judging, and blinded human annotation. In parallel, empirical comparison should remain structural until multiple lawfully obtained, timestamped cascades and a validated content measure are available. Only after those conditions are met would it be reasonable to test claims about generalisable topology effects, real-world firestorm timing, or predictive correspondence with social-media misinformation.

8 Conclusion

This thesis examined how persona composition and network conditions affect outputs from two factual-degradation instruments and interpreted the design using Pfeffer’s firestorm framework. The evidence supports a bounded measurement-sensitivity study. It does not support a model of real Twitter users, and it does not test whether an online firestorm occurred.

8.1 Answers to the research questions

RQ1: measurement. The continuous and dual-discrete auditors produced different levels and weakly aligned network-condition rankings. Dual-discrete live-cell mean MI exceeded continuous live-cell mean MI on all eight network-condition aggregates in both identity arms. The mean topology gaps were 0.8201 for H and 1.0450 for He. These values show whole-pipeline sensitivity, including different missing-item penalties. They do not show that one auditor measured the true amount of misinformation. Continuous and discrete scores must therefore remain separate.

RQ2: persona composition. Named composition described the results more consistently than the pooled H/He label. Homogeneous conspiracy-family means were 2.2194 continuous and 3.1691 dual-discrete. Science and environment means were 0.1118 and 0.8687. The conspiracy-heavy `mix_00` also exceeded conspiracy-free `mix_02` under both instruments. These contrasts apply to the executed prompts and model. They do not establish a general effect of human diversity.

RQ3: network conditions and survival. Bundled network conditions altered same-regime H/He contrasts, live-cell mean MI, and scorable-cell survival, but no universal ranking emerged. Continuous He–H differences changed sign across the eight graphs. Dual-discrete differences were positive on all eight. Dead-cell rates also varied by condition, and live-only means describe surviving cascades. The observed k^* values are eight-tick, instrument-specific outcomes. They are not empirical ignition times.

RQ4: constructed-graph stress test. The 63-node D-net used four configurations and eight article-level cells on a constructed hashtag graph. Conspiracy-only H exceeded the mixed assignment for both articles under both instruments. This recurrence is not a replication. Structural correspondence remained partial. Breadth met the selected tolerance, whereas depth and structural virality did not. The Digital Twin Fidelity Score was 0.2754 and the implemented check failed. Homogeneous one-community modularity is $Q = 0$. D-net is therefore a documented stress test, not a validated retweet cascade or a source of empirical Twitter MPR.

8.2 Contributions and boundary

The thesis contributes a theory-to-design ledger, a 1,728-cell climate factorial, a measurement-sensitivity analysis, a composition-conditioned interpretation, and a documented 63-node stress test. The canonical harvest contains 288 configurations, 1,728 cells, and 292 hatched-dead cells. It uses `gpt-4o-mini`, eight processing ticks, and one retained run per configuration. The C5 “combined” result is a table union. It is not a physical supergraph. Its execution is traceable but not exactly reproducible: the selected metadata record 64,270 HTTP-error responses, failed rewrites can propagate unchanged text, and failed audits remain unscored. No auditor parse/all-correct fallback was observed. Restricting the analysis to zero-error runs reverses the small continuous He–H contrast (-0.0295 rather than $+0.1549$); the dual-discrete contrast remains positive. Accordingly, the continuous collapsed-arm direction is not a robust headline conclusion. No human factuality, realism, or persona-validity evaluation was conducted.

Pfeffer, Zorbach, and Carley identified seven Outlook factors (Pfeffer et al., 2014). The six-knob vocabulary in this thesis is a computational remapping. Cross-media dynamics remained held. Empirical timing was unavailable. The engine’s ternary actions did not implement the original binary-choice factor. These omissions prevent a seven-factor confirmation.

8.3 Outlook

The next study should prioritise a smaller, preregistered validation design over a larger unreplicated grid. Independent graph and model seeds should support uncertainty estimates. Matched persona compositions should isolate diversity from prevalence. Independent human coders should assess factuality and realism under an approved protocol. Longer horizons should test censoring at tick eight. Lawfully obtained timestamped interaction data would permit a genuine temporal and cascade comparison. Until independent replication, human validation, longer-horizon analysis, and empirical temporal data are available, the testbed should be used only for mechanism-oriented simulation and measurement-sensitivity analysis. This revision does not assign or guarantee a grade.

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A Reproducibility materials

This appendix records the executable specification of the Phase 2 campaign. Paths are relative to the repository root. The file `thesisExperiment/configs/grid_phase2.json` provides the canonical machine-readable design. If the appendix conflicts with a generated artefact, readers should follow this file and the raw run metadata. Configurations recorded `graphRandomSeed: 42`, but only the echo-chamber and polarised builders consumed it; ER, small-world, scale-free, and action sampling used unseeded `Math.random`. The parser retained one complete positive-call directory per configuration. Continuous and dual-discrete misinformation indices are separate instruments and must not be averaged.

A.1 Configuration matrix

Table A.1: Phase 2 factorial configuration matrix. A cell is one configuration–article pair. D-net and probes are outside the 1,728-cell factorial grid.

Slice	Instrument	Identity arm	Topologies	Identity settings	Configs	Article cells
T2c H	continuous	H	8	12 perso- nas	$8 \times 12 = 96$	$96 \times 6 = 576$
T2d H	dual; dis- crete head- line	H	8	12 perso- nas	$8 \times 12 = 96$	$96 \times 6 = 576$
T2c He	continuous	He	8	6 mixes	$8 \times 6 = 48$	$48 \times 6 = 288$
T2d He	dual; dis- crete head- line	He	8	6 mixes	$8 \times 6 = 48$	$48 \times 6 = 288$
T2 total	two instru- ments	H and He	8	18 set- tings	288	1,728
D-net	continuous or dual	H or He	custom	2 assign- ments	4	8
Probes	field checks	selected	custom or chain	selected	5 configs	5 parsed rows

All T2 configurations use `gpt-4o-mini` for both the propagating agent and auditor, eight nodes, eight processing ticks, a configured maximum of eight hops, a single-node single-time

seed into `node_0`, an inbox cap of four, an always-active schedule, relation evolution with `trustDelta=0.05`, and six seed articles. Linear-chain and ring configurations use action weights 0.10/0.85/0.05 for forward/reinterpret/drop and trust threshold 0.08. The other six main-grid generators use 0.35/0.50/0.15 and threshold 0.15. Output is isolated under `thesisExperiment/runs_phase2/`; Phase 1 `runs/`, `results/tables/`, and `configs/full/` are not inputs to the Phase 2 headlines.

Configuration names have the grammar shown in Listing A.1. The parser matches the longest known topology token before interpreting the remainder as a persona or mix.

Listing A.1: Phase 2 configuration-name grammar.

```
T2{c|d}_{H|He}_{topology}_{persona-id|mix-id}

c  -> miScoringMode = "continuous"
d  -> miScoringMode = "dual" (headline is discrete)
H  -> one homogeneous persona
He -> one heterogeneous eight-seat mix
```

Examples:

```
T2c_H_linear_chain_climate_scientist
T2d_He_polarized_mix_00
```

A.2 Network topology definitions

Table A.2: The eight directed, eight-node T2 topology generators.

Identifier	Parameters	Construction
<code>linear_chain</code>	edge trust 0.7	Directed edges $v_i \rightarrow v_{i+1}$ for $i = 0, \dots, 6$.
<code>ring</code>	edge trust 0.7	Linear chain plus $v_7 \rightarrow v_0$.
<code>random_er</code>	$p = 0.42$; recorded <code>minSeedOutDegree=2</code> is ignored	Each ordered non-self pair is sampled with unseeded <code>Math.random</code> ; no seed-degree guard is applied; edge trust is $0.3 + U(0, 0.5)$.
<code>small_world</code>	$k = 4, \beta = 0.1$	Directed ring lattice to two neighbours on each side; forward-side edges are rewired with probability β .
<code>scale_free</code>	Barabási–Albert $m = 2$	A fully connected $m+1$ seed is extended by preferential attachment; selected old–new pairs receive edges in both directions.
<code>echo_chamber</code>	2 chambers; $p_{\text{in}} = 0.75$, $p_{\text{out}} = 0.08$; trust 0.85/0.15; seed out- degree 2	Alternating chamber membership; dense high-trust within-chamber and sparse low-trust cross-chamber directed edges.

Table A.2 continued

Identifier	Parameters	Construction
polarized	$p_{\text{in}} = 0.75$, $p_{\text{out}} = 0.05$; trust 0.88/0.10; seed out-degree 2	Two tightly connected halves with sparse, low-trust cross-cluster edges.
hierarchical	branching factor 3; down/up trust 0.85/0.30	Rooted three-ary tree with parent–child and weaker child–parent edges.

Homophily-adjusted generators calculate trust from the Jaccard overlap of persona tags and clamp it to $[0.05, 0.95]$. The stored `graph_topology.json`, rather than the generator label alone, is the authoritative realised graph. D-net is not a ninth T2 generator: it is a 63-node, 228-directed-edge custom graph copied from the reconstructed hashtag co-occurrence object.

A.3 Persona and article inventory

A.3.1 Personas

Table A.3: Merged Phase 2 persona inventory. “H” marks membership in the 12-person homogeneous grid; the other five personas are available to heterogeneous mixes or graph cluster pools.

Persona identifier	Family/type	H grid	Operational role
conspiracy_believer	conspiracy	yes	high-centrality chemtrails account
conspiracy_haarp_weather	conspiracy	yes	HAARP and weather-warfare frame
conspiracy_depopulation	conspiracy	yes	depopulation and Gates-target frame
conspiracy_climate_pigggyback	conspiracy	yes	climate-action hashtag hijack
climate_action_advocate	climate action	yes	mitigation, justice, accountability
climate_justice_youth	climate action	yes	youth and intergenerational justice
mitigation_first_policy	climate action	yes	policy and SRM-governance analysis
environmental_concern	environment	yes	general ecological-risk frame
ozone_stratosphere_specialist	environment	yes	atmospheric chemistry and ozone
biodiversity_food_security	environment	yes	ecosystems, rainfall, and crops

Table A.3 continued

Persona identifier	Family/type	H grid	Operational role
climate_scientist	expert	yes	assessed-science correction
science_journalist	expert/media	yes	sourced factual summary
conspiracy_peripheralskywatcher	conspiracy	no	low-centrality first-person sky report
climate_action_sweden_scopex	climate action	no	Swedish SCoPEX consent protest
platform_moderator_toxicity	institution	no	procedural harm reduction
caregiver_air_quality	env./health	no	household health and air quality
hashtag_attention_communicator	communication	no	high-visibility hashtag amplification

These are theory-faithful reductions of discourse roles, not HDBSCAN centroids estimated from the unavailable 814,924 tweets. The full prompt text and metadata are in `thesisExperiment/personas/merged.json`; compact per-condition files are in `thesisExperiment/personas/phase2/`. The six heterogeneous mixes, in node-seat order, are:

Listing A.2: Heterogeneous eight-seat persona mixes.

```

mix_00 = [conspiracy_believer, conspiracy_haarp_weather,
          conspiracy_depopulation, conspiracy_climate_piggyback,
          climate_action_advocate, climate_justice_youth,
          mitigation_first_policy, environmental_concern]
mix_01 = [conspiracy_depopulation, conspiracy_climate_piggyback,
          climate_action_advocate, climate_justice_youth,
          mitigation_first_policy, environmental_concern,
          ozone_stratosphere_specialist, biodiversity_food_security]
mix_02 = [climate_action_advocate, climate_justice_youth,
          mitigation_first_policy, environmental_concern,
          ozone_stratosphere_specialist, biodiversity_food_security,
          climate_scientist, science_journalist]
mix_03 = [mitigation_first_policy, platform_moderator_toxicity,
          ozone_stratosphere_specialist, climate_action_sweden_scopex,
          biodiversity_food_security, climate_scientist,
          conspiracy_believer, climate_justice_youth]
mix_04 = [climate_action_advocate, conspiracy_peripheral_skywatcher,
          science_journalist, biodiversity_food_security,
          climate_action_sweden_scopex, platform_moderator_toxicity,
          ozone_stratosphere_specialist, conspiracy_depopulation]
mix_05 = [climate_scientist, climate_action_advocate,
          conspiracy_climate_piggyback, biodiversity_food_security,
          platform_moderator_toxicity, science_journalist,
          ozone_stratosphere_specialist, conspiracy_peripheral_skywatcher]

```

These declared orders are realised for linear, ring, ER, small-world, and scale-free generators. Echo-chamber, polarised, and hierarchical configurations assign from filtered cluster pools and, for five of six mixes, realise only four or five unique personas. Polarised assignment alternates by node index and does not align with the graph’s two contiguous halves.

A.3.2 Articles

Table A.4: Merged article inventory. Every article contains five yes/no ground-truth questions. “Core” marks the six articles used in every T2 configuration.

Article identifier	Domain	Use	Short topic
scopex_2017	geoengineering	core, D-net	SCoPEX measurement proposal
chemtrails_gates_2018_2021	misinformation	core, D-net	chemtrails, SCoPEX, and Gates
sai_geoengineering	geoengineering	core	SAI proposal versus deployment
paris_agreement	policy	core	Paris treaty and NDCs
climate_consensus	science	core	anthropogenic-warming consensus
polar_bears	science	core	sea ice and hunting habitat
spice_uk_cancelled	geoengineering	library	cancelled SPICE field test
glaciers_retreat	science	library	global glacier retreat
co2_fertilization	misinformation	library	plant growth versus climate harm
sea_level_rise	science	library	thermal expansion and land ice
net_zero	policy	library	emissions-removals balance
attribution_extremes	science	library	probabilistic event attribution
paris_agreement_2015	policy	library	expanded Paris explainer
consensus_97_percent	science	library	conditional 97% consensus
glacier_retreat	science	library	WGMS glacier mass balance
polar_bears_sea_ice	science	library	IUCN and ice-season evidence
co2_plant_food	science	library	fertilisation and ocean acidification
sai_solar_geoenginee ring	geoengineering	library	SAI risks and termination shock
net_zero_2050	policy	library	carbon budgets and mid-century pathways
extreme_weather_attr ibution	science	library	likelihood and intensity

Table A.4 continued

Article identifier	Domain	Use	Short topic
climatefever_polar_bears	science	library	ClimateFEVER evidence labels
climatefever_co2_claims	science	library	ocean-degassing and greenhouse claims
volcanic_vs_human_co2	science	library	volcanic versus anthropogenic emissions
great_barrier_reef_bleaching	science	library	marine heat and bleaching

The source text, exact questions, expected Boolean answers, and source URLs are stored in `thesisExperiment/articles/merged.json`. Scholarly provenance for the six core articles is summarised in Table 4.1. The inventories above identify experimental inputs; they do not reproduce the full stimuli.

A.4 Prompt templates and IFD fields

Each reinterpretation uses the selected persona’s `systemPrompt` and the fixed user template below. Forward actions make no rewrite call. Agent and auditor outputs are model outputs, not human ratings.

Listing A.3: Rewrite user prompt from `src/SimulationNode.js`.

```
Here is the information you received. Rewrite it through your
perspective in 100-200 words. Return only the rewritten text,
no preamble:
```

```
<incoming content>
```

Listing A.4: Discrete and continuous auditor templates from `src/Auditor.js`.

DISCRETE SYSTEM

You are a fact-checker. For each question, the expected correct answer is provided.

Evaluate the given text and return a score for each question:

1 = text correctly states the expected answer

0 = text does not contain enough information to address the question

-1 = text contradicts or distorts the expected answer

Return only a JSON object: {"answers": [1, 0, -1, ...]}

CONTINUOUS SYSTEM

You are a fact-checker. For each question, the expected correct answer is provided.

Rate how accurately the given text addresses each question on a scale from 0.0 to 1.0:

1.0 = text fully and correctly states the expected answer

0.5 = text does not address this question (missing or neutral)
 0.0 = text directly contradicts or inverts the expected answer
 Intermediate values represent partial accuracy.
 Return only a JSON object: {"scores": [1.0, 0.5, 0.8, ...]}

USER, BOTH MODES

Text:

<text>

Questions (with expected answers):

1. <question> (expected: Yes|No)

...

IFD_SCORE_QUERY: Return only the JSON with values 1, 0, or -1.

CONTINUOUS_SCORE_QUERY: Return only the JSON with float scores 0.0 to 1.0.

For $m = 5$ questions, discrete scores $s_i \in \{-1, 0, 1\}$ produce

$$MI_d = n_{\text{missing}} + n_{\text{incorrect}}, \quad CR = n_{\text{correct}}/m, \quad MR = n_{\text{missing}}/m, \quad IR = n_{\text{incorrect}}/m.$$

Continuous scores $q_i \in [0, 1]$ produce

$$CR = \frac{1}{m} \sum_i q_i, \quad MI_c = m(1 - CR).$$

For descriptive CR/MR/IR counts in continuous mode, $q_i \geq 0.67$ is correct, $q_i \leq 0.33$ is incorrect, and intermediate values are missing. In dual mode two calls are made in parallel. The top-level IFD and `misinfoIndex` mirror the discrete result; the sidecar stores both results, absolute gap $|MI_d - MI_c|$, and Pearson agreement after mapping $-1, 0, 1$ to $0, 0.5, 1$.

Table A.5: Fields in an event's `ifd` object.

Field	Type/range	Definition
<code>mi</code>	$[0, 5]$	Discrete count or continuous $m(1 - CR)$.
<code>cr, mr, ir</code>	$[0, 1]$	Correct, missing, and incorrect rates.
<code>cms</code>	non-negative	Discrete $IR/(CR + 10^{-9})$; continuous $(1 - CR)/(CR + 10^{-9})$.
<code>ie</code>	non-negative	Entropy-like score over the mode's rate terms.
<code>scores</code>	array length 5	Per-question discrete or continuous scores.
<code>mode</code>	string	<code>discrete</code> , <code>continuous</code> , or <code>dual</code> .
<code>dual.discrete</code>	object	Complete discrete IFD object.
<code>dual.continuous</code>	object	Complete continuous IFD object.
<code>dual.gap</code>	float	Absolute MI difference; a diagnostic, not a third MPR.
<code>dual.agreement</code>	$[-1, 1]$ or null	Pearson correlation of aligned question scores.

On auditor JSON parse failure the current engine defaults discrete questions to 1 and continuous questions to 1.0, emitting a warning. This is a consequential implementation

choice and is retained here for reproducibility.

A.5 Raw run schema

Listing A.5: Raw run directory layout.

```
runs_phase2/<experimentName>_<YYYY-MM-DD_HH-MM-SS>/
  metadata.json
  state.json
  graph_topology.json
  human_eval_template.csv
  results_<articleId>.json
  nodes/node_0.json ... nodes/node_7.json
```

T2 runs contain six `results_*.json` files. D-net contains two and stores 63 `nodes/user_*.json` files; probes contain one result. The parser selects the newest complete directory with `llmUsage.calls>0` for each `experimentName`.

Table A.6: Raw JSON schema by file.

Object	Fields and interpretation
<code>metadata.json</code>	<code>experimentId</code> , <code>status</code> , <code>timestamp</code> , full config, <code>llmUsage</code> , and per-article <code>results</code> . <code>config.experimentName</code> is the reliable name when the optional top-level name is null.
<code>llmUsage</code>	<code>calls</code> , <code>promptTokens</code> , <code>completionTokens</code> , <code>totalTokens</code> , <code>errors</code> , <code>model</code> , and <code>estimatedUsd</code> . No credential is stored.
<code>state.json</code>	<code>experimentId</code> , <code>phase</code> , <code>status</code> , propagation progress, audit progress, <code>error</code> , and <code>lastUpdated</code> .
<code>node_*.json</code>	<code>nodeId</code> , <code>personaId</code> , <code>modelId</code> , outgoing relations, node params, inbox, event history, and action stats.
History event	<code>tick</code> , <code>articleId</code> , <code>sourceNodeId</code> , <code>hops</code> , <code>action</code> , <code>contentIn</code> , <code>contentOut</code> , <code>misinfoIndex</code> , <code>ifd</code> , <code>frameAnalysis</code> , <code>reason</code> , <code>chainTrust</code> , <code>provenance</code> , and <code>timestamp</code> .
<code>results_<article>.json</code>	<code>nodeSummaries</code> , <code>metrics</code> , <code>provenanceMetrics</code> , and <code>networkEvolution</code> . Typical metrics include network MI over time, IFD metrics, structural virality, critical mass, and persona IFD; dual runs add dual metrics.
<code>graph_topology.json</code>	realised node records and directed <code>{from,to,trust}</code> edges.

A.6 Parsed table schema

The canonical wide schema is shared by `TH_rows.csv`, `The_rows.csv`, `continuous.csv`, `dual_discrete.csv`, `dead_cells.csv`, and `all_rows.csv`.

Table A.7: Canonical parsed-cell fields, in CSV column order.

Field	Meaning
<code>runDir</code>	Selected timestamped raw directory.
<code>experimentName</code>	Stable configuration identifier.
<code>slice, arm</code>	T2 slice and H/He arm.
<code>topology, rest</code>	Generator/custom label and persona or mix identifier.
<code>miScoringMode</code>	<code>continuous</code> or <code>dual</code> .
<code>articleId</code>	Seed article; together with experiment name forms the cell key.
<code>status, llmCalls</code>	Run completion state and real model-call count.
<code>nEvents, nScored</code>	Collected events and events with non-null MI.
<code>dead</code>	True iff <code>nScored</code> ≤ 1.
<code>meanMI</code>	Mean event <code>misinfoIndex</code> ; the cell headline.
<code>meanNodeMPR</code>	Mean of per-node mean MI; shown beside, not merged with, <code>meanMI</code> .
<code>maxMI</code>	Maximum scored-event MI.
<code>meanDiscreteMI</code>	Mean discrete MI where available.
<code>meanContinuousMI</code>	Mean continuous MI, including dual sidecar where available.
<code>meanDualGap</code>	Mean absolute dual gap; not an MPR.
<code>meanAgreement</code>	Mean dual per-event Pearson agreement.
<code>kStarDiscrete</code>	First tick at which mean discrete MI exceeds 3 and remains above 3.
<code>kStarContinuous</code>	Continuous equivalent of k^* .
<code>kStarCensoredDiscr</code>	True when discrete k^* occurs at the last observed tick.
<code>ete</code>	
<code>hatchStatus,</code> <code>hatchNote</code>	Live/dead rendering status and interpretation warning.

The file `dual_gap.csv` is narrow. Its fields are `experimentName`, `articleId`, `topology`, `meanDualGap`, `meanAgreement`, `meanDiscreteMI`, and `meanContinuousMI`. Join it to the wide dual table with experiment name and article identifier as the composite key. Pair T2c with T2d with topology, persona or mix, and article identifier as the composite key. Never join by row number. Dead cells remain in the ledger and are excluded from live means rather than imputed as zero.

A.7 D-net mapping

The empirical object is `thesisExperiment/data/derived/debnath_hashtag_cascade.json`, schema `thesisExperiment.debnath_hashtag_cascade.v1`. It is a hashtag co-occurrence graph, not a retweet or `conversation_id` cascade; its `retweets[]` array is an importer edge container. It has 63 nodes, 228 directed edges, seed `user_chemtrails_hub`,

no empirical MPR, and no empirical time series. D-net scores only `scopex_2017` and `chemtrails_gates_2018_2021`.

Table A.8: D-net identity mapping counts.

Layer	Conspiracy	Climate action	Environmental	Expert/other
Empirical cluster identity	28	19	14	2
Intended BP family	28	19	13	3
Mixed D-net persona family	28	19	13	3
Homogeneous D-net	63	0	0	0

The mixed assignment exactly matches the reconstructed intended persona ID for 56 of 63 seats. The seven coarsened assignments preserve discourse type:

Table A.9: D-net seats whose mixed persona ID is coarser than the reconstruct mapping.

Node	Hashtag	Intended BP	Mixed seat
<code>user_depopulation_amp</code>	<code>#depopulation</code>	<code>conspiracy_depopulation</code>	<code>conspiracy_believer</code>
<code>user_climateaction_piggyback_amp</code>	<code>#climateaction</code>	<code>conspiracy_climate_piggyback</code>	<code>conspiracy_believer</code>
<code>user_climateaction_piggyback_peri_1</code>	<code>#climateaction</code>	<code>conspiracy_climate_piggyback</code>	<code>conspiracy_believer</code>
<code>user_biodiversity_hub</code>	<code>#biodiversity</code>	<code>biodiversity_food_security</code>	<code>environmental_concern</code>
<code>user_biodiversity_peri_1</code>	<code>#biodiversity</code>	<code>biodiversity_food_security</code>	<code>environmental_concern</code>
<code>user_foodsecurity_amp</code>	<code>#foodsecurity</code>	<code>biodiversity_food_security</code>	<code>environmental_concern</code>
<code>user_mitigation_amp</code>	<code>#mitigation</code>	<code>mitigation_first_policy</code>	<code>climate_action_advocate</code>

The complete 63-row node map is retained in `thesisExperiment/analysis_full/PFE_FFER_DEBNATH_VS_SIM.md`; duplicating it here would create a second canonical mapping. Homogeneous D-net deliberately overwrites all 63 seats with `conspiracy_believer`. H and He use the same edge set, so clustering is fixed while label-dependent homophily changes.

A.8 Reproduction commands

Run the following from the repository root. Parsing and plotting consume existing runs; the D-net and campaign runners can make paid API calls and are shown only for provenance.

Listing A.6: Parse, compare, replot, and compile commands.

```
# Rebuild canonical Phase 2 CSVs and parse summaries.
node thesisExperiment/scripts/parse_phase2.js
```

```

# Recompute the structural Debnath/D-net comparison only if its
# empirical topology has changed.
node thesisExperiment/scripts/compare_phase2.js

# Replot both analysis layers.
python3 thesisExperiment/analysis_phase2/plot_phase2.py
python3 thesisExperiment/analysis_full/analyze_full.py

# Recompute the Pfeffer/Debnath mapping overlay without new LLM runs.
node thesisExperiment/scripts/pfeffer_debnath_vs_sim.js

# Compile the existing standalone Phase 2 LaTeX chapter.
make -C thesisExperiment/latex_phase2
# Equivalent:
cd thesisExperiment/latex_phase2
pdflatex -interaction=nonstopmode -halt-on-error main
biber main
pdflatex -interaction=nonstopmode -halt-on-error main
pdflatex -interaction=nonstopmode -halt-on-error main

# Potentially paid execution commands; normally skip because harvest
# is complete.
node thesisExperiment/scripts/run_dnet.js
node thesisExperiment/scripts/master_phase2.js --skip-probe

```

This appendix is compiled with the thesis master `thesisExperiment/thesis_final/main.tex`. Rebuild from `thesisExperiment/thesis_final/` with `latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex`, as recorded in `thesisExperiment/thesis_final/BUILD.md`.

A.9 Artifact index

Table A.10: Canonical Phase 2 artefact paths.

Path	Purpose
<code>thesisExperiment/configs/grid_phase2.json</code>	machine-readable design and inventories
<code>thesisExperiment/configs/phase2/</code>	288 T2 configs, four D-net configs, index, and probes
<code>thesisExperiment/personas/merged.json</code>	17-person merged library
<code>thesisExperiment/personas/phase2/</code>	exact H and He persona files
<code>thesisExperiment/articles/merged.json</code>	24 articles with questions and expected answers

Table A.10 continued

Path	Purpose
thesisExperiment/runs_phase2/	raw Phase 2 runs and status records
thesisExperiment/results_phase2/tables/	canonical parsed CSVs
thesisExperiment/results_phase2/summary.json	harvest counts and generated-at stamp
thesisExperiment/results_phase2/PARSE.md	human-readable parse ledger
thesisExperiment/results_phase2/debnath_compare.json	structural comparison
thesisExperiment/analysis_phase2/	first analysis and plots
thesisExperiment/analysis_full/	C1–C5 tables, plots, notes, and key numbers
thesisExperiment/analysis_full/PFEFFER_DEBNATH_VS_SIM.md	full D-net map and observable audit
thesisExperiment/data/derived/debnath_hashtag_cascade.json	63-node reconstructed custom graph
thesisExperiment/latex_phase2/	standalone Phase 2 results chapter
thesisExperiment/AGENT_HANDOFF.md	detailed schema and harvest handoff
thesisExperiment/CONSOLIDATION_LOG.md	branch provenance, overlap resolutions, and canonical 292-dead-cell decision
thesisExperiment/CONSOLIDATED_OVERLAPS/	retained non-canonical conflict copies

A.10 Data availability and consolidation provenance

A public persistent DOI was not present in the repository when this statement was written and is not invented here. The present PDF therefore documents a GitHub-hosted, commit-addressable harvest rather than a DOI-backed release. The public repository URL is https://github.com/RajGM/LLM_Society. The `origin/main` commit recorded here is 042023488cefb343e739ad4b4bdba5d23171ff48 (author date 2026-09-19 17:12:23 UTC). No LICENSE file was found at the repository root. The README licence badge claims MIT; `package.json` lists ISC. Those claims do not substitute for a licence file. **[USER INPUT REQUIRED: choose and deposit a licence file; add an archive DOI only if one exists]** SHA-256 checksums of canonical harvest index files, computed from the working tree used to compile this PDF:

```
thesisExperiment/results_phase2/summary.json
198f9efd40a424a561a6f85a2cca88e071992a9bda305a93a99d4c0cdb285b27

thesisExperiment/results_phase2/PARSE.md
21b030d54897aba7832a21d975b09dd5b49099198563544e65a2cb2f23f4f95b

thesisExperiment/CONSOLIDATION_LOG.md
275262b044c551109594e12912091de23394d812dd1009ec96d0ce772d81fe70

thesisExperiment/analysis_full/key_numbers.json
55a4d4e6e4253d088d6fb5ec780f5d38388c51f414baa9b45f89ec30ac0c3835

thesisExperiment/data/derived/debnath_hashtag_cascade.json
35a6afa4afb63b5da7d2084892469aea601094c1d5bc0784c2e0ed0268ccf673
```

API credentials must not appear in configurations, manuscripts, terminal transcripts, issues, manifests, committed files, or printed `.env` contents.

The consolidation record is `thesisExperiment/CONSOLIDATION_LOG.md`. It identifies source branches, merge order, retained overlap copies, official artefact locations, and the resolution of the historical 290-versus-292 dead-cell discrepancy. The canonical parse ledger contains 292 dead T2 cells. The repository retains older 290-cell analysis copies under `thesisExperiment/CONSOLIDATED_OVERLAPS/analysis_phase2_hatch290/` for provenance; these copies are non-canonical and must not be used in place of the 292-cell tables.

